R Notebook

```
library(latex2exp)
library(ramify)
```

```
## Warning: package 'ramify' was built under R version 4.3.3
```

```
##
## Attaching package: 'ramify'
```

```
## The following object is masked from 'package:graphics':
##
##     clip
```

```
library(imageseg)
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
## Warning: package 'lattice' was built under R version 4.3.3
```

```
library(tools)
library(xfun)
```

```
## Warning: package 'xfun' was built under R version 4.3.3
```

```
##
## Attaching package: 'xfun'
```

```
## The following objects are masked from 'package:tools':
##
##     file_ext, Rcmd
```

```
## The following objects are masked from 'package:base':
##
##     attr, isFALSE
```

```
library(EBImage)
```

```
##
## Attaching package: 'EBImage'
```

```
## The following object is masked from 'package:ramify':
##
##     resize
```

```r
library(stringi)
```

```
## Warning: package 'stringi' was built under R version 4.3.3
```

```r
library(ROCR)
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
library(reticulate)
```

```
## Warning: package 'reticulate' was built under R version 4.3.3
```

```r
library(tensorflow)
```

```
## Warning: package 'tensorflow' was built under R version 4.3.3
```

```
##
## Attaching package: 'tensorflow'
```

```
## The following object is masked from 'package:caret':
##
##     train
```

```r
library(keras)
```

```
## Warning: package 'keras' was built under R version 4.3.3
```

```
##
## Attaching package: 'keras'
```

```
## The following object is masked from 'package:EBImage':
##
##     normalize
```

```
library(keras3)
```

```
## Warning: package 'keras3' was built under R version 4.3.3
```

```
## Registered S3 methods overwritten by 'keras3':
##   method                               from
##   as.data.frame.keras_training_history keras
##   plot.keras_training_history          keras
##   print.keras_training_history         keras
##   r_to_py.R6ClassGenerator             keras
```

```
##
## Attaching package: 'keras3'
```

```
## The following objects are masked from 'package:keras':
## 
##     %<-active%, %py_class%, activation_elu, activation_exponential,
##     activation_gelu, activation_hard_sigmoid, activation_linear,
##     activation_relu, activation_selu, activation_sigmoid,
##     activation_softmax, activation_softplus, activation_softsign,
##     activation_tanh, adapt, application_densenet121,
##     application_densenet169, application_densenet201,
##     application_efficientnet_b0, application_efficientnet_b1,
##     application_efficientnet_b2, application_efficientnet_b3,
##     application_efficientnet_b4, application_efficientnet_b5,
##     application_efficientnet_b6, application_efficientnet_b7,
##     application_inception_resnet_v2, application_inception_v3,
##     application_mobilenet, application_mobilenet_v2,
##     application_mobilenet_v3_large, application_mobilenet_v3_small,
##     application_nasnetlarge, application_nasnetmobile,
##     application_resnet101, application_resnet101_v2,
##     application_resnet152, application_resnet152_v2,
##     application_resnet50, application_resnet50_v2, application_vgg16,
##     application_vgg19, application_xception, bidirectional,
##     callback_backup_and_restore, callback_csv_logger,
##     callback_early_stopping, callback_lambda,
##     callback_learning_rate_scheduler, callback_model_checkpoint,
##     callback_reduce_lr_on_plateau, callback_remote_monitor,
##     callback_tensorboard, clone_model, constraint_maxnorm,
##     constraint_minmaxnorm, constraint_nonneg, constraint_unitnorm,
##     count_params, custom_metric, dataset_boston_housing,
##     dataset_cifar10, dataset_cifar100, dataset_fashion_mnist,
##     dataset_imdb, dataset_imdb_word_index, dataset_mnist,
##     dataset_reuters, dataset_reuters_word_index, freeze_weights,
##     from_config, get_config, get_file, get_layer, get_vocabulary,
##     get_weights, image_array_save, image_dataset_from_directory,
##     image_load, image_to_array, imagenet_decode_predictions,
##     imagenet_preprocess_input, initializer_constant,
##     initializer_glorot_normal, initializer_glorot_uniform,
##     initializer_he_normal, initializer_he_uniform,
##     initializer_identity, initializer_lecun_normal,
##     initializer_lecun_uniform, initializer_ones,
##     initializer_orthogonal, initializer_random_normal,
##     initializer_random_uniform, initializer_truncated_normal,
##     initializer_variance_scaling, initializer_zeros, install_keras,
##     keras, keras_model, keras_model_sequential, Layer,
##     layer_activation, layer_activation_elu,
##     layer_activation_leaky_relu, layer_activation_parametric_relu,
##     layer_activation_relu, layer_activation_softmax,
##     layer_activity_regularization, layer_add, layer_additive_attention,
##     layer_alpha_dropout, layer_attention, layer_average,
##     layer_average_pooling_1d, layer_average_pooling_2d,
##     layer_average_pooling_3d, layer_batch_normalization,
##     layer_category_encoding, layer_center_crop, layer_concatenate,
##     layer_conv_1d, layer_conv_1d_transpose, layer_conv_2d,
##     layer_conv_2d_transpose, layer_conv_3d, layer_conv_3d_transpose,
##     layer_conv_lstm_1d, layer_conv_lstm_2d, layer_conv_lstm_3d,
##     layer_cropping_1d, layer_cropping_2d, layer_cropping_3d,
##     layer_dense, layer_depthwise_conv_1d, layer_depthwise_conv_2d,
```

```
##     layer_discretization, layer_dot, layer_dropout, layer_embedding,
##     layer_flatten, layer_gaussian_dropout, layer_gaussian_noise,
##     layer_global_average_pooling_1d, layer_global_average_pooling_2d,
##     layer_global_average_pooling_3d, layer_global_max_pooling_1d,
##     layer_global_max_pooling_2d, layer_global_max_pooling_3d,
##     layer_gru, layer_hashing, layer_input, layer_integer_lookup,
##     layer_lambda, layer_layer_normalization, layer_lstm, layer_masking,
##     layer_max_pooling_1d, layer_max_pooling_2d, layer_max_pooling_3d,
##     layer_maximum, layer_minimum, layer_multi_head_attention,
##     layer_multiply, layer_normalization, layer_permute,
##     layer_random_brightness, layer_random_contrast, layer_random_crop,
##     layer_random_flip, layer_random_rotation, layer_random_translation,
##     layer_random_zoom, layer_repeat_vector, layer_rescaling,
##     layer_reshape, layer_resizing, layer_rnn, layer_separable_conv_1d,
##     layer_separable_conv_2d, layer_simple_rnn,
##     layer_spatial_dropout_1d, layer_spatial_dropout_2d,
##     layer_spatial_dropout_3d, layer_string_lookup, layer_subtract,
##     layer_text_vectorization, layer_unit_normalization,
##     layer_upsampling_1d, layer_upsampling_2d, layer_upsampling_3d,
##     layer_zero_padding_1d, layer_zero_padding_2d,
##     layer_zero_padding_3d, learning_rate_schedule_cosine_decay,
##     learning_rate_schedule_cosine_decay_restarts,
##     learning_rate_schedule_exponential_decay,
##     learning_rate_schedule_inverse_time_decay,
##     learning_rate_schedule_piecewise_constant_decay,
##     learning_rate_schedule_polynomial_decay, loss_binary_crossentropy,
##     loss_categorical_crossentropy, loss_categorical_hinge,
##     loss_cosine_similarity, loss_hinge, loss_huber, loss_kl_divergence,
##     loss_mean_absolute_error, loss_mean_absolute_percentage_error,
##     loss_mean_squared_error, loss_mean_squared_logarithmic_error,
##     loss_poisson, loss_sparse_categorical_crossentropy,
##     loss_squared_hinge, mark_active, metric_auc,
##     metric_binary_accuracy, metric_binary_crossentropy,
##     metric_categorical_accuracy, metric_categorical_crossentropy,
##     metric_categorical_hinge, metric_cosine_similarity,
##     metric_false_negatives, metric_false_positives, metric_hinge,
##     metric_mean, metric_mean_absolute_error,
##     metric_mean_absolute_percentage_error, metric_mean_iou,
##     metric_mean_squared_error, metric_mean_squared_logarithmic_error,
##     metric_mean_wrapper, metric_poisson, metric_precision,
##     metric_precision_at_recall, metric_recall,
##     metric_recall_at_precision, metric_root_mean_squared_error,
##     metric_sensitivity_at_specificity,
##     metric_sparse_categorical_accuracy,
##     metric_sparse_categorical_crossentropy,
##     metric_sparse_top_k_categorical_accuracy,
##     metric_specificity_at_sensitivity, metric_squared_hinge,
##     metric_sum, metric_top_k_categorical_accuracy,
##     metric_true_negatives, metric_true_positives, new_callback_class,
##     new_layer_class, new_learning_rate_schedule_class, new_loss_class,
##     new_metric_class, new_model_class, normalize, optimizer_adadelta,
##     optimizer_adagrad, optimizer_adam, optimizer_adamax,
##     optimizer_ftrl, optimizer_nadam, optimizer_rmsprop, optimizer_sgd,
##     pad_sequences, pop_layer, predict_on_batch, regularizer_l1,
##     regularizer_l1_l2, regularizer_l2, regularizer_orthogonal,
##     set_vocabulary, set_weights, shape, test_on_batch,
```

```
##     text_dataset_from_directory, time_distributed,
##     timeseries_dataset_from_array, to_categorical, train_on_batch,
##     unfreeze_weights, use_backend, with_custom_object_scope, zip_lists
```

```
## The following objects are masked from 'package:tensorflow':
##
##     set_random_seed, shape
```

```
## The following object is masked from 'package:EBImage':
##
##     normalize
```

```
library(imager)
```

```
## Warning: package 'imager' was built under R version 4.3.3
```

```
## Loading required package: magrittr
```

```
##
## Attaching package: 'imager'
```

```
## The following object is masked from 'package:magrittr':
##
##     add
```

```
## The following object is masked from 'package:pROC':
##
##     ci
```

```
## The following objects are masked from 'package:EBImage':
##
##     channel, dilate, display, erode, resize, watershed
```

```
## The following objects are masked from 'package:ramify':
##
##     fill, resize
```

```
## The following objects are masked from 'package:stats':
##
##     convolve, spectrum
```

```
## The following object is masked from 'package:graphics':
##
##     frame
```

```
## The following object is masked from 'package:base':
##
##     save.image
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:imager':
##
##     where
```

```
## The following object is masked from 'package:EBImage':
##
##     combine
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(purrr)
```

```
##
## Attaching package: 'purrr'
```

```
## The following object is masked from 'package:magrittr':
##
##     set_names
```

```
## The following object is masked from 'package:EBImage':
##
##     transpose
```

```
## The following object is masked from 'package:caret':
##
##     lift
```

```
## The following object is masked from 'package:ramify':
##
##     flatten
```

```
library(skimr)
library(tm)
```

```
## Warning: package 'tm' was built under R version 4.3.3
```

```
## Loading required package: NLP
```

```
## Warning: package 'NLP' was built under R version 4.3.3
```

```
##
## Attaching package: 'NLP'
```

```
## The following object is masked from 'package:ggplot2':
##
##     annotate
```

```
library(RWeka)
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.3.3
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:purrr':
##
##     discard
```

```
library(cutpointr)
```

```
##
## Attaching package: 'cutpointr'
```

```
## The following objects are masked from 'package:pROC':
##
##     auc, roc
```

```
## The following objects are masked from 'package:caret':
##
##     precision, recall, sensitivity, specificity
```

```
library(magick)
```

```
## Warning: package 'magick' was built under R version 4.3.3
```

```
## Linking to ImageMagick 6.9.12.98
## Enabled features: cairo, freetype, fftw, ghostscript, heic, lcms, pango, raw, rsvg, webp
## Disabled features: fontconfig, x11
```

```
library(superml)
```

```
## Warning: package 'superml' was built under R version 4.3.3
```

```
## Loading required package: R6
```

```
library(grid)
```

```
##
## Attaching package: 'grid'
```

```
## The following object is masked from 'package:imager':
##
##     depth
```

```
library(crayon)
```

```
## Warning: package 'crayon' was built under R version 4.3.3
```

```
##
## Attaching package: 'crayon'
```

```
## The following object is masked from 'package:ggplot2':
##
##     %+%
```

```
library(stringr)
```

```
##
## Attaching package: 'stringr'
```

```
## The following object is masked from 'package:imager':
##
##     boundary
```

```
library(cowplot)
```

```
## Warning: package 'cowplot' was built under R version 4.3.3
```

```
##
## Attaching package: 'cowplot'
```

```
## The following object is masked from 'package:imager':
##
##     draw_text
```

```
library(patchwork)
```

```
## Warning: package 'patchwork' was built under R version 4.3.3
```

```
##
## Attaching package: 'patchwork'
```

```
## The following object is masked from 'package:cowplot':
##
##     align_plots
```

```
# Set the working directory
setwd("C:/Ngoc/MastersThesis/Draft")

main_folder <- "MRI_brain_images/Axial"
class_folders <- list.dirs(main_folder, full.names = FALSE) # -> class_folders[1] is an empty
string
class_folders <- stri_remove_empty(class_folders, na_empty = TRUE)
```

A/ Find training and test set AUC for a built - from - scratch CNN model:

Unlike some preprocessing steps in tabular data (like normalization / standardization, filling in missing values, …), format conversion and resizing do not involve calculations and transformations based on data distributions that may affect the learning process of the model. These two steps are just used to ensure consistent input format and size for the model to work properly on the dataset. Therefore, instead of:

- Checking for any inconsistencies in the format of the images based on just the training set, then based on the result, convert all images from the training, validation and test set to the same format,
- As well as finding the range of image height and width from just the training set, then based on the result, resize all images to the same size,

We can:

- Check for any inconsistencies in the format of the images based on the whole dataset, then based on the result, convert all images from the training, validation and test set to the same format,
- As well as finding the range of image height and width from the whole dataset, then based on the result, resize all images to the same size.

For this approach, we only need to apply resizing and format conversion steps once at the begining then save the processed images for future code runs, instead of having to re - run these steps every time we split the train - validation - test set differently. This is practical and helps simplify the coding process.

1. Check for any inconsistencies in the format of the images:

```
# ############## Turn to comments from the second run onwards ###############
# num_gray = 0
# num_rgb = 0
# num_other = 0
#
# for (class_folder in class_folders){
#   directory <- file.path(main_folder, class_folder)
#   files <- list.files(directory, full.names = TRUE, pattern = "\\.jpg$")
#   for (file in files){
#     tryCatch(
#       {
#         img <- image_read(file)
#         info <- image_info(img)
#         if (info$colorspace == "Gray"){
#           num_gray = num_gray + 1
#         }
#         else if (info$colorspace == "sRGB"){
#             num_rgb = num_rgb + 1
#         }
#         else {
#           num_other = num_other + 1
#           cat("Image format: ", info$colorspace, "\n", file, "\n\n")
#         }
#       },
#       error = function(e)
#       {
#       cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#   }
# }
#
# cat("Number of Grayscale images:", num_gray, "\n")
# cat("Number of RGB images:", num_rgb, "\n")
# cat("Number of other types of images:", num_other, "\n")
#
# # -> The dataset contains 1026 images in Grayscale format and 2543 images in RGB format
```

2. Convert all images with RGB format into Grayscale format:

```
# ############## Turn to comments from the second run onwards ###############
# for (class_folder in class_folders){
#   directory <- file.path(main_folder, class_folder)
#   files <- list.files(directory, full.names = TRUE, pattern = "\\.jpg$")
#   for (file in files){
#     tryCatch(
#       {
#         img <- image_read(file)
#         info <- image_info(img)
#         if (info$colorspace == "sRGB"){
#           # Turn RGB images into Grayscale, only keep values from the Red channel
#           img_gray <- image_channel(img, channel = "Red")
#
#           # Convert a magick object into a cimg image
#           image <- magick2cimg(img_gray)
#
#           # Save the change into the original dataset
#           save.image(image, file)
#         }
#       },
#       error = function(e)
#       {
#       cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#   }
# }
```

3. Find the range of image height and width in the dataset:

```
# ############# Turn to comments from the second run onwards ##############
# min_width <- Inf
# max_width <- 0
# min_height <- Inf
# max_height <- 0
# widths <- list()
# heights <- list()
#
# for (class_folder in class_folders){
#   directory <- file.path(main_folder, class_folder)
#   files <- list.files(directory, full.names = TRUE, pattern = "\\.jpg$")
#   for (file in files){
#     tryCatch(
#       {
#         img <- load.image(file)
#         width <- dim(img)[1]
#         height <- dim(img)[2]
#         widths <- c(widths, width)
#         heights <- c(heights, height)
#
#         if (width < min_width){
#           min_width <- width
#         }
#         if (width > max_width){
#           max_width <- width
#         }
#         if (height < min_height){
#           min_height <- height
#         }
#         if (height > max_height){
#           max_height <- height
#         }
#       },
#       error = function(e)
#       {
#       cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#   }
# }
#
# cat("The range of image width:", min_width, " - ", max_width, "pixels \n")
# cat("The range of image height:", min_height, " - ", max_height, "pixels \n")
```

```
# # Distribution of image widths and heights
# hist(unlist(widths), main = "Distribution of Image Widths",
#      xlab = "Width (pixels)", ylab = "Frequency", col = "lightblue",
#      xlim = c(0, 2000), ylim = c(0, 2000), border = "black")
#
# hist(unlist(heights), main = "Distribution of Image Heights",
#      xlab = "Height (pixels)", ylab = "Frequency", col = "lightblue",
#      xlim = c(0, 2500), ylim = c(0, 2000), border = "black")
```

4. Resize all images into size 200 x 200 pixels:

```
# ############## Turn to comments from the second run onwards ###############
# resize <- function(directory)
# {
#   files <- list.files(directory, full.names = TRUE, pattern = "\\.jpg$")
#   for (file in files)
#   {
#     tryCatch(
#       {
#       img <- image_read(file)
#
#       # Resize images and adds padding
#       img_resized <- image_resize(img, geometry = "200x200", filter = "Cubic")
#       img_resized_padding <- image_extent(img_resized, geometry = "200x200", gravity = "cen
ter", color = "black")
#
#       # Convert a magick object into a cimg image
#       image <- magick2cimg(img_resized_padding)
#
#       # Save the change into the original dataset
#       save.image(image, file)
#       },
#       error = function(e)
#       {
#       cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#   }
# }
#
# for (class_folder in class_folders) {
#   resize(file.path(main_folder, class_folder))
# }
```

5. Get the list of images and the list of labels:

```
# Get the list of image paths and the list of categorical labels
image_paths <- list()
labels <- list()
for (class_folder in class_folders) {
  images <- list.files(file.path(main_folder, class_folder), full.names = TRUE, pattern =
"\\.jpg$")
  image_paths <- c(image_paths, images)
  labels <- c(labels, rep(class_folder, length(images)))
}

# Load the images
X <- lapply(unlist(image_paths), load.image)
```

6. **Split training - test - validation set:**

```
set.seed(1)

# Shuffle X and labels in the same order to make sure the label values appear in a random ord
er
shuffle_indices <- sample(1:length(X))
X <- X[shuffle_indices]
labels <- labels[shuffle_indices]

# Split train : test : val as 70 : 20 : 10
train_indices_temp <- createDataPartition(unlist(labels), p = 0.8, list = FALSE)
test_indices <- setdiff(seq_along(unlist(labels)), train_indices_temp)
# -> seq_along(x) creates a sequence of integers with the same length as x

X_train_temp <- X[train_indices_temp]
labels_train_temp <- labels[train_indices_temp]
X_test <- X[test_indices]
labels_test <- labels[test_indices]

val_indices <- createDataPartition(unlist(labels_train_temp), p = 0.125, list = FALSE)
train_indices <- setdiff(seq_along(unlist(labels_train_temp)), val_indices)

X_train <- X_train_temp[train_indices]
labels_train <- labels_train_temp[train_indices]
X_val <- X_train_temp[val_indices]
labels_val <- labels_train_temp[val_indices]
# (train: 0.875 x 0.8 = 0.7 ; val: 0.125 x 0.8 = 0.1)

# Examine the size of the training, validation, and test sets
cat("Train set's size: X_train =", length(X_train), ", y_train =", length(labels_train),
"\n")
```

```
## Train set's size: X_train = 2499 , y_train = 2499
```

```
cat("Validation set's size: X_val =", length(X_val), ", y_val =", length(labels_val), "\n")
```

```
## Validation set's size: X_val = 358 , y_val = 358
```

```
cat("Test set's size: X_test =", length(X_test), ", y_test =", length(labels_test), "\n")
```

```
## Test set's size: X_test = 712 , y_test = 712
```

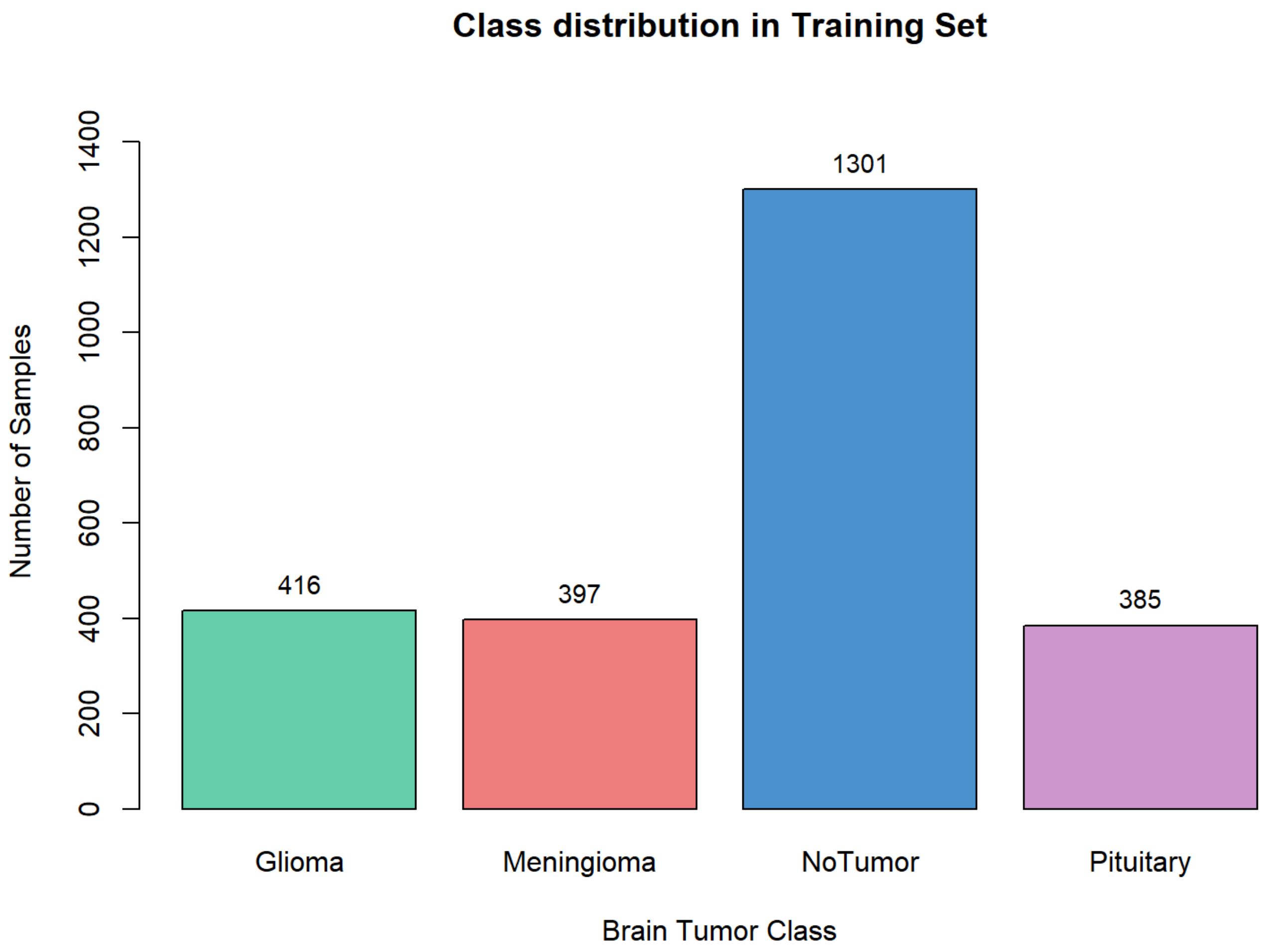
```
# Check the pixel value range of training images
cat("Pixel value range of training images: [", min(unlist(X_train)), ", ", max(unlist(X_trai
n)), "] \n", sep = "")
```

```
## Pixel value range of training images: [0, 1]
```

```
# -> All training images are already normalized
```

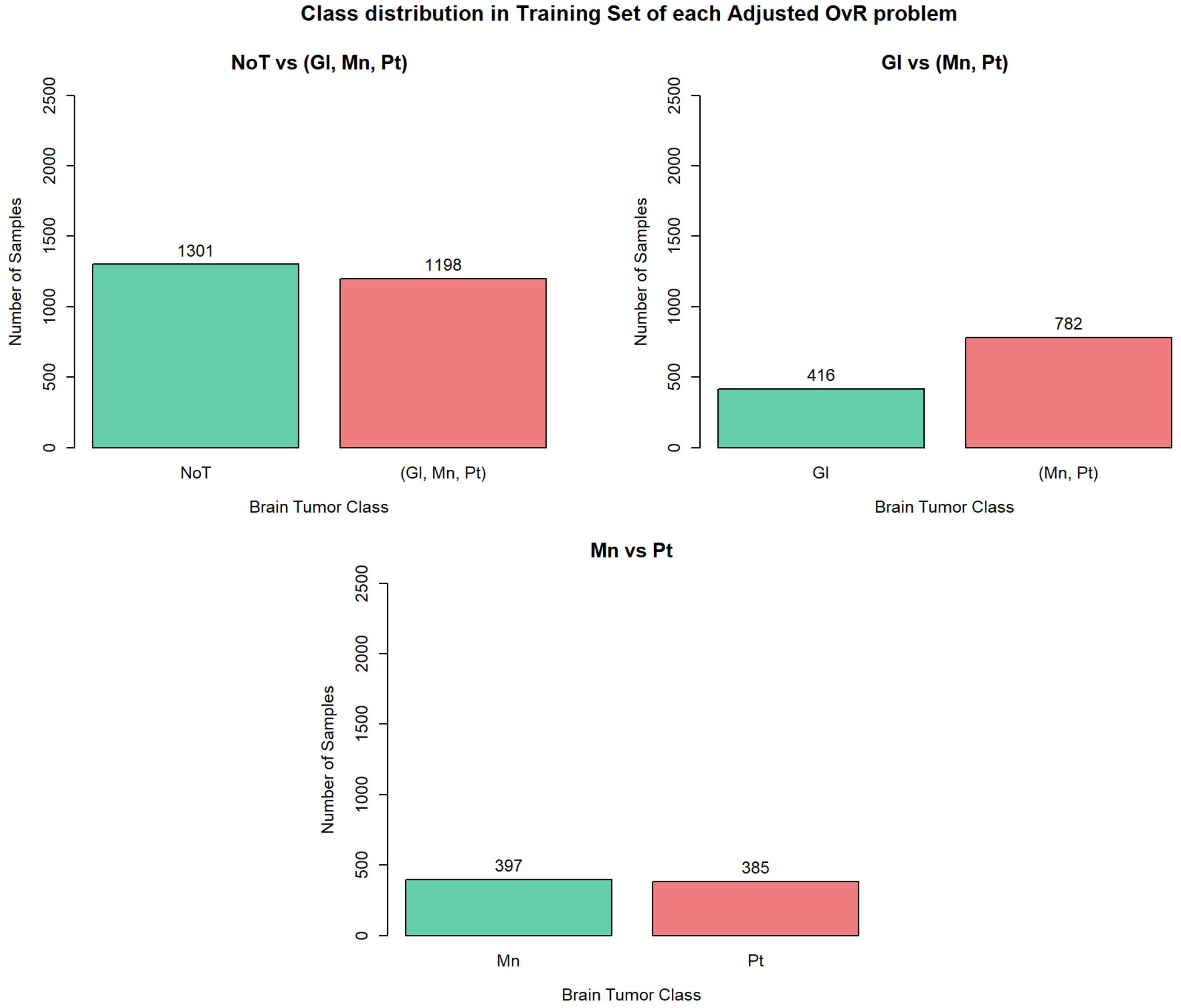
7. Check for class distribution in training set:

```
count <- as.numeric(table(unlist(labels_train)))
bp <- barplot(table(unlist(labels_train)), main = "Class distribution in Training Set", col =
c("aquamarine3", "lightcoral", "steelblue3", "plum3"), ylim = c(0, 1500), xlab = "Brain Tumor
Class", ylab = "Number of Samples")
text(x = bp, y = count, label = count, cex = 0.9, pos = 3, col = "black")
```



8. Apply other preprocessing steps:

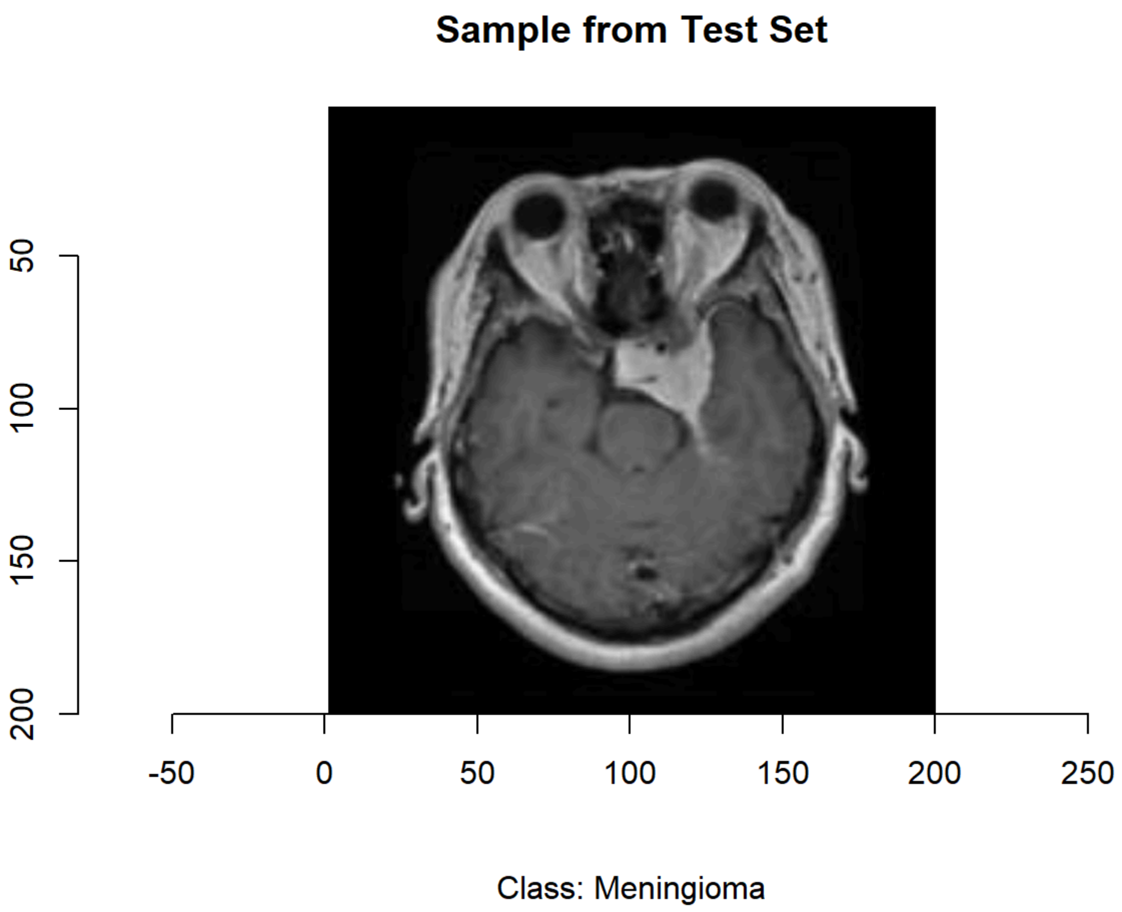
```
class_counts <- sort(table(unlist(labels_train)), decreasing = TRUE)  # Sort the class names
in decreasing order to minimize data imbalance in each Adjusted OvR problem
class <- names(class_counts)

label_map <- c("NoTumor" = "NoT", "Glioma" = "Gl", "Meningioma" = "Mn", "Pituitary" = "Pt")
# Shorten the labels

layout_ <- matrix(c(1, 1, 2, 2,
                    0, 3, 3, 0), nrow = 2, byrow = TRUE)

layout(layout_) # Create 3 subplots with 2 rows, centering the 3rd subplot
par(oma = c(1,1,2,1), mar = c(4,4,4,4)) # mar changes the spacing of each plot from (bottom,
left, top, right)

# Check for class imbalance in training set of each model in OvR approach
for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  labels_train_ovr_plt <- unlist(labels_train)[unlist(labels_train) %in% remaining_classes]

  # Convert the response variable to binary
  labels_train_ovr_plt <- ifelse(labels_train_ovr_plt == class[i], 1, 0)  # Returns result in
numeric format
  labels_train_ovr_plt  <- factor(labels_train_ovr_plt, levels = c("1", "0"))  # Convert to f
actor, arrange the order of labels to be positive before negative for the barplot order

  # Check for class imbalance
  count <- as.numeric(table(labels_train_ovr_plt))
  neg_labels <- paste(label_map[remaining_classes[-1]], collapse = ", ")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    custom_labels <- ifelse(names(table(labels_train_ovr_plt)) == "1", paste0(label_map[class
[i]]), paste0(neg_labels))
    plot_title <- paste0(label_map[class[i]], " vs ", neg_labels)
    }
  else {
    # Other models
    custom_labels <- ifelse(names(table(labels_train_ovr_plt)) == "1", paste0(label_map[class
[i]]), paste0("(", neg_labels, ")"))
    plot_title <- paste0(label_map[class[i]], " vs ", "(", neg_labels, ")")
    }
  bp <- barplot(table(labels_train_ovr_plt), names.arg = custom_labels, main = plot_title, ce
x.main = 1.4, cex.axis = 1.2, cex.lab = 1.2, cex.names = 1.2, col = c("aquamarine3", "lightco
ral"), ylim = c(0, 2500), xlab = "Brain Tumor Class", ylab = "Number of Samples")
  text(x = bp, y = count, label = count, cex = 1.2, pos = 3, col = "black")
}

mtext(expression(bold("Class distribution in Training Set of each Adjusted OvR problem \n")),
side = 3, line = -1.5, outer = TRUE)
```




```
# -> Class pairs: All class pairs are not heavily imbalanced
# (Gl vs (Mn, Pt) only has mild imbalance with the ratio between the two classes being around
35:65)
```

```
# Reshape the images
xtr <- do.call(abind, X_train)
xv <- do.call(abind, X_val)
xte <- do.call(abind, X_test)

X_train_ <- aperm(xtr, perm = c(4,1,2,3))
X_val_ <- aperm(xv, perm = c(4,1,2,3))
X_test_ <- aperm(xte, perm = c(4,1,2,3))
# aperm() transposes the array so that the order of the dimensions specified in "dim" is chan
ged to a new order in "perm": (300, 300, 1, length(X))

## Show some samples in the dataset
# Display the 1005th image from the Training set
plot(as.cimg(X_train_[1005,,,]))
title(main = "Sample from Training Set", sub = paste0("Class: ", unlist(labels_train)[1005]))
```


```
# Display the 420th image from the Test set
plot(as.cimg(X_test_[420,,,]))
title(main = "Sample from Test Set", sub = paste0("Class: ", unlist(labels_test)[420]))
```


Sample from Test Set
50
100
150
200
-50
0
50
100
150
200
250
Class: Meningioma

```
# Create list of OvR training sets
X_train_ovr <- c()
labels_train_ovr <- c()

# Add all models in Adjusted OvR approach to the list
for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  labels_train_ovr_ <- unlist(labels_train)[unlist(labels_train) %in% remaining_classes]
  idx_tr <- which(unlist(labels_train) %in% remaining_classes)
  X_train_ovr_ <- X_train_[idx_tr,,, , drop = FALSE]

  # Convert the response variable to binary
  labels_train_ovr_ <- ifelse(labels_train_ovr_ == class[i], 1, 0)  # Returns result in numer
ic format

  # Add to list
  X_train_ovr <- append(X_train_ovr, list(X_train_ovr_))
  labels_train_ovr <- append(labels_train_ovr, list(labels_train_ovr_))
}

# Create list of OvR test sets
X_test_ovr <- c()
labels_test_ovr <- c()

for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  labels_test_ovr_ <- unlist(labels_test)[unlist(labels_test) %in% remaining_classes]
  idx_te <- which(unlist(labels_test) %in% remaining_classes)
  X_test_ovr_ <- X_test_[idx_te,,, , drop = FALSE]

  # Convert the response variable to binary
  labels_test_ovr_  <- ifelse(labels_test_ovr_ == class[i], 1, 0)  # Returns result in numeri
c format

  # Add to list
  X_test_ovr <- append(X_test_ovr, list(X_test_ovr_))
  labels_test_ovr <- append(labels_test_ovr, list(labels_test_ovr_))
}

# Create list of OvR validation sets
X_val_ovr <- c()
labels_val_ovr <- c()

for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  labels_val_ovr_ <- unlist(labels_val)[unlist(labels_val) %in% remaining_classes]
  idx_v <- which(unlist(labels_val) %in% remaining_classes)
  X_val_ovr_ <- X_val_[idx_v,,, , drop = FALSE]

  # Convert the response variable to binary
  labels_val_ovr_  <- ifelse(labels_val_ovr_ == class[i], 1, 0)  # Returns result in numeric
format

  # Add to list
  X_val_ovr <- append(X_val_ovr, list(X_val_ovr_))
```

```
  labels_val_ovr <- append(labels_val_ovr, list(labels_val_ovr_))
}
```

9. Build the CNN models:

```
my_models_ovr <- c()

for(i in 1:(length(class) - 1)){
  # Initialize the Sequential model
  my_model <- keras_model_sequential()  # All OvR models here have the same structure so ther
e's actually no need to create a loop

  # Block 1
  my_model %>%
    layer_conv_2d(filters = 16, input_shape = c(200, 200, 1), kernel_size = c(3, 3), strides
= c(1, 1), padding = 'same', name = 'Block_1_Conv_1') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_conv_2d(filters = 32, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', n
ame = 'Block_1_Conv_2') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bl
ock_1_MaxPool') %>%
    layer_dropout(rate = 0.2)

  # Block 2
  my_model %>%
    layer_conv_2d(filters = 32, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', n
ame = 'Block_2_Conv_1') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_conv_2d(filters = 64, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', n
ame = 'Block_2_Conv_2') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bl
ock_2_MaxPool') %>%
    layer_dropout(rate = 0.2)

  # Block 3
  my_model %>%
    layer_conv_2d(filters = 64, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', n
ame = 'Block_3_Conv_1') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_conv_2d(filters = 128, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same',
name = 'Block_3_Conv_2') %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bl
ock_3_MaxPool') %>%
    layer_dropout(rate = 0.2)

  # Flatten
  my_model %>%
    layer_flatten()

  # Fully connected (FC) layer 1
  my_model %>%
```

```
    layer_dense(units = 64) %>%
    layer_batch_normalization() %>%
    layer_activation('relu') %>%
    layer_dropout(rate = 0.4)

  # Fully connected (FC) layer 2
  my_model %>%
    layer_dense(units = 1) %>%
    layer_activation('sigmoid')

  my_models_ovr <- append(my_models_ovr, list(my_model))
}

# Skip or keep summary (only 4 models max and if all 4 have the same hyperparameters then we only need to print 1 summary?)

# Summary of each model structure
cat(crayon::bold("Structure of each OvR model: \n"))
```

```
## Structure of each OvR model:
```

```
summary(my_models_ovr[[1]])
```

```
## Model: "sequential"
## ________________________________________________________________________________
##  Layer (type)                  Output Shape                Param #    Trainable
## ================================================================================
##  Block_1_Conv_1 (Conv2D)       (None, 200, 200, 16)        160        Y
##  batch_normalization (BatchNor  (None, 200, 200, 16)       64         Y
##  malization)
##  activation (Activation)       (None, 200, 200, 16)        0          Y
##  Block_1_Conv_2 (Conv2D)       (None, 200, 200, 32)        4640       Y
##  batch_normalization_1 (BatchN  (None, 200, 200, 32)       128        Y
##  ormalization)
##  activation_1 (Activation)     (None, 200, 200, 32)        0          Y
##  Block_1_MaxPool (MaxPooling2D  (None, 100, 100, 32)       0          Y
##  )
##  dropout (Dropout)             (None, 100, 100, 32)        0          Y
##  Block_2_Conv_1 (Conv2D)       (None, 100, 100, 32)        9248       Y
##  batch_normalization_2 (BatchN  (None, 100, 100, 32)       128        Y
##  ormalization)
##  activation_2 (Activation)     (None, 100, 100, 32)        0          Y
##  Block_2_Conv_2 (Conv2D)       (None, 100, 100, 64)        18496      Y
##  batch_normalization_3 (BatchN  (None, 100, 100, 64)       256        Y
##  ormalization)
##  activation_3 (Activation)     (None, 100, 100, 64)        0          Y
##  Block_2_MaxPool (MaxPooling2D  (None, 50, 50, 64)         0          Y
##  )
##  dropout_1 (Dropout)           (None, 50, 50, 64)          0          Y
##  Block_3_Conv_1 (Conv2D)       (None, 50, 50, 64)          36928      Y
##  batch_normalization_4 (BatchN  (None, 50, 50, 64)         256        Y
##  ormalization)
##  activation_4 (Activation)     (None, 50, 50, 64)          0          Y
##  Block_3_Conv_2 (Conv2D)       (None, 50, 50, 128)         73856      Y
##  batch_normalization_5 (BatchN  (None, 50, 50, 128)        512        Y
##  ormalization)
##  activation_5 (Activation)     (None, 50, 50, 128)         0          Y
##  Block_3_MaxPool (MaxPooling2D  (None, 25, 25, 128)        0          Y
##  )
##  dropout_2 (Dropout)           (None, 25, 25, 128)         0          Y
##  flatten (Flatten)             (None, 80000)               0          Y
##  dense (Dense)                 (None, 64)                  5120064    Y
##  batch_normalization_6 (BatchN  (None, 64)                 256        Y
##  ormalization)
##  activation_6 (Activation)     (None, 64)                  0          Y
##  dropout_3 (Dropout)           (None, 64)                  0          Y
##  dense_1 (Dense)               (None, 1)                   65         Y
##  activation_7 (Activation)     (None, 1)                   0          Y
## ================================================================================
## Total params: 5,265,057
## Trainable params: 5,264,257
## Non-trainable params: 800
## ________________________________________________________________________________
```

10. Compile the models:

```
pos_class <- head(class, -1)  # head(class, -1) is class without the last element

lr_ovr <- c()
lr_ovr <- ifelse(pos_class == "NoTumor", 0.01, ifelse(pos_class == "Glioma", 0.005, 0.005))

for(i in 1:(length(class) - 1)){
  my_models_ovr[[i]] %>%
    compile(loss = 'binary_crossentropy', optimizer = optimizer_sgd(learning_rate = lr_ovr
[i], momentum = 0.8), metrics = list('AUC'))
}
```

11. Train the models:

```
############# Turn to comments from the second run onwards ##############
history_list <- c()
epoch_ovr <- c()
epoch_ovr <- ifelse(pos_class == "NoTumor", 10, ifelse(pos_class == "Glioma", 20, 25))

for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")
  neg_labels_ <- paste0(label_map[remaining_classes[-1]], collapse = "-")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    msg <- paste0("Adjusted OvR model ", label_map[class[i]], " vs ", neg_labels, ": \n")
    cat(crayon::bold(msg))
    }
  else {
    # Other models
    msg <- paste0("Adjusted OvR model ", label_map[class[i]], " vs ", "(", neg_labels, "):
\n")
    cat(crayon::bold(msg))
  }

  # Create a callback ModelCheckpoint for each OvR problem. This callback will save the model
weights when finding one better than the current best
  checkpoint_filepath <- paste0("braintumor_best_model_weights_", label_map[class[i]], "_vs
_", neg_labels_, ".h5")

  mc <- callback_model_checkpoint(filepath = checkpoint_filepath, monitor = "val_loss", save_
best_only = TRUE, save_weights_only = TRUE, save_freq = "epoch", verbose = 0)

  # Train the model
  history <- my_models_ovr[[i]] %>%
    fit(X_train_ovr[[i]], labels_train_ovr[[i]], batch_size = 16, epochs = epoch_ovr[i], verb
ose = 1, validation_data = list(X_val_ovr[[i]], labels_val_ovr[[i]]), callbacks = list(mc))
  history_list <- append(history_list, list(history))
  cat("\n")
}
```

```
## Adjusted OvR model NoT vs (Gl, Mn, Pt):
## Epoch 1/10
##
##   1/157 [..............................] - ETA: 8:35 - loss: 0.7571 - auc: 0.6719
##   2/157 [..............................] - ETA: 8s - loss: 0.5782 - auc: 0.7922
##   3/157 [..............................] - ETA: 8s - loss: 0.4941 - auc: 0.8507
##   4/157 [..............................] - ETA: 8s - loss: 0.5192 - auc: 0.8480
##   5/157 [..............................] - ETA: 8s - loss: 0.4935 - auc: 0.8471
##   6/157 [>.............................] - ETA: 8s - loss: 0.4433 - auc: 0.8773
##   7/157 [>.............................] - ETA: 7s - loss: 0.4715 - auc: 0.8684
##   8/157 [>.............................] - ETA: 7s - loss: 0.4505 - auc: 0.8771
##   9/157 [>.............................] - ETA: 7s - loss: 0.4376 - auc: 0.8832
##  10/157 [>.............................] - ETA: 7s - loss: 0.4320 - auc: 0.8840
##  11/157 [=>............................] - ETA: 7s - loss: 0.4190 - auc: 0.8914
##  12/157 [=>............................] - ETA: 7s - loss: 0.4387 - auc: 0.8867
##  13/157 [=>............................] - ETA: 7s - loss: 0.4463 - auc: 0.8812
##  14/157 [=>............................] - ETA: 7s - loss: 0.4315 - auc: 0.8892
##  15/157 [=>............................] - ETA: 7s - loss: 0.4291 - auc: 0.8908
##  16/157 [==>...........................] - ETA: 7s - loss: 0.4211 - auc: 0.8949
##  17/157 [==>...........................] - ETA: 7s - loss: 0.4046 - auc: 0.9037
##  18/157 [==>...........................] - ETA: 7s - loss: 0.3944 - auc: 0.9082
##  19/157 [==>...........................] - ETA: 7s - loss: 0.3838 - auc: 0.9133
##  20/157 [==>...........................] - ETA: 7s - loss: 0.3742 - auc: 0.9180
##  21/157 [===>..........................] - ETA: 7s - loss: 0.3645 - auc: 0.9225
##  22/157 [===>..........................] - ETA: 7s - loss: 0.3577 - auc: 0.9260
##  23/157 [===>..........................] - ETA: 7s - loss: 0.3572 - auc: 0.9256
##  24/157 [===>..........................] - ETA: 7s - loss: 0.3556 - auc: 0.9267
##  25/157 [===>..........................] - ETA: 7s - loss: 0.3472 - auc: 0.9305
##  26/157 [===>..........................] - ETA: 6s - loss: 0.3416 - auc: 0.9328
##  27/157 [====>.........................] - ETA: 6s - loss: 0.3426 - auc: 0.9313
##  28/157 [====>.........................] - ETA: 6s - loss: 0.3421 - auc: 0.9317
##  29/157 [====>.........................] - ETA: 6s - loss: 0.3398 - auc: 0.9320
##  30/157 [====>.........................] - ETA: 6s - loss: 0.3343 - auc: 0.9346
##  31/157 [====>.........................] - ETA: 6s - loss: 0.3286 - auc: 0.9370
##  32/157 [=====>........................] - ETA: 6s - loss: 0.3320 - auc: 0.9356
##  33/157 [=====>........................] - ETA: 6s - loss: 0.3304 - auc: 0.9363
##  34/157 [=====>........................] - ETA: 6s - loss: 0.3297 - auc: 0.9365
##  35/157 [=====>........................] - ETA: 6s - loss: 0.3235 - auc: 0.9391
##  36/157 [=====>........................] - ETA: 6s - loss: 0.3384 - auc: 0.9376
##  37/157 [======>.......................] - ETA: 6s - loss: 0.3385 - auc: 0.9378
##  38/157 [======>.......................] - ETA: 6s - loss: 0.3321 - auc: 0.9400
##  39/157 [======>.......................] - ETA: 6s - loss: 0.3261 - auc: 0.9422
##  40/157 [======>.......................] - ETA: 6s - loss: 0.3266 - auc: 0.9414
##  41/157 [======>.......................] - ETA: 6s - loss: 0.3287 - auc: 0.9399
##  42/157 [=======>......................] - ETA: 6s - loss: 0.3324 - auc: 0.9380
##  43/157 [=======>......................] - ETA: 6s - loss: 0.3279 - auc: 0.9398
##  44/157 [=======>......................] - ETA: 6s - loss: 0.3305 - auc: 0.9381
##  45/157 [=======>......................] - ETA: 5s - loss: 0.3332 - auc: 0.9372
##  46/157 [=======>......................] - ETA: 5s - loss: 0.3328 - auc: 0.9374
##  47/157 [=======>......................] - ETA: 5s - loss: 0.3327 - auc: 0.9375
##  48/157 [========>.....................] - ETA: 5s - loss: 0.3306 - auc: 0.9381
##  49/157 [========>.....................] - ETA: 5s - loss: 0.3292 - auc: 0.9385
##  50/157 [========>.....................] - ETA: 5s - loss: 0.3274 - auc: 0.9391
##  51/157 [========>.....................] - ETA: 5s - loss: 0.3245 - auc: 0.9402
##  52/157 [========>.....................] - ETA: 5s - loss: 0.3244 - auc: 0.9401
```

```
##  53/157 [=========>....................] - ETA: 5s - loss: 0.3212 - auc: 0.9411
##  54/157 [=========>....................] - ETA: 5s - loss: 0.3206 - auc: 0.9413
##  55/157 [=========>....................] - ETA: 5s - loss: 0.3184 - auc: 0.9421
##  56/157 [=========>....................] - ETA: 5s - loss: 0.3157 - auc: 0.9430
##  57/157 [=========>....................] - ETA: 5s - loss: 0.3204 - auc: 0.9415
##  58/157 [==========>...................] - ETA: 5s - loss: 0.3167 - auc: 0.9429
##  59/157 [==========>...................] - ETA: 5s - loss: 0.3202 - auc: 0.9415
##  60/157 [==========>...................] - ETA: 5s - loss: 0.3206 - auc: 0.9413
##  61/157 [==========>...................] - ETA: 5s - loss: 0.3203 - auc: 0.9414
##  62/157 [==========>...................] - ETA: 5s - loss: 0.3170 - auc: 0.9426
##  63/157 [===========>..................] - ETA: 5s - loss: 0.3195 - auc: 0.9415
##  64/157 [===========>..................] - ETA: 4s - loss: 0.3218 - auc: 0.9408
##  65/157 [===========>..................] - ETA: 4s - loss: 0.3195 - auc: 0.9416
##  66/157 [===========>..................] - ETA: 4s - loss: 0.3183 - auc: 0.9420
##  67/157 [===========>..................] - ETA: 4s - loss: 0.3171 - auc: 0.9424
##  68/157 [===========>..................] - ETA: 4s - loss: 0.3154 - auc: 0.9431
##  69/157 [============>.................] - ETA: 4s - loss: 0.3132 - auc: 0.9439
##  70/157 [============>.................] - ETA: 4s - loss: 0.3107 - auc: 0.9448
##  71/157 [============>.................] - ETA: 4s - loss: 0.3102 - auc: 0.9451
##  72/157 [============>.................] - ETA: 4s - loss: 0.3071 - auc: 0.9462
##  73/157 [============>.................] - ETA: 4s - loss: 0.3058 - auc: 0.9467
##  74/157 [=============>................] - ETA: 4s - loss: 0.3030 - auc: 0.9476
##  75/157 [=============>................] - ETA: 4s - loss: 0.3023 - auc: 0.9478
##  76/157 [=============>................] - ETA: 4s - loss: 0.3124 - auc: 0.9452
##  77/157 [=============>................] - ETA: 4s - loss: 0.3115 - auc: 0.9455
##  78/157 [=============>................] - ETA: 4s - loss: 0.3113 - auc: 0.9456
##  79/157 [==============>...............] - ETA: 4s - loss: 0.3090 - auc: 0.9463
##  80/157 [==============>...............] - ETA: 4s - loss: 0.3069 - auc: 0.9471
##  81/157 [==============>...............] - ETA: 4s - loss: 0.3045 - auc: 0.9480
##  82/157 [==============>...............] - ETA: 4s - loss: 0.3040 - auc: 0.9482
##  83/157 [==============>...............] - ETA: 3s - loss: 0.3025 - auc: 0.9487
##  84/157 [===============>..............] - ETA: 3s - loss: 0.3043 - auc: 0.9476
##  85/157 [===============>..............] - ETA: 3s - loss: 0.3051 - auc: 0.9470
##  86/157 [===============>..............] - ETA: 3s - loss: 0.3036 - auc: 0.9475
##  87/157 [===============>..............] - ETA: 3s - loss: 0.3017 - auc: 0.9481
##  88/157 [===============>..............] - ETA: 3s - loss: 0.3012 - auc: 0.9482
##  89/157 [================>.............] - ETA: 3s - loss: 0.3005 - auc: 0.9485
##  90/157 [================>.............] - ETA: 3s - loss: 0.3039 - auc: 0.9474
##  91/157 [================>.............] - ETA: 3s - loss: 0.3031 - auc: 0.9476
##  92/157 [================>.............] - ETA: 3s - loss: 0.3030 - auc: 0.9474
##  93/157 [================>.............] - ETA: 3s - loss: 0.3023 - auc: 0.9476
##  94/157 [================>.............] - ETA: 3s - loss: 0.3040 - auc: 0.9469
##  95/157 [=================>............] - ETA: 3s - loss: 0.3034 - auc: 0.9471
##  96/157 [=================>............] - ETA: 3s - loss: 0.3016 - auc: 0.9478
##  97/157 [=================>............] - ETA: 3s - loss: 0.3018 - auc: 0.9477
##  98/157 [=================>............] - ETA: 3s - loss: 0.3024 - auc: 0.9473
##  99/157 [=================>............] - ETA: 3s - loss: 0.3025 - auc: 0.9472
## 100/157 [==================>...........] - ETA: 3s - loss: 0.3014 - auc: 0.9476
## 101/157 [==================>...........] - ETA: 2s - loss: 0.2998 - auc: 0.9481
## 102/157 [==================>...........] - ETA: 2s - loss: 0.2994 - auc: 0.9482
## 103/157 [==================>...........] - ETA: 2s - loss: 0.3005 - auc: 0.9479
## 104/157 [==================>...........] - ETA: 2s - loss: 0.3009 - auc: 0.9476
## 105/157 [===================>..........] - ETA: 2s - loss: 0.3014 - auc: 0.9474
## 106/157 [===================>..........] - ETA: 2s - loss: 0.2999 - auc: 0.9479
## 107/157 [===================>..........] - ETA: 2s - loss: 0.2996 - auc: 0.9480
## 108/157 [===================>..........] - ETA: 2s - loss: 0.2977 - auc: 0.9487
```

```
## 109/157 [===================>..........] - ETA: 2s - loss: 0.2964 - auc: 0.9492
## 110/157 [====================>.........] - ETA: 2s - loss: 0.2955 - auc: 0.9494
## 111/157 [====================>.........] - ETA: 2s - loss: 0.2945 - auc: 0.9497
## 112/157 [====================>.........] - ETA: 2s - loss: 0.2932 - auc: 0.9501
## 113/157 [====================>.........] - ETA: 2s - loss: 0.2941 - auc: 0.9497
## 114/157 [====================>.........] - ETA: 2s - loss: 0.2931 - auc: 0.9500
## 115/157 [====================>.........] - ETA: 2s - loss: 0.2912 - auc: 0.9506
## 116/157 [=====================>........] - ETA: 2s - loss: 0.2901 - auc: 0.9510
## 117/157 [=====================>........] - ETA: 2s - loss: 0.2880 - auc: 0.9516
## 118/157 [=====================>........] - ETA: 2s - loss: 0.2864 - auc: 0.9522
## 119/157 [=====================>........] - ETA: 2s - loss: 0.2856 - auc: 0.9524
## 120/157 [=====================>........] - ETA: 1s - loss: 0.2855 - auc: 0.9524
## 121/157 [======================>.......] - ETA: 1s - loss: 0.2851 - auc: 0.9524
## 122/157 [======================>.......] - ETA: 1s - loss: 0.2841 - auc: 0.9527
## 123/157 [======================>.......] - ETA: 1s - loss: 0.2832 - auc: 0.9530
## 124/157 [======================>.......] - ETA: 1s - loss: 0.2833 - auc: 0.9529
## 125/157 [======================>.......] - ETA: 1s - loss: 0.2834 - auc: 0.9529
## 126/157 [=======================>......] - ETA: 1s - loss: 0.2815 - auc: 0.9535
## 127/157 [=======================>......] - ETA: 1s - loss: 0.2813 - auc: 0.9535
## 128/157 [=======================>......] - ETA: 1s - loss: 0.2811 - auc: 0.9536
## 129/157 [=======================>......] - ETA: 1s - loss: 0.2796 - auc: 0.9540
## 130/157 [=======================>......] - ETA: 1s - loss: 0.2784 - auc: 0.9544
## 131/157 [========================>.....] - ETA: 1s - loss: 0.2788 - auc: 0.9542
## 132/157 [========================>.....] - ETA: 1s - loss: 0.2785 - auc: 0.9543
## 133/157 [========================>.....] - ETA: 1s - loss: 0.2798 - auc: 0.9539
## 134/157 [========================>.....] - ETA: 1s - loss: 0.2804 - auc: 0.9536
## 135/157 [========================>.....] - ETA: 1s - loss: 0.2794 - auc: 0.9539
## 136/157 [========================>.....] - ETA: 1s - loss: 0.2784 - auc: 0.9542
## 137/157 [=========================>....] - ETA: 1s - loss: 0.2787 - auc: 0.9541
## 138/157 [=========================>....] - ETA: 1s - loss: 0.2780 - auc: 0.9544
## 139/157 [=========================>....] - ETA: 0s - loss: 0.2766 - auc: 0.9548
## 140/157 [=========================>....] - ETA: 0s - loss: 0.2755 - auc: 0.9552
## 141/157 [=========================>....] - ETA: 0s - loss: 0.2762 - auc: 0.9550
## 142/157 [==========================>...] - ETA: 0s - loss: 0.2750 - auc: 0.9554
## 143/157 [==========================>...] - ETA: 0s - loss: 0.2756 - auc: 0.9552
## 144/157 [==========================>...] - ETA: 0s - loss: 0.2753 - auc: 0.9553
## 145/157 [==========================>...] - ETA: 0s - loss: 0.2738 - auc: 0.9557
## 146/157 [==========================>...] - ETA: 0s - loss: 0.2727 - auc: 0.9561
## 147/157 [===========================>..] - ETA: 0s - loss: 0.2713 - auc: 0.9565
## 148/157 [===========================>..] - ETA: 0s - loss: 0.2721 - auc: 0.9562
## 149/157 [===========================>..] - ETA: 0s - loss: 0.2718 - auc: 0.9564
## 150/157 [===========================>..] - ETA: 0s - loss: 0.2712 - auc: 0.9566
## 151/157 [===========================>..] - ETA: 0s - loss: 0.2698 - auc: 0.9570
## 152/157 [============================>.] - ETA: 0s - loss: 0.2685 - auc: 0.9574
## 153/157 [============================>.] - ETA: 0s - loss: 0.2687 - auc: 0.9573
## 154/157 [============================>.] - ETA: 0s - loss: 0.2703 - auc: 0.9566
## 155/157 [============================>.] - ETA: 0s - loss: 0.2703 - auc: 0.9566
## 156/157 [============================>.] - ETA: 0s - loss: 0.2695 - auc: 0.9569
## 157/157 [==============================] - ETA: 0s - loss: 0.2693 - auc: 0.9569
## 157/157 [==============================] - 12s 58ms/step - loss: 0.2693 - auc: 0.9569 - va
l_loss: 2.0103 - val_auc: 0.6596
## Epoch 2/10
##
##   1/157 [..............................] - ETA: 8s - loss: 0.0286 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0386 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0532 - auc: 1.0000
```

```
##   4/157 [..............................] - ETA: 8s - loss: 0.0514 - auc: 1.0000
##   5/157 [..............................] - ETA: 8s - loss: 0.0623 - auc: 1.0000
##   6/157 [>.............................] - ETA: 8s - loss: 0.0925 - auc: 0.9983
##   7/157 [>.............................] - ETA: 7s - loss: 0.0955 - auc: 0.9981
##   8/157 [>.............................] - ETA: 7s - loss: 0.1035 - auc: 0.9963
##   9/157 [>.............................] - ETA: 7s - loss: 0.1062 - auc: 0.9956
##  10/157 [>.............................] - ETA: 7s - loss: 0.1141 - auc: 0.9917
##  11/157 [=>............................] - ETA: 7s - loss: 0.1081 - auc: 0.9926
##  12/157 [=>............................] - ETA: 7s - loss: 0.1061 - auc: 0.9934
##  13/157 [=>............................] - ETA: 7s - loss: 0.1235 - auc: 0.9883
##  14/157 [=>............................] - ETA: 7s - loss: 0.1302 - auc: 0.9876
##  15/157 [=>............................] - ETA: 7s - loss: 0.1281 - auc: 0.9881
##  16/157 [==>...........................] - ETA: 7s - loss: 0.1277 - auc: 0.9892
##  17/157 [==>...........................] - ETA: 7s - loss: 0.1251 - auc: 0.9899
##  18/157 [==>...........................] - ETA: 7s - loss: 0.1300 - auc: 0.9892
##  19/157 [==>...........................] - ETA: 7s - loss: 0.1281 - auc: 0.9895
##  20/157 [==>...........................] - ETA: 7s - loss: 0.1262 - auc: 0.9899
##  21/157 [===>..........................] - ETA: 7s - loss: 0.1267 - auc: 0.9904
##  22/157 [===>..........................] - ETA: 7s - loss: 0.1235 - auc: 0.9908
##  23/157 [===>..........................] - ETA: 7s - loss: 0.1299 - auc: 0.9902
##  24/157 [===>..........................] - ETA: 7s - loss: 0.1318 - auc: 0.9897
##  25/157 [===>..........................] - ETA: 7s - loss: 0.1276 - auc: 0.9904
##  26/157 [===>..........................] - ETA: 6s - loss: 0.1301 - auc: 0.9892
##  27/157 [====>.........................] - ETA: 6s - loss: 0.1270 - auc: 0.9897
##  28/157 [====>.........................] - ETA: 6s - loss: 0.1239 - auc: 0.9904
##  29/157 [====>.........................] - ETA: 6s - loss: 0.1228 - auc: 0.9907
##  30/157 [====>.........................] - ETA: 6s - loss: 0.1239 - auc: 0.9907
##  31/157 [====>.........................] - ETA: 6s - loss: 0.1216 - auc: 0.9912
##  32/157 [=====>........................] - ETA: 6s - loss: 0.1201 - auc: 0.9915
##  33/157 [=====>........................] - ETA: 6s - loss: 0.1180 - auc: 0.9918
##  34/157 [=====>........................] - ETA: 6s - loss: 0.1224 - auc: 0.9915
##  35/157 [=====>........................] - ETA: 6s - loss: 0.1238 - auc: 0.9916
##  36/157 [=====>........................] - ETA: 6s - loss: 0.1269 - auc: 0.9911
##  37/157 [======>.......................] - ETA: 6s - loss: 0.1249 - auc: 0.9915
##  38/157 [======>.......................] - ETA: 6s - loss: 0.1275 - auc: 0.9910
##  39/157 [======>.......................] - ETA: 6s - loss: 0.1304 - auc: 0.9901
##  40/157 [======>.......................] - ETA: 6s - loss: 0.1322 - auc: 0.9896
##  41/157 [======>.......................] - ETA: 6s - loss: 0.1361 - auc: 0.9889
##  42/157 [=======>......................] - ETA: 6s - loss: 0.1335 - auc: 0.9894
##  43/157 [=======>......................] - ETA: 6s - loss: 0.1333 - auc: 0.9893
##  44/157 [=======>......................] - ETA: 6s - loss: 0.1316 - auc: 0.9896
##  45/157 [=======>......................] - ETA: 5s - loss: 0.1298 - auc: 0.9900
##  46/157 [=======>......................] - ETA: 5s - loss: 0.1404 - auc: 0.9875
##  47/157 [=======>......................] - ETA: 5s - loss: 0.1488 - auc: 0.9858
##  48/157 [========>.....................] - ETA: 5s - loss: 0.1471 - auc: 0.9862
##  49/157 [========>.....................] - ETA: 5s - loss: 0.1468 - auc: 0.9865
##  50/157 [========>.....................] - ETA: 5s - loss: 0.1468 - auc: 0.9865
##  51/157 [========>.....................] - ETA: 5s - loss: 0.1460 - auc: 0.9865
##  52/157 [========>.....................] - ETA: 5s - loss: 0.1441 - auc: 0.9869
##  53/157 [=========>....................] - ETA: 5s - loss: 0.1455 - auc: 0.9868
##  54/157 [=========>....................] - ETA: 5s - loss: 0.1445 - auc: 0.9870
##  55/157 [=========>....................] - ETA: 5s - loss: 0.1481 - auc: 0.9861
##  56/157 [=========>....................] - ETA: 5s - loss: 0.1472 - auc: 0.9864
##  57/157 [=========>....................] - ETA: 5s - loss: 0.1462 - auc: 0.9868
##  58/157 [==========>...................] - ETA: 5s - loss: 0.1539 - auc: 0.9851
##  59/157 [==========>...................] - ETA: 5s - loss: 0.1554 - auc: 0.9848
```

```
##  60/157 [==========>...................] - ETA: 5s - loss: 0.1556 - auc: 0.9848
##  61/157 [==========>...................] - ETA: 5s - loss: 0.1588 - auc: 0.9844
##  62/157 [==========>...................] - ETA: 5s - loss: 0.1572 - auc: 0.9847
##  63/157 [===========>..................] - ETA: 5s - loss: 0.1575 - auc: 0.9845
##  64/157 [===========>..................] - ETA: 4s - loss: 0.1579 - auc: 0.9844
##  65/157 [===========>..................] - ETA: 4s - loss: 0.1565 - auc: 0.9847
##  66/157 [===========>..................] - ETA: 4s - loss: 0.1554 - auc: 0.9849
##  67/157 [===========>..................] - ETA: 4s - loss: 0.1545 - auc: 0.9851
##  68/157 [===========>..................] - ETA: 4s - loss: 0.1546 - auc: 0.9851
##  69/157 [============>.................] - ETA: 4s - loss: 0.1552 - auc: 0.9849
##  70/157 [============>.................] - ETA: 4s - loss: 0.1548 - auc: 0.9851
##  71/157 [============>.................] - ETA: 4s - loss: 0.1539 - auc: 0.9853
##  72/157 [============>.................] - ETA: 4s - loss: 0.1570 - auc: 0.9848
##  73/157 [============>.................] - ETA: 4s - loss: 0.1586 - auc: 0.9845
##  74/157 [=============>................] - ETA: 4s - loss: 0.1591 - auc: 0.9844
##  75/157 [=============>................] - ETA: 4s - loss: 0.1574 - auc: 0.9847
##  76/157 [=============>................] - ETA: 4s - loss: 0.1573 - auc: 0.9848
##  77/157 [=============>................] - ETA: 4s - loss: 0.1562 - auc: 0.9850
##  78/157 [=============>................] - ETA: 4s - loss: 0.1551 - auc: 0.9852
##  79/157 [==============>...............] - ETA: 4s - loss: 0.1543 - auc: 0.9853
##  80/157 [==============>...............] - ETA: 4s - loss: 0.1543 - auc: 0.9853
##  81/157 [==============>...............] - ETA: 4s - loss: 0.1535 - auc: 0.9855
##  82/157 [==============>...............] - ETA: 4s - loss: 0.1525 - auc: 0.9857
##  83/157 [==============>...............] - ETA: 3s - loss: 0.1522 - auc: 0.9858
##  84/157 [===============>..............] - ETA: 3s - loss: 0.1511 - auc: 0.9861
##  85/157 [===============>..............] - ETA: 3s - loss: 0.1503 - auc: 0.9862
##  86/157 [===============>..............] - ETA: 3s - loss: 0.1491 - auc: 0.9864
##  87/157 [===============>..............] - ETA: 3s - loss: 0.1486 - auc: 0.9865
##  88/157 [===============>..............] - ETA: 3s - loss: 0.1480 - auc: 0.9867
##  89/157 [================>.............] - ETA: 3s - loss: 0.1469 - auc: 0.9868
##  90/157 [================>.............] - ETA: 3s - loss: 0.1472 - auc: 0.9868
##  91/157 [================>.............] - ETA: 3s - loss: 0.1471 - auc: 0.9868
##  92/157 [================>.............] - ETA: 3s - loss: 0.1463 - auc: 0.9870
##  93/157 [================>.............] - ETA: 3s - loss: 0.1520 - auc: 0.9856
##  94/157 [================>.............] - ETA: 3s - loss: 0.1510 - auc: 0.9859
##  95/157 [=================>............] - ETA: 3s - loss: 0.1525 - auc: 0.9856
##  96/157 [=================>............] - ETA: 3s - loss: 0.1522 - auc: 0.9857
##  97/157 [=================>............] - ETA: 3s - loss: 0.1529 - auc: 0.9856
##  98/157 [=================>............] - ETA: 3s - loss: 0.1526 - auc: 0.9858
##  99/157 [=================>............] - ETA: 3s - loss: 0.1520 - auc: 0.9859
## 100/157 [==================>...........] - ETA: 3s - loss: 0.1521 - auc: 0.9860
## 101/157 [==================>...........] - ETA: 2s - loss: 0.1533 - auc: 0.9857
## 102/157 [==================>...........] - ETA: 2s - loss: 0.1530 - auc: 0.9858
## 103/157 [==================>...........] - ETA: 2s - loss: 0.1551 - auc: 0.9852
## 104/157 [==================>...........] - ETA: 2s - loss: 0.1549 - auc: 0.9853
## 105/157 [===================>..........] - ETA: 2s - loss: 0.1547 - auc: 0.9854
## 106/157 [===================>..........] - ETA: 2s - loss: 0.1544 - auc: 0.9854
## 107/157 [===================>..........] - ETA: 2s - loss: 0.1545 - auc: 0.9854
## 108/157 [===================>..........] - ETA: 2s - loss: 0.1539 - auc: 0.9855
## 109/157 [===================>..........] - ETA: 2s - loss: 0.1531 - auc: 0.9857
## 110/157 [====================>.........] - ETA: 2s - loss: 0.1521 - auc: 0.9859
## 111/157 [====================>.........] - ETA: 2s - loss: 0.1537 - auc: 0.9855
## 112/157 [====================>.........] - ETA: 2s - loss: 0.1529 - auc: 0.9857
## 113/157 [====================>.........] - ETA: 2s - loss: 0.1520 - auc: 0.9859
## 114/157 [====================>.........] - ETA: 2s - loss: 0.1518 - auc: 0.9859
## 115/157 [====================>.........] - ETA: 2s - loss: 0.1524 - auc: 0.9858
```

```
## 116/157 [=====================>........] - ETA: 2s - loss: 0.1519 - auc: 0.9859
## 117/157 [=====================>........] - ETA: 2s - loss: 0.1511 - auc: 0.9861
## 118/157 [=====================>........] - ETA: 2s - loss: 0.1526 - auc: 0.9857
## 119/157 [=====================>........] - ETA: 2s - loss: 0.1525 - auc: 0.9857
## 120/157 [=====================>........] - ETA: 1s - loss: 0.1518 - auc: 0.9858
## 121/157 [======================>.......] - ETA: 1s - loss: 0.1509 - auc: 0.9860
## 122/157 [======================>.......] - ETA: 1s - loss: 0.1514 - auc: 0.9859
## 123/157 [======================>.......] - ETA: 1s - loss: 0.1507 - auc: 0.9860
## 124/157 [======================>.......] - ETA: 1s - loss: 0.1502 - auc: 0.9861
## 125/157 [======================>.......] - ETA: 1s - loss: 0.1496 - auc: 0.9863
## 126/157 [=======================>......] - ETA: 1s - loss: 0.1490 - auc: 0.9864
## 127/157 [=======================>......] - ETA: 1s - loss: 0.1484 - auc: 0.9865
## 128/157 [=======================>......] - ETA: 1s - loss: 0.1477 - auc: 0.9866
## 129/157 [=======================>......] - ETA: 1s - loss: 0.1478 - auc: 0.9866
## 130/157 [=======================>......] - ETA: 1s - loss: 0.1478 - auc: 0.9867
## 131/157 [========================>.....] - ETA: 1s - loss: 0.1481 - auc: 0.9866
## 132/157 [========================>.....] - ETA: 1s - loss: 0.1480 - auc: 0.9867
## 133/157 [========================>.....] - ETA: 1s - loss: 0.1473 - auc: 0.9868
## 134/157 [========================>.....] - ETA: 1s - loss: 0.1472 - auc: 0.9869
## 135/157 [========================>.....] - ETA: 1s - loss: 0.1471 - auc: 0.9869
## 136/157 [========================>.....] - ETA: 1s - loss: 0.1464 - auc: 0.9870
## 137/157 [=========================>....] - ETA: 1s - loss: 0.1476 - auc: 0.9866
## 138/157 [=========================>....] - ETA: 1s - loss: 0.1467 - auc: 0.9868
## 139/157 [=========================>....] - ETA: 0s - loss: 0.1481 - auc: 0.9865
## 140/157 [=========================>....] - ETA: 0s - loss: 0.1484 - auc: 0.9865
## 141/157 [=========================>....] - ETA: 0s - loss: 0.1484 - auc: 0.9865
## 142/157 [==========================>...] - ETA: 0s - loss: 0.1476 - auc: 0.9867
## 143/157 [==========================>...] - ETA: 0s - loss: 0.1472 - auc: 0.9867
## 144/157 [==========================>...] - ETA: 0s - loss: 0.1466 - auc: 0.9868
## 145/157 [==========================>...] - ETA: 0s - loss: 0.1474 - auc: 0.9865
## 146/157 [==========================>...] - ETA: 0s - loss: 0.1472 - auc: 0.9865
## 147/157 [===========================>..] - ETA: 0s - loss: 0.1474 - auc: 0.9865
## 148/157 [===========================>..] - ETA: 0s - loss: 0.1472 - auc: 0.9866
## 149/157 [===========================>..] - ETA: 0s - loss: 0.1474 - auc: 0.9865
## 150/157 [===========================>..] - ETA: 0s - loss: 0.1473 - auc: 0.9865
## 151/157 [===========================>..] - ETA: 0s - loss: 0.1472 - auc: 0.9865
## 152/157 [============================>.] - ETA: 0s - loss: 0.1473 - auc: 0.9864
## 153/157 [============================>.] - ETA: 0s - loss: 0.1480 - auc: 0.9863
## 154/157 [============================>.] - ETA: 0s - loss: 0.1475 - auc: 0.9864
## 155/157 [============================>.] - ETA: 0s - loss: 0.1475 - auc: 0.9865
## 156/157 [============================>.] - ETA: 0s - loss: 0.1472 - auc: 0.9865
## 157/157 [==============================] - 9s 56ms/step - loss: 0.1470 - auc: 0.9866 - val
_loss: 1.1989 - val_auc: 0.8914
## Epoch 3/10
##
##   1/157 [..............................] - ETA: 7s - loss: 0.0618 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0877 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0882 - auc: 1.0000
##   4/157 [..............................] - ETA: 8s - loss: 0.1444 - auc: 0.9911
##   5/157 [..............................] - ETA: 8s - loss: 0.1293 - auc: 0.9933
##   6/157 [>.............................] - ETA: 8s - loss: 0.1271 - auc: 0.9936
##   7/157 [>.............................] - ETA: 7s - loss: 0.1258 - auc: 0.9938
##   8/157 [>.............................] - ETA: 7s - loss: 0.1157 - auc: 0.9952
##   9/157 [>.............................] - ETA: 7s - loss: 0.1289 - auc: 0.9923
##  10/157 [>.............................] - ETA: 7s - loss: 0.1191 - auc: 0.9938
##  11/157 [=>............................] - ETA: 7s - loss: 0.1291 - auc: 0.9910
```

```
## 12/157 [=>............................] - ETA: 7s - loss: 0.1203 - auc: 0.9925
## 13/157 [=>............................] - ETA: 7s - loss: 0.1265 - auc: 0.9910
## 14/157 [=>............................] - ETA: 7s - loss: 0.1300 - auc: 0.9905
## 15/157 [=>............................] - ETA: 7s - loss: 0.1230 - auc: 0.9916
## 16/157 [==>...........................] - ETA: 7s - loss: 0.1187 - auc: 0.9923
## 17/157 [==>...........................] - ETA: 7s - loss: 0.1129 - auc: 0.9932
## 18/157 [==>...........................] - ETA: 7s - loss: 0.1128 - auc: 0.9934
## 19/157 [==>...........................] - ETA: 7s - loss: 0.1104 - auc: 0.9936
## 20/157 [==>...........................] - ETA: 7s - loss: 0.1065 - auc: 0.9941
## 21/157 [===>..........................] - ETA: 7s - loss: 0.1029 - auc: 0.9945
## 22/157 [===>..........................] - ETA: 7s - loss: 0.1001 - auc: 0.9949
## 23/157 [===>..........................] - ETA: 7s - loss: 0.0981 - auc: 0.9952
## 24/157 [===>..........................] - ETA: 7s - loss: 0.1011 - auc: 0.9950
## 25/157 [===>..........................] - ETA: 7s - loss: 0.1021 - auc: 0.9949
## 26/157 [===>..........................] - ETA: 6s - loss: 0.1001 - auc: 0.9951
## 27/157 [====>.........................] - ETA: 6s - loss: 0.1048 - auc: 0.9941
## 28/157 [====>.........................] - ETA: 6s - loss: 0.1071 - auc: 0.9940
## 29/157 [====>.........................] - ETA: 6s - loss: 0.1044 - auc: 0.9943
## 30/157 [====>.........................] - ETA: 6s - loss: 0.1040 - auc: 0.9944
## 31/157 [====>.........................] - ETA: 6s - loss: 0.1041 - auc: 0.9943
## 32/157 [=====>........................] - ETA: 6s - loss: 0.1030 - auc: 0.9945
## 33/157 [=====>........................] - ETA: 6s - loss: 0.1006 - auc: 0.9948
## 34/157 [=====>........................] - ETA: 6s - loss: 0.1157 - auc: 0.9915
## 35/157 [=====>........................] - ETA: 6s - loss: 0.1134 - auc: 0.9918
## 36/157 [=====>........................] - ETA: 6s - loss: 0.1170 - auc: 0.9914
## 37/157 [======>.......................] - ETA: 6s - loss: 0.1199 - auc: 0.9915
## 38/157 [======>.......................] - ETA: 6s - loss: 0.1181 - auc: 0.9918
## 39/157 [======>.......................] - ETA: 6s - loss: 0.1185 - auc: 0.9916
## 40/157 [======>.......................] - ETA: 6s - loss: 0.1182 - auc: 0.9916
## 41/157 [======>.......................] - ETA: 6s - loss: 0.1168 - auc: 0.9919
## 42/157 [=======>......................] - ETA: 6s - loss: 0.1154 - auc: 0.9920
## 43/157 [=======>......................] - ETA: 6s - loss: 0.1142 - auc: 0.9922
## 44/157 [=======>......................] - ETA: 6s - loss: 0.1127 - auc: 0.9924
## 45/157 [=======>......................] - ETA: 5s - loss: 0.1124 - auc: 0.9925
## 46/157 [=======>......................] - ETA: 5s - loss: 0.1106 - auc: 0.9928
## 47/157 [=======>......................] - ETA: 5s - loss: 0.1160 - auc: 0.9916
## 48/157 [========>.....................] - ETA: 5s - loss: 0.1190 - auc: 0.9913
## 49/157 [========>.....................] - ETA: 5s - loss: 0.1288 - auc: 0.9885
## 50/157 [========>.....................] - ETA: 5s - loss: 0.1274 - auc: 0.9888
## 51/157 [========>.....................] - ETA: 5s - loss: 0.1271 - auc: 0.9890
## 52/157 [========>.....................] - ETA: 5s - loss: 0.1257 - auc: 0.9892
## 53/157 [=========>....................] - ETA: 5s - loss: 0.1241 - auc: 0.9894
## 54/157 [=========>....................] - ETA: 5s - loss: 0.1262 - auc: 0.9890
## 55/157 [=========>....................] - ETA: 5s - loss: 0.1252 - auc: 0.9893
## 56/157 [=========>....................] - ETA: 5s - loss: 0.1241 - auc: 0.9895
## 57/157 [=========>....................] - ETA: 5s - loss: 0.1230 - auc: 0.9897
## 58/157 [==========>...................] - ETA: 5s - loss: 0.1216 - auc: 0.9899
## 59/157 [==========>...................] - ETA: 5s - loss: 0.1211 - auc: 0.9900
## 60/157 [==========>...................] - ETA: 5s - loss: 0.1207 - auc: 0.9900
## 61/157 [==========>...................] - ETA: 5s - loss: 0.1197 - auc: 0.9902
## 62/157 [==========>...................] - ETA: 5s - loss: 0.1189 - auc: 0.9904
## 63/157 [===========>..................] - ETA: 5s - loss: 0.1205 - auc: 0.9902
## 64/157 [===========>..................] - ETA: 4s - loss: 0.1209 - auc: 0.9902
## 65/157 [===========>..................] - ETA: 4s - loss: 0.1221 - auc: 0.9901
## 66/157 [===========>..................] - ETA: 4s - loss: 0.1207 - auc: 0.9904
## 67/157 [===========>..................] - ETA: 4s - loss: 0.1207 - auc: 0.9904
```

```
##  68/157 [===========>..................] - ETA: 4s - loss: 0.1196 - auc: 0.9906
##  69/157 [============>.................] - ETA: 4s - loss: 0.1180 - auc: 0.9908
##  70/157 [============>.................] - ETA: 4s - loss: 0.1174 - auc: 0.9909
##  71/157 [============>.................] - ETA: 4s - loss: 0.1165 - auc: 0.9911
##  72/157 [============>.................] - ETA: 4s - loss: 0.1165 - auc: 0.9911
##  73/157 [============>.................] - ETA: 4s - loss: 0.1153 - auc: 0.9912
##  74/157 [=============>................] - ETA: 4s - loss: 0.1145 - auc: 0.9914
##  75/157 [=============>................] - ETA: 4s - loss: 0.1136 - auc: 0.9916
##  76/157 [=============>................] - ETA: 4s - loss: 0.1133 - auc: 0.9916
##  77/157 [=============>................] - ETA: 4s - loss: 0.1126 - auc: 0.9917
##  78/157 [=============>................] - ETA: 4s - loss: 0.1152 - auc: 0.9914
##  79/157 [==============>...............] - ETA: 4s - loss: 0.1151 - auc: 0.9914
##  80/157 [==============>...............] - ETA: 4s - loss: 0.1186 - auc: 0.9907
##  81/157 [==============>...............] - ETA: 4s - loss: 0.1178 - auc: 0.9908
##  82/157 [==============>...............] - ETA: 3s - loss: 0.1200 - auc: 0.9905
##  83/157 [==============>...............] - ETA: 3s - loss: 0.1238 - auc: 0.9896
##  84/157 [===============>..............] - ETA: 3s - loss: 0.1228 - auc: 0.9897
##  85/157 [===============>..............] - ETA: 3s - loss: 0.1220 - auc: 0.9899
##  86/157 [===============>..............] - ETA: 3s - loss: 0.1214 - auc: 0.9901
##  87/157 [===============>..............] - ETA: 3s - loss: 0.1205 - auc: 0.9903
##  88/157 [===============>..............] - ETA: 3s - loss: 0.1201 - auc: 0.9904
##  89/157 [================>.............] - ETA: 3s - loss: 0.1194 - auc: 0.9905
##  90/157 [================>.............] - ETA: 3s - loss: 0.1188 - auc: 0.9906
##  91/157 [================>.............] - ETA: 3s - loss: 0.1183 - auc: 0.9907
##  92/157 [================>.............] - ETA: 3s - loss: 0.1178 - auc: 0.9909
##  93/157 [================>.............] - ETA: 3s - loss: 0.1172 - auc: 0.9910
##  94/157 [================>.............] - ETA: 3s - loss: 0.1160 - auc: 0.9911
##  95/157 [=================>............] - ETA: 3s - loss: 0.1170 - auc: 0.9909
##  96/157 [=================>............] - ETA: 3s - loss: 0.1170 - auc: 0.9909
##  97/157 [=================>............] - ETA: 3s - loss: 0.1171 - auc: 0.9910
##  98/157 [=================>............] - ETA: 3s - loss: 0.1164 - auc: 0.9911
##  99/157 [=================>............] - ETA: 3s - loss: 0.1158 - auc: 0.9912
## 100/157 [==================>...........] - ETA: 3s - loss: 0.1151 - auc: 0.9913
## 101/157 [==================>...........] - ETA: 2s - loss: 0.1151 - auc: 0.9913
## 102/157 [==================>...........] - ETA: 2s - loss: 0.1151 - auc: 0.9913
## 103/157 [==================>...........] - ETA: 2s - loss: 0.1144 - auc: 0.9914
## 104/157 [==================>...........] - ETA: 2s - loss: 0.1138 - auc: 0.9915
## 105/157 [===================>..........] - ETA: 2s - loss: 0.1138 - auc: 0.9916
## 106/157 [===================>..........] - ETA: 2s - loss: 0.1132 - auc: 0.9917
## 107/157 [===================>..........] - ETA: 2s - loss: 0.1125 - auc: 0.9917
## 108/157 [===================>..........] - ETA: 2s - loss: 0.1127 - auc: 0.9917
## 109/157 [===================>..........] - ETA: 2s - loss: 0.1141 - auc: 0.9915
## 110/157 [====================>.........] - ETA: 2s - loss: 0.1134 - auc: 0.9916
## 111/157 [====================>.........] - ETA: 2s - loss: 0.1129 - auc: 0.9917
## 112/157 [====================>.........] - ETA: 2s - loss: 0.1138 - auc: 0.9916
## 113/157 [====================>.........] - ETA: 2s - loss: 0.1132 - auc: 0.9917
## 114/157 [====================>.........] - ETA: 2s - loss: 0.1129 - auc: 0.9918
## 115/157 [====================>.........] - ETA: 2s - loss: 0.1122 - auc: 0.9918
## 116/157 [=====================>........] - ETA: 2s - loss: 0.1117 - auc: 0.9919
## 117/157 [=====================>........] - ETA: 2s - loss: 0.1121 - auc: 0.9920
## 118/157 [=====================>........] - ETA: 2s - loss: 0.1124 - auc: 0.9919
## 119/157 [=====================>........] - ETA: 2s - loss: 0.1117 - auc: 0.9920
## 120/157 [=====================>........] - ETA: 1s - loss: 0.1132 - auc: 0.9918
## 121/157 [======================>.......] - ETA: 1s - loss: 0.1152 - auc: 0.9915
## 122/157 [======================>.......] - ETA: 1s - loss: 0.1151 - auc: 0.9915
## 123/157 [======================>.......] - ETA: 1s - loss: 0.1146 - auc: 0.9916
```

```
## 124/157 [======================>.......] - ETA: 1s - loss: 0.1145 - auc: 0.9916
## 125/157 [======================>.......] - ETA: 1s - loss: 0.1142 - auc: 0.9916
## 126/157 [=======================>......] - ETA: 1s - loss: 0.1134 - auc: 0.9917
## 127/157 [=======================>......] - ETA: 1s - loss: 0.1160 - auc: 0.9913
## 128/157 [=======================>......] - ETA: 1s - loss: 0.1155 - auc: 0.9914
## 129/157 [=======================>......] - ETA: 1s - loss: 0.1154 - auc: 0.9914
## 130/157 [=======================>......] - ETA: 1s - loss: 0.1155 - auc: 0.9914
## 131/157 [========================>.....] - ETA: 1s - loss: 0.1148 - auc: 0.9915
## 132/157 [========================>.....] - ETA: 1s - loss: 0.1145 - auc: 0.9915
## 133/157 [========================>.....] - ETA: 1s - loss: 0.1147 - auc: 0.9915
## 134/157 [========================>.....] - ETA: 1s - loss: 0.1143 - auc: 0.9916
## 135/157 [========================>.....] - ETA: 1s - loss: 0.1143 - auc: 0.9916
## 136/157 [========================>.....] - ETA: 1s - loss: 0.1139 - auc: 0.9917
## 137/157 [=========================>....] - ETA: 1s - loss: 0.1146 - auc: 0.9916
## 138/157 [=========================>....] - ETA: 1s - loss: 0.1140 - auc: 0.9917
## 139/157 [=========================>....] - ETA: 0s - loss: 0.1141 - auc: 0.9917
## 140/157 [=========================>....] - ETA: 0s - loss: 0.1138 - auc: 0.9918
## 141/157 [=========================>....] - ETA: 0s - loss: 0.1136 - auc: 0.9918
## 142/157 [==========================>...] - ETA: 0s - loss: 0.1171 - auc: 0.9912
## 143/157 [==========================>...] - ETA: 0s - loss: 0.1165 - auc: 0.9913
## 144/157 [==========================>...] - ETA: 0s - loss: 0.1161 - auc: 0.9914
## 145/157 [==========================>...] - ETA: 0s - loss: 0.1164 - auc: 0.9913
## 146/157 [==========================>...] - ETA: 0s - loss: 0.1159 - auc: 0.9914
## 147/157 [===========================>..] - ETA: 0s - loss: 0.1158 - auc: 0.9914
## 148/157 [===========================>..] - ETA: 0s - loss: 0.1159 - auc: 0.9914
## 149/157 [===========================>..] - ETA: 0s - loss: 0.1153 - auc: 0.9915
## 150/157 [===========================>..] - ETA: 0s - loss: 0.1159 - auc: 0.9914
## 151/157 [===========================>..] - ETA: 0s - loss: 0.1169 - auc: 0.9912
## 152/157 [============================>.] - ETA: 0s - loss: 0.1166 - auc: 0.9913
## 153/157 [============================>.] - ETA: 0s - loss: 0.1163 - auc: 0.9913
## 154/157 [============================>.] - ETA: 0s - loss: 0.1156 - auc: 0.9914
## 155/157 [============================>.] - ETA: 0s - loss: 0.1152 - auc: 0.9915
## 156/157 [============================>.] - ETA: 0s - loss: 0.1149 - auc: 0.9915
## 157/157 [==============================] - 9s 55ms/step - loss: 0.1161 - auc: 0.9914 - val
_loss: 0.5734 - val_auc: 0.9860
## Epoch 4/10
##
##   1/157 [..............................] - ETA: 7s - loss: 0.0689 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.1107 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0844 - auc: 1.0000
##   4/157 [..............................] - ETA: 8s - loss: 0.1032 - auc: 0.9951
##   5/157 [..............................] - ETA: 8s - loss: 0.0914 - auc: 0.9968
##   6/157 [>.............................] - ETA: 8s - loss: 0.1651 - auc: 0.9848
##   7/157 [>.............................] - ETA: 7s - loss: 0.1485 - auc: 0.9882
##   8/157 [>.............................] - ETA: 7s - loss: 0.1471 - auc: 0.9883
##   9/157 [>.............................] - ETA: 7s - loss: 0.1400 - auc: 0.9900
##  10/157 [>.............................] - ETA: 7s - loss: 0.1386 - auc: 0.9891
##  11/157 [=>............................] - ETA: 7s - loss: 0.1317 - auc: 0.9902
##  12/157 [=>............................] - ETA: 7s - loss: 0.1314 - auc: 0.9905
##  13/157 [=>............................] - ETA: 7s - loss: 0.1308 - auc: 0.9907
##  14/157 [=>............................] - ETA: 7s - loss: 0.1293 - auc: 0.9908
##  15/157 [=>............................] - ETA: 7s - loss: 0.1346 - auc: 0.9897
##  16/157 [==>...........................] - ETA: 7s - loss: 0.1350 - auc: 0.9897
##  17/157 [==>...........................] - ETA: 7s - loss: 0.1302 - auc: 0.9907
##  18/157 [==>...........................] - ETA: 7s - loss: 0.1275 - auc: 0.9912
##  19/157 [==>...........................] - ETA: 7s - loss: 0.1239 - auc: 0.9917
```

```
##  20/157 [==>...........................] - ETA: 7s - loss: 0.1336 - auc: 0.9902
##  21/157 [===>..........................] - ETA: 7s - loss: 0.1321 - auc: 0.9903
##  22/157 [===>..........................] - ETA: 7s - loss: 0.1326 - auc: 0.9903
##  23/157 [===>..........................] - ETA: 7s - loss: 0.1455 - auc: 0.9864
##  24/157 [===>..........................] - ETA: 7s - loss: 0.1433 - auc: 0.9869
##  25/157 [===>..........................] - ETA: 7s - loss: 0.1654 - auc: 0.9815
##  26/157 [===>..........................] - ETA: 7s - loss: 0.1609 - auc: 0.9823
##  27/157 [====>.........................] - ETA: 7s - loss: 0.1596 - auc: 0.9829
##  28/157 [====>.........................] - ETA: 7s - loss: 0.1630 - auc: 0.9823
##  29/157 [====>.........................] - ETA: 7s - loss: 0.1584 - auc: 0.9832
##  30/157 [====>.........................] - ETA: 7s - loss: 0.1573 - auc: 0.9837
##  31/157 [====>.........................] - ETA: 7s - loss: 0.1603 - auc: 0.9833
##  32/157 [=====>........................] - ETA: 7s - loss: 0.1570 - auc: 0.9841
##  33/157 [=====>........................] - ETA: 7s - loss: 0.1586 - auc: 0.9842
##  34/157 [=====>........................] - ETA: 7s - loss: 0.1571 - auc: 0.9845
##  35/157 [=====>........................] - ETA: 7s - loss: 0.1556 - auc: 0.9848
##  36/157 [=====>........................] - ETA: 6s - loss: 0.1521 - auc: 0.9855
##  37/157 [======>.......................] - ETA: 6s - loss: 0.1494 - auc: 0.9860
##  38/157 [======>.......................] - ETA: 6s - loss: 0.1480 - auc: 0.9863
##  39/157 [======>.......................] - ETA: 6s - loss: 0.1448 - auc: 0.9869
##  40/157 [======>.......................] - ETA: 6s - loss: 0.1426 - auc: 0.9873
##  41/157 [======>.......................] - ETA: 6s - loss: 0.1409 - auc: 0.9875
##  42/157 [=======>......................] - ETA: 6s - loss: 0.1382 - auc: 0.9880
##  43/157 [=======>......................] - ETA: 6s - loss: 0.1372 - auc: 0.9883
##  44/157 [=======>......................] - ETA: 6s - loss: 0.1358 - auc: 0.9886
##  45/157 [=======>......................] - ETA: 6s - loss: 0.1336 - auc: 0.9890
##  46/157 [=======>......................] - ETA: 6s - loss: 0.1337 - auc: 0.9890
##  47/157 [=======>......................] - ETA: 6s - loss: 0.1319 - auc: 0.9893
##  48/157 [========>.....................] - ETA: 6s - loss: 0.1319 - auc: 0.9895
##  49/157 [========>.....................] - ETA: 6s - loss: 0.1325 - auc: 0.9896
##  50/157 [========>.....................] - ETA: 6s - loss: 0.1310 - auc: 0.9899
##  51/157 [========>.....................] - ETA: 5s - loss: 0.1303 - auc: 0.9901
##  52/157 [========>.....................] - ETA: 5s - loss: 0.1288 - auc: 0.9904
##  53/157 [=========>....................] - ETA: 5s - loss: 0.1266 - auc: 0.9907
##  54/157 [=========>....................] - ETA: 5s - loss: 0.1263 - auc: 0.9908
##  55/157 [=========>....................] - ETA: 5s - loss: 0.1245 - auc: 0.9911
##  56/157 [=========>....................] - ETA: 5s - loss: 0.1238 - auc: 0.9911
##  57/157 [=========>....................] - ETA: 5s - loss: 0.1225 - auc: 0.9913
##  58/157 [==========>...................] - ETA: 5s - loss: 0.1211 - auc: 0.9916
##  59/157 [==========>...................] - ETA: 5s - loss: 0.1194 - auc: 0.9918
##  60/157 [==========>...................] - ETA: 5s - loss: 0.1180 - auc: 0.9920
##  61/157 [==========>...................] - ETA: 5s - loss: 0.1204 - auc: 0.9917
##  62/157 [==========>...................] - ETA: 5s - loss: 0.1188 - auc: 0.9919
##  63/157 [===========>..................] - ETA: 5s - loss: 0.1185 - auc: 0.9920
##  64/157 [===========>..................] - ETA: 5s - loss: 0.1190 - auc: 0.9918
##  65/157 [===========>..................] - ETA: 5s - loss: 0.1174 - auc: 0.9920
##  66/157 [===========>..................] - ETA: 5s - loss: 0.1161 - auc: 0.9922
##  67/157 [===========>..................] - ETA: 4s - loss: 0.1170 - auc: 0.9922
##  68/157 [===========>..................] - ETA: 4s - loss: 0.1239 - auc: 0.9907
##  69/157 [============>.................] - ETA: 4s - loss: 0.1233 - auc: 0.9908
##  70/157 [============>.................] - ETA: 4s - loss: 0.1223 - auc: 0.9910
##  71/157 [============>.................] - ETA: 4s - loss: 0.1208 - auc: 0.9912
##  72/157 [============>.................] - ETA: 4s - loss: 0.1206 - auc: 0.9912
##  73/157 [============>.................] - ETA: 4s - loss: 0.1206 - auc: 0.9913
##  74/157 [=============>................] - ETA: 4s - loss: 0.1196 - auc: 0.9915
##  75/157 [=============>................] - ETA: 4s - loss: 0.1183 - auc: 0.9917
```

```
##  76/157 [=============>................] - ETA: 4s - loss: 0.1190 - auc: 0.9915
##  77/157 [=============>................] - ETA: 4s - loss: 0.1182 - auc: 0.9916
##  78/157 [=============>................] - ETA: 4s - loss: 0.1174 - auc: 0.9917
##  79/157 [==============>...............] - ETA: 4s - loss: 0.1166 - auc: 0.9918
##  80/157 [==============>...............] - ETA: 4s - loss: 0.1154 - auc: 0.9920
##  81/157 [==============>...............] - ETA: 4s - loss: 0.1142 - auc: 0.9922
##  82/157 [==============>...............] - ETA: 4s - loss: 0.1132 - auc: 0.9923
##  83/157 [==============>...............] - ETA: 4s - loss: 0.1130 - auc: 0.9924
##  84/157 [===============>..............] - ETA: 4s - loss: 0.1123 - auc: 0.9925
##  85/157 [===============>..............] - ETA: 3s - loss: 0.1147 - auc: 0.9919
##  86/157 [===============>..............] - ETA: 3s - loss: 0.1135 - auc: 0.9920
##  87/157 [===============>..............] - ETA: 3s - loss: 0.1129 - auc: 0.9921
##  88/157 [===============>..............] - ETA: 3s - loss: 0.1118 - auc: 0.9923
##  89/157 [================>.............] - ETA: 3s - loss: 0.1113 - auc: 0.9924
##  90/157 [================>.............] - ETA: 3s - loss: 0.1105 - auc: 0.9925
##  91/157 [================>.............] - ETA: 3s - loss: 0.1095 - auc: 0.9926
##  92/157 [================>.............] - ETA: 3s - loss: 0.1091 - auc: 0.9927
##  93/157 [================>.............] - ETA: 3s - loss: 0.1094 - auc: 0.9926
##  94/157 [================>.............] - ETA: 3s - loss: 0.1085 - auc: 0.9928
##  95/157 [=================>............] - ETA: 3s - loss: 0.1077 - auc: 0.9929
##  96/157 [=================>............] - ETA: 3s - loss: 0.1079 - auc: 0.9929
##  97/157 [=================>............] - ETA: 3s - loss: 0.1075 - auc: 0.9929
##  98/157 [=================>............] - ETA: 3s - loss: 0.1065 - auc: 0.9930
##  99/157 [=================>............] - ETA: 3s - loss: 0.1071 - auc: 0.9930
## 100/157 [==================>...........] - ETA: 3s - loss: 0.1062 - auc: 0.9931
## 101/157 [==================>...........] - ETA: 3s - loss: 0.1111 - auc: 0.9921
## 102/157 [==================>...........] - ETA: 3s - loss: 0.1111 - auc: 0.9921
## 103/157 [==================>...........] - ETA: 2s - loss: 0.1103 - auc: 0.9922
## 104/157 [==================>...........] - ETA: 2s - loss: 0.1106 - auc: 0.9922
## 105/157 [===================>..........] - ETA: 2s - loss: 0.1102 - auc: 0.9923
## 106/157 [===================>..........] - ETA: 2s - loss: 0.1095 - auc: 0.9924
## 107/157 [===================>..........] - ETA: 2s - loss: 0.1091 - auc: 0.9925
## 108/157 [===================>..........] - ETA: 2s - loss: 0.1108 - auc: 0.9921
## 109/157 [===================>..........] - ETA: 2s - loss: 0.1108 - auc: 0.9922
## 110/157 [====================>.........] - ETA: 2s - loss: 0.1100 - auc: 0.9923
## 111/157 [====================>.........] - ETA: 2s - loss: 0.1092 - auc: 0.9924
## 112/157 [====================>.........] - ETA: 2s - loss: 0.1094 - auc: 0.9924
## 113/157 [====================>.........] - ETA: 2s - loss: 0.1089 - auc: 0.9924
## 114/157 [====================>.........] - ETA: 2s - loss: 0.1081 - auc: 0.9926
## 115/157 [====================>.........] - ETA: 2s - loss: 0.1073 - auc: 0.9927
## 116/157 [=====================>........] - ETA: 2s - loss: 0.1065 - auc: 0.9928
## 117/157 [=====================>........] - ETA: 2s - loss: 0.1058 - auc: 0.9929
## 118/157 [=====================>........] - ETA: 2s - loss: 0.1051 - auc: 0.9929
## 119/157 [=====================>........] - ETA: 2s - loss: 0.1048 - auc: 0.9930
## 120/157 [=====================>........] - ETA: 2s - loss: 0.1041 - auc: 0.9931
## 121/157 [======================>.......] - ETA: 1s - loss: 0.1033 - auc: 0.9932
## 122/157 [======================>.......] - ETA: 1s - loss: 0.1037 - auc: 0.9932
## 123/157 [======================>.......] - ETA: 1s - loss: 0.1035 - auc: 0.9932
## 124/157 [======================>.......] - ETA: 1s - loss: 0.1042 - auc: 0.9931
## 125/157 [======================>.......] - ETA: 1s - loss: 0.1052 - auc: 0.9930
## 126/157 [=======================>......] - ETA: 1s - loss: 0.1045 - auc: 0.9930
## 127/157 [=======================>......] - ETA: 1s - loss: 0.1041 - auc: 0.9931
## 128/157 [=======================>......] - ETA: 1s - loss: 0.1040 - auc: 0.9931
## 129/157 [=======================>......] - ETA: 1s - loss: 0.1033 - auc: 0.9932
## 130/157 [=======================>......] - ETA: 1s - loss: 0.1025 - auc: 0.9933
## 131/157 [========================>.....] - ETA: 1s - loss: 0.1020 - auc: 0.9934
```

```
## 132/157 [========================>.....] - ETA: 1s - loss: 0.1016 - auc: 0.9935
## 133/157 [========================>.....] - ETA: 1s - loss: 0.1011 - auc: 0.9935
## 134/157 [========================>.....] - ETA: 1s - loss: 0.1006 - auc: 0.9936
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0999 - auc: 0.9937
## 136/157 [========================>.....] - ETA: 1s - loss: 0.1005 - auc: 0.9936
## 137/157 [=========================>....] - ETA: 1s - loss: 0.1004 - auc: 0.9936
## 138/157 [=========================>....] - ETA: 1s - loss: 0.1001 - auc: 0.9937
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0994 - auc: 0.9938
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0991 - auc: 0.9938
## 141/157 [=========================>....] - ETA: 0s - loss: 0.1065 - auc: 0.9923
## 142/157 [==========================>...] - ETA: 0s - loss: 0.1060 - auc: 0.9924
## 143/157 [==========================>...] - ETA: 0s - loss: 0.1054 - auc: 0.9924
## 144/157 [==========================>...] - ETA: 0s - loss: 0.1051 - auc: 0.9925
## 145/157 [==========================>...] - ETA: 0s - loss: 0.1055 - auc: 0.9924
## 146/157 [==========================>...] - ETA: 0s - loss: 0.1054 - auc: 0.9925
## 147/157 [===========================>..] - ETA: 0s - loss: 0.1053 - auc: 0.9925
## 148/157 [===========================>..] - ETA: 0s - loss: 0.1049 - auc: 0.9925
## 149/157 [===========================>..] - ETA: 0s - loss: 0.1045 - auc: 0.9926
## 150/157 [===========================>..] - ETA: 0s - loss: 0.1039 - auc: 0.9927
## 151/157 [===========================>..] - ETA: 0s - loss: 0.1033 - auc: 0.9928
## 152/157 [============================>.] - ETA: 0s - loss: 0.1028 - auc: 0.9928
## 153/157 [============================>.] - ETA: 0s - loss: 0.1027 - auc: 0.9928
## 154/157 [============================>.] - ETA: 0s - loss: 0.1023 - auc: 0.9929
## 155/157 [============================>.] - ETA: 0s - loss: 0.1034 - auc: 0.9928
## 156/157 [============================>.] - ETA: 0s - loss: 0.1037 - auc: 0.9928
## 157/157 [==============================] - 9s 56ms/step - loss: 0.1044 - auc: 0.9927 - val
_loss: 0.9034 - val_auc: 0.9893
## Epoch 5/10
##
##   1/157 [..............................] - ETA: 8s - loss: 0.0279 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.2048 - auc: 0.9821
##   3/157 [..............................] - ETA: 8s - loss: 0.1720 - auc: 0.9870
##   4/157 [..............................] - ETA: 8s - loss: 0.1482 - auc: 0.9863
##   5/157 [..............................] - ETA: 8s - loss: 0.1334 - auc: 0.9887
##   6/157 [>.............................] - ETA: 8s - loss: 0.1448 - auc: 0.9885
##   7/157 [>.............................] - ETA: 8s - loss: 0.1372 - auc: 0.9912
##   8/157 [>.............................] - ETA: 8s - loss: 0.1259 - auc: 0.9930
##   9/157 [>.............................] - ETA: 7s - loss: 0.1171 - auc: 0.9937
##  10/157 [>.............................] - ETA: 7s - loss: 0.1091 - auc: 0.9944
##  11/157 [=>............................] - ETA: 7s - loss: 0.1037 - auc: 0.9951
##  12/157 [=>............................] - ETA: 7s - loss: 0.1324 - auc: 0.9877
##  13/157 [=>............................] - ETA: 7s - loss: 0.1234 - auc: 0.9888
##  14/157 [=>............................] - ETA: 7s - loss: 0.1175 - auc: 0.9895
##  15/157 [=>............................] - ETA: 7s - loss: 0.1128 - auc: 0.9904
##  16/157 [==>...........................] - ETA: 7s - loss: 0.1070 - auc: 0.9912
##  17/157 [==>...........................] - ETA: 7s - loss: 0.1031 - auc: 0.9917
##  18/157 [==>...........................] - ETA: 7s - loss: 0.0987 - auc: 0.9920
##  19/157 [==>...........................] - ETA: 7s - loss: 0.1032 - auc: 0.9909
##  20/157 [==>...........................] - ETA: 7s - loss: 0.1011 - auc: 0.9911
##  21/157 [===>..........................] - ETA: 7s - loss: 0.0975 - auc: 0.9917
##  22/157 [===>..........................] - ETA: 7s - loss: 0.0955 - auc: 0.9921
##  23/157 [===>..........................] - ETA: 7s - loss: 0.0998 - auc: 0.9916
##  24/157 [===>..........................] - ETA: 7s - loss: 0.0989 - auc: 0.9919
##  25/157 [===>..........................] - ETA: 7s - loss: 0.0974 - auc: 0.9922
##  26/157 [===>..........................] - ETA: 7s - loss: 0.0958 - auc: 0.9924
##  27/157 [====>.........................] - ETA: 6s - loss: 0.0947 - auc: 0.9928
```

```
## 28/157 [====>.........................] - ETA: 6s - loss: 0.0942 - auc: 0.9929
## 29/157 [====>.........................] - ETA: 6s - loss: 0.0923 - auc: 0.9931
## 30/157 [====>.........................] - ETA: 6s - loss: 0.0939 - auc: 0.9930
## 31/157 [====>.........................] - ETA: 6s - loss: 0.0919 - auc: 0.9933
## 32/157 [=====>........................] - ETA: 6s - loss: 0.0908 - auc: 0.9934
## 33/157 [=====>........................] - ETA: 6s - loss: 0.0886 - auc: 0.9937
## 34/157 [=====>........................] - ETA: 6s - loss: 0.0881 - auc: 0.9939
## 35/157 [=====>........................] - ETA: 6s - loss: 0.0876 - auc: 0.9940
## 36/157 [=====>........................] - ETA: 6s - loss: 0.0858 - auc: 0.9942
## 37/157 [======>.......................] - ETA: 6s - loss: 0.0845 - auc: 0.9945
## 38/157 [======>.......................] - ETA: 6s - loss: 0.0863 - auc: 0.9944
## 39/157 [======>.......................] - ETA: 6s - loss: 0.0849 - auc: 0.9945
## 40/157 [======>.......................] - ETA: 6s - loss: 0.0833 - auc: 0.9947
## 41/157 [======>.......................] - ETA: 6s - loss: 0.0841 - auc: 0.9947
## 42/157 [=======>......................] - ETA: 6s - loss: 0.0830 - auc: 0.9948
## 43/157 [=======>......................] - ETA: 6s - loss: 0.0893 - auc: 0.9940
## 44/157 [=======>......................] - ETA: 6s - loss: 0.0919 - auc: 0.9937
## 45/157 [=======>......................] - ETA: 6s - loss: 0.0905 - auc: 0.9938
## 46/157 [=======>......................] - ETA: 5s - loss: 0.0925 - auc: 0.9937
## 47/157 [=======>......................] - ETA: 5s - loss: 0.0928 - auc: 0.9938
## 48/157 [========>.....................] - ETA: 5s - loss: 0.0935 - auc: 0.9938
## 49/157 [========>.....................] - ETA: 5s - loss: 0.0929 - auc: 0.9940
## 50/157 [========>.....................] - ETA: 5s - loss: 0.0915 - auc: 0.9941
## 51/157 [========>.....................] - ETA: 5s - loss: 0.0918 - auc: 0.9942
## 52/157 [========>.....................] - ETA: 5s - loss: 0.0954 - auc: 0.9936
## 53/157 [=========>....................] - ETA: 5s - loss: 0.0944 - auc: 0.9937
## 54/157 [=========>....................] - ETA: 5s - loss: 0.0945 - auc: 0.9938
## 55/157 [=========>....................] - ETA: 5s - loss: 0.0947 - auc: 0.9938
## 56/157 [=========>....................] - ETA: 5s - loss: 0.0947 - auc: 0.9938
## 57/157 [=========>....................] - ETA: 5s - loss: 0.0942 - auc: 0.9940
## 58/157 [==========>...................] - ETA: 5s - loss: 0.0929 - auc: 0.9941
## 59/157 [==========>...................] - ETA: 5s - loss: 0.0921 - auc: 0.9942
## 60/157 [==========>...................] - ETA: 5s - loss: 0.0910 - auc: 0.9944
## 61/157 [==========>...................] - ETA: 5s - loss: 0.0919 - auc: 0.9943
## 62/157 [==========>...................] - ETA: 5s - loss: 0.0906 - auc: 0.9944
## 63/157 [===========>..................] - ETA: 5s - loss: 0.0901 - auc: 0.9945
## 64/157 [===========>..................] - ETA: 5s - loss: 0.0911 - auc: 0.9944
## 65/157 [===========>..................] - ETA: 4s - loss: 0.0969 - auc: 0.9936
## 66/157 [===========>..................] - ETA: 4s - loss: 0.0964 - auc: 0.9937
## 67/157 [===========>..................] - ETA: 4s - loss: 0.0956 - auc: 0.9938
## 68/157 [===========>..................] - ETA: 4s - loss: 0.0944 - auc: 0.9939
## 69/157 [============>.................] - ETA: 4s - loss: 0.0933 - auc: 0.9941
## 70/157 [============>.................] - ETA: 4s - loss: 0.0930 - auc: 0.9941
## 71/157 [============>.................] - ETA: 4s - loss: 0.0925 - auc: 0.9942
## 72/157 [============>.................] - ETA: 4s - loss: 0.0962 - auc: 0.9929
## 73/157 [============>.................] - ETA: 4s - loss: 0.0955 - auc: 0.9931
## 74/157 [=============>................] - ETA: 4s - loss: 0.0959 - auc: 0.9931
## 75/157 [=============>................] - ETA: 4s - loss: 0.0950 - auc: 0.9932
## 76/157 [=============>................] - ETA: 4s - loss: 0.0946 - auc: 0.9933
## 77/157 [=============>................] - ETA: 4s - loss: 0.0939 - auc: 0.9935
## 78/157 [=============>................] - ETA: 4s - loss: 0.0940 - auc: 0.9935
## 79/157 [==============>...............] - ETA: 4s - loss: 0.0938 - auc: 0.9935
## 80/157 [==============>...............] - ETA: 4s - loss: 0.0935 - auc: 0.9936
## 81/157 [==============>...............] - ETA: 4s - loss: 0.0931 - auc: 0.9937
## 82/157 [==============>...............] - ETA: 4s - loss: 0.0934 - auc: 0.9937
## 83/157 [==============>...............] - ETA: 3s - loss: 0.0926 - auc: 0.9938
```

```
##  84/157 [===============>..............] - ETA: 3s - loss: 0.0940 - auc: 0.9937
##  85/157 [===============>..............] - ETA: 3s - loss: 0.0931 - auc: 0.9938
##  86/157 [===============>..............] - ETA: 3s - loss: 0.0928 - auc: 0.9939
##  87/157 [===============>..............] - ETA: 3s - loss: 0.0930 - auc: 0.9939
##  88/157 [===============>..............] - ETA: 3s - loss: 0.0926 - auc: 0.9940
##  89/157 [================>.............] - ETA: 3s - loss: 0.0918 - auc: 0.9941
##  90/157 [================>.............] - ETA: 3s - loss: 0.0914 - auc: 0.9941
##  91/157 [================>.............] - ETA: 3s - loss: 0.0908 - auc: 0.9942
##  92/157 [================>.............] - ETA: 3s - loss: 0.0904 - auc: 0.9943
##  93/157 [================>.............] - ETA: 3s - loss: 0.0907 - auc: 0.9943
##  94/157 [================>.............] - ETA: 3s - loss: 0.0914 - auc: 0.9942
##  95/157 [=================>............] - ETA: 3s - loss: 0.0936 - auc: 0.9940
##  96/157 [=================>............] - ETA: 3s - loss: 0.0929 - auc: 0.9941
##  97/157 [=================>............] - ETA: 3s - loss: 0.0920 - auc: 0.9942
##  98/157 [=================>............] - ETA: 3s - loss: 0.0923 - auc: 0.9941
##  99/157 [=================>............] - ETA: 3s - loss: 0.0922 - auc: 0.9942
## 100/157 [==================>...........] - ETA: 3s - loss: 0.0914 - auc: 0.9943
## 101/157 [==================>...........] - ETA: 3s - loss: 0.0910 - auc: 0.9943
## 102/157 [==================>...........] - ETA: 2s - loss: 0.0910 - auc: 0.9944
## 103/157 [==================>...........] - ETA: 2s - loss: 0.0903 - auc: 0.9944
## 104/157 [==================>...........] - ETA: 2s - loss: 0.0901 - auc: 0.9945
## 105/157 [===================>..........] - ETA: 2s - loss: 0.0897 - auc: 0.9945
## 106/157 [===================>..........] - ETA: 2s - loss: 0.0891 - auc: 0.9946
## 107/157 [===================>..........] - ETA: 2s - loss: 0.0886 - auc: 0.9947
## 108/157 [===================>..........] - ETA: 2s - loss: 0.0880 - auc: 0.9947
## 109/157 [===================>..........] - ETA: 2s - loss: 0.0874 - auc: 0.9948
## 110/157 [====================>.........] - ETA: 2s - loss: 0.0870 - auc: 0.9949
## 111/157 [====================>.........] - ETA: 2s - loss: 0.0875 - auc: 0.9948
## 112/157 [====================>.........] - ETA: 2s - loss: 0.0872 - auc: 0.9948
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0881 - auc: 0.9948
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0874 - auc: 0.9948
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0867 - auc: 0.9949
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0861 - auc: 0.9950
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0855 - auc: 0.9950
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0879 - auc: 0.9943
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0876 - auc: 0.9943
## 120/157 [=====================>........] - ETA: 1s - loss: 0.0872 - auc: 0.9944
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0866 - auc: 0.9944
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0861 - auc: 0.9945
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0862 - auc: 0.9945
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0861 - auc: 0.9945
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0856 - auc: 0.9946
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0863 - auc: 0.9946
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0860 - auc: 0.9946
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0860 - auc: 0.9946
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0856 - auc: 0.9947
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0851 - auc: 0.9947
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0854 - auc: 0.9947
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0849 - auc: 0.9948
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0844 - auc: 0.9948
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0854 - auc: 0.9947
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0849 - auc: 0.9948
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0844 - auc: 0.9949
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0839 - auc: 0.9949
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0833 - auc: 0.9950
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0828 - auc: 0.9950
```

```
## 140/157 [==========================>...] - ETA: 0s - loss: 0.0826 - auc: 0.9950
## 141/157 [==========================>...] - ETA: 0s - loss: 0.0843 - auc: 0.9949
## 142/157 [===========================>..] - ETA: 0s - loss: 0.0843 - auc: 0.9949
## 143/157 [===========================>..] - ETA: 0s - loss: 0.0838 - auc: 0.9950
## 144/157 [===========================>..] - ETA: 0s - loss: 0.0834 - auc: 0.9951
## 145/157 [===========================>..] - ETA: 0s - loss: 0.0829 - auc: 0.9951
## 146/157 [===========================>..] - ETA: 0s - loss: 0.0833 - auc: 0.9951
## 147/157 [============================>.] - ETA: 0s - loss: 0.0828 - auc: 0.9951
## 148/157 [============================>.] - ETA: 0s - loss: 0.0836 - auc: 0.9950
## 149/157 [============================>.] - ETA: 0s - loss: 0.0833 - auc: 0.9951
## 150/157 [============================>.] - ETA: 0s - loss: 0.0828 - auc: 0.9951
## 151/157 [============================>.] - ETA: 0s - loss: 0.0833 - auc: 0.9951
## 152/157 [=============================>.] - ETA: 0s - loss: 0.0840 - auc: 0.9950
## 153/157 [=============================>.] - ETA: 0s - loss: 0.0835 - auc: 0.9951
## 154/157 [=============================>.] - ETA: 0s - loss: 0.0830 - auc: 0.9951
## 155/157 [=============================>.] - ETA: 0s - loss: 0.0828 - auc: 0.9951
## 156/157 [=============================>.] - ETA: 0s - loss: 0.0824 - auc: 0.9952
## 157/157 [==============================] - 9s 56ms/step - loss: 0.0824 - auc: 0.9952 - val
_loss: 0.0604 - val_auc: 1.0000
## Epoch 6/10
##
##   1/157 [..............................] - ETA: 8s - loss: 0.0551 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0624 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0465 - auc: 1.0000
##   4/157 [..............................] - ETA: 8s - loss: 0.0906 - auc: 0.9980
##   5/157 [..............................] - ETA: 8s - loss: 0.0833 - auc: 0.9987
##   6/157 [>.............................] - ETA: 8s - loss: 0.0727 - auc: 0.9991
##   7/157 [>.............................] - ETA: 8s - loss: 0.0776 - auc: 0.9974
##   8/157 [>.............................] - ETA: 8s - loss: 0.0724 - auc: 0.9980
##   9/157 [>.............................] - ETA: 8s - loss: 0.0705 - auc: 0.9983
##  10/157 [>.............................] - ETA: 7s - loss: 0.0649 - auc: 0.9986
##  11/157 [=>............................] - ETA: 7s - loss: 0.0615 - auc: 0.9988
##  12/157 [=>............................] - ETA: 7s - loss: 0.0580 - auc: 0.9990
##  13/157 [=>............................] - ETA: 7s - loss: 0.0654 - auc: 0.9982
##  14/157 [=>............................] - ETA: 7s - loss: 0.0633 - auc: 0.9984
##  15/157 [=>............................] - ETA: 7s - loss: 0.0626 - auc: 0.9985
##  16/157 [==>...........................] - ETA: 7s - loss: 0.0806 - auc: 0.9961
##  17/157 [==>...........................] - ETA: 7s - loss: 0.0773 - auc: 0.9964
##  18/157 [==>...........................] - ETA: 7s - loss: 0.0741 - auc: 0.9967
##  19/157 [==>...........................] - ETA: 7s - loss: 0.0727 - auc: 0.9969
##  20/157 [==>...........................] - ETA: 7s - loss: 0.0727 - auc: 0.9969
##  21/157 [===>..........................] - ETA: 7s - loss: 0.0701 - auc: 0.9972
##  22/157 [===>..........................] - ETA: 7s - loss: 0.0676 - auc: 0.9973
##  23/157 [===>..........................] - ETA: 7s - loss: 0.0650 - auc: 0.9975
##  24/157 [===>..........................] - ETA: 7s - loss: 0.0634 - auc: 0.9976
##  25/157 [===>..........................] - ETA: 7s - loss: 0.0630 - auc: 0.9976
##  26/157 [===>..........................] - ETA: 7s - loss: 0.0633 - auc: 0.9976
##  27/157 [====>.........................] - ETA: 7s - loss: 0.0629 - auc: 0.9977
##  28/157 [====>.........................] - ETA: 7s - loss: 0.0643 - auc: 0.9977
##  29/157 [====>.........................] - ETA: 6s - loss: 0.0624 - auc: 0.9978
##  30/157 [====>.........................] - ETA: 6s - loss: 0.0642 - auc: 0.9978
##  31/157 [====>.........................] - ETA: 6s - loss: 0.0625 - auc: 0.9979
##  32/157 [=====>........................] - ETA: 6s - loss: 0.0613 - auc: 0.9980
##  33/157 [=====>........................] - ETA: 6s - loss: 0.0622 - auc: 0.9979
##  34/157 [=====>........................] - ETA: 6s - loss: 0.0676 - auc: 0.9971
##  35/157 [=====>........................] - ETA: 6s - loss: 0.0671 - auc: 0.9972
```

```
##  36/157 [=====>........................] - ETA: 6s - loss: 0.0666 - auc: 0.9972
##  37/157 [======>.......................] - ETA: 6s - loss: 0.0677 - auc: 0.9971
##  38/157 [======>.......................] - ETA: 6s - loss: 0.0683 - auc: 0.9971
##  39/157 [======>.......................] - ETA: 6s - loss: 0.0673 - auc: 0.9972
##  40/157 [======>.......................] - ETA: 6s - loss: 0.0684 - auc: 0.9972
##  41/157 [======>.......................] - ETA: 6s - loss: 0.0695 - auc: 0.9971
##  42/157 [=======>......................] - ETA: 6s - loss: 0.0681 - auc: 0.9972
##  43/157 [=======>......................] - ETA: 6s - loss: 0.0673 - auc: 0.9973
##  44/157 [=======>......................] - ETA: 6s - loss: 0.0671 - auc: 0.9973
##  45/157 [=======>......................] - ETA: 6s - loss: 0.0668 - auc: 0.9974
##  46/157 [=======>......................] - ETA: 6s - loss: 0.0736 - auc: 0.9966
##  47/157 [=======>......................] - ETA: 5s - loss: 0.0723 - auc: 0.9968
##  48/157 [========>.....................] - ETA: 5s - loss: 0.0713 - auc: 0.9968
##  49/157 [========>.....................] - ETA: 5s - loss: 0.0710 - auc: 0.9969
##  50/157 [========>.....................] - ETA: 5s - loss: 0.0699 - auc: 0.9970
##  51/157 [========>.....................] - ETA: 5s - loss: 0.0690 - auc: 0.9971
##  52/157 [========>.....................] - ETA: 5s - loss: 0.0678 - auc: 0.9972
##  53/157 [=========>....................] - ETA: 5s - loss: 0.0672 - auc: 0.9972
##  54/157 [=========>....................] - ETA: 5s - loss: 0.0667 - auc: 0.9973
##  55/157 [=========>....................] - ETA: 5s - loss: 0.0659 - auc: 0.9974
##  56/157 [=========>....................] - ETA: 5s - loss: 0.0656 - auc: 0.9974
##  57/157 [=========>....................] - ETA: 5s - loss: 0.0645 - auc: 0.9975
##  58/157 [==========>...................] - ETA: 5s - loss: 0.0646 - auc: 0.9975
##  59/157 [==========>...................] - ETA: 5s - loss: 0.0640 - auc: 0.9976
##  60/157 [==========>...................] - ETA: 5s - loss: 0.0638 - auc: 0.9976
##  61/157 [==========>...................] - ETA: 5s - loss: 0.0632 - auc: 0.9976
##  62/157 [==========>...................] - ETA: 5s - loss: 0.0635 - auc: 0.9976
##  63/157 [===========>..................] - ETA: 5s - loss: 0.0639 - auc: 0.9976
##  64/157 [===========>..................] - ETA: 5s - loss: 0.0630 - auc: 0.9976
##  65/157 [===========>..................] - ETA: 5s - loss: 0.0622 - auc: 0.9977
##  66/157 [===========>..................] - ETA: 4s - loss: 0.0615 - auc: 0.9977
##  67/157 [===========>..................] - ETA: 4s - loss: 0.0610 - auc: 0.9978
##  68/157 [===========>..................] - ETA: 4s - loss: 0.0603 - auc: 0.9978
##  69/157 [============>.................] - ETA: 4s - loss: 0.0600 - auc: 0.9979
##  70/157 [============>.................] - ETA: 4s - loss: 0.0594 - auc: 0.9979
##  71/157 [============>.................] - ETA: 4s - loss: 0.0607 - auc: 0.9978
##  72/157 [============>.................] - ETA: 4s - loss: 0.0600 - auc: 0.9979
##  73/157 [============>.................] - ETA: 4s - loss: 0.0597 - auc: 0.9979
##  74/157 [=============>................] - ETA: 4s - loss: 0.0591 - auc: 0.9980
##  75/157 [=============>................] - ETA: 4s - loss: 0.0584 - auc: 0.9980
##  76/157 [=============>................] - ETA: 4s - loss: 0.0582 - auc: 0.9981
##  77/157 [=============>................] - ETA: 4s - loss: 0.0577 - auc: 0.9981
##  78/157 [=============>................] - ETA: 4s - loss: 0.0573 - auc: 0.9981
##  79/157 [==============>...............] - ETA: 4s - loss: 0.0567 - auc: 0.9982
##  80/157 [==============>...............] - ETA: 4s - loss: 0.0562 - auc: 0.9982
##  81/157 [==============>...............] - ETA: 4s - loss: 0.0556 - auc: 0.9982
##  82/157 [==============>...............] - ETA: 4s - loss: 0.0558 - auc: 0.9982
##  83/157 [==============>...............] - ETA: 4s - loss: 0.0574 - auc: 0.9981
##  84/157 [===============>..............] - ETA: 3s - loss: 0.0573 - auc: 0.9981
##  85/157 [===============>..............] - ETA: 3s - loss: 0.0571 - auc: 0.9981
##  86/157 [===============>..............] - ETA: 3s - loss: 0.0567 - auc: 0.9982
##  87/157 [===============>..............] - ETA: 3s - loss: 0.0564 - auc: 0.9982
##  88/157 [===============>..............] - ETA: 3s - loss: 0.0560 - auc: 0.9982
##  89/157 [================>.............] - ETA: 3s - loss: 0.0555 - auc: 0.9982
##  90/157 [================>.............] - ETA: 3s - loss: 0.0552 - auc: 0.9983
##  91/157 [================>.............] - ETA: 3s - loss: 0.0550 - auc: 0.9983
```

```
##  92/157 [================>.............] - ETA: 3s - loss: 0.0545 - auc: 0.9983
##  93/157 [================>.............] - ETA: 3s - loss: 0.0546 - auc: 0.9983
##  94/157 [================>.............] - ETA: 3s - loss: 0.0541 - auc: 0.9984
##  95/157 [=================>............] - ETA: 3s - loss: 0.0536 - auc: 0.9984
##  96/157 [=================>............] - ETA: 3s - loss: 0.0537 - auc: 0.9984
##  97/157 [=================>............] - ETA: 3s - loss: 0.0539 - auc: 0.9984
##  98/157 [=================>............] - ETA: 3s - loss: 0.0536 - auc: 0.9984
##  99/157 [=================>............] - ETA: 3s - loss: 0.0532 - auc: 0.9984
## 100/157 [==================>...........] - ETA: 3s - loss: 0.0531 - auc: 0.9985
## 101/157 [==================>...........] - ETA: 3s - loss: 0.0526 - auc: 0.9985
## 102/157 [==================>...........] - ETA: 2s - loss: 0.0525 - auc: 0.9985
## 103/157 [==================>...........] - ETA: 2s - loss: 0.0521 - auc: 0.9985
## 104/157 [==================>...........] - ETA: 2s - loss: 0.0519 - auc: 0.9985
## 105/157 [===================>..........] - ETA: 2s - loss: 0.0515 - auc: 0.9986
## 106/157 [===================>..........] - ETA: 2s - loss: 0.0527 - auc: 0.9984
## 107/157 [===================>..........] - ETA: 2s - loss: 0.0525 - auc: 0.9985
## 108/157 [===================>..........] - ETA: 2s - loss: 0.0523 - auc: 0.9985
## 109/157 [===================>..........] - ETA: 2s - loss: 0.0522 - auc: 0.9985
## 110/157 [====================>.........] - ETA: 2s - loss: 0.0525 - auc: 0.9985
## 111/157 [====================>.........] - ETA: 2s - loss: 0.0522 - auc: 0.9985
## 112/157 [====================>.........] - ETA: 2s - loss: 0.0518 - auc: 0.9985
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0522 - auc: 0.9985
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0519 - auc: 0.9985
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0515 - auc: 0.9986
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0521 - auc: 0.9985
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0523 - auc: 0.9985
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0520 - auc: 0.9986
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0516 - auc: 0.9986
## 120/157 [=====================>........] - ETA: 2s - loss: 0.0520 - auc: 0.9986
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0517 - auc: 0.9986
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0514 - auc: 0.9986
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0523 - auc: 0.9985
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0519 - auc: 0.9985
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0521 - auc: 0.9985
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0519 - auc: 0.9986
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0516 - auc: 0.9986
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0513 - auc: 0.9986
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0509 - auc: 0.9986
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0506 - auc: 0.9986
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0503 - auc: 0.9986
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0502 - auc: 0.9987
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0501 - auc: 0.9987
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0498 - auc: 0.9987
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0506 - auc: 0.9987
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0509 - auc: 0.9986
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0506 - auc: 0.9987
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0504 - auc: 0.9987
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0502 - auc: 0.9987
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0499 - auc: 0.9987
## 141/157 [=========================>....] - ETA: 0s - loss: 0.0496 - auc: 0.9987
## 142/157 [==========================>...] - ETA: 0s - loss: 0.0495 - auc: 0.9987
## 143/157 [==========================>...] - ETA: 0s - loss: 0.0498 - auc: 0.9987
## 144/157 [==========================>...] - ETA: 0s - loss: 0.0504 - auc: 0.9987
## 145/157 [==========================>...] - ETA: 0s - loss: 0.0502 - auc: 0.9987
## 146/157 [==========================>...] - ETA: 0s - loss: 0.0501 - auc: 0.9987
## 147/157 [===========================>..] - ETA: 0s - loss: 0.0500 - auc: 0.9987
```

```
## 148/157 [===========================>..] - ETA: 0s - loss: 0.0497 - auc: 0.9987
## 149/157 [===========================>..] - ETA: 0s - loss: 0.0497 - auc: 0.9987
## 150/157 [===========================>..] - ETA: 0s - loss: 0.0496 - auc: 0.9987
## 151/157 [===========================>..] - ETA: 0s - loss: 0.0495 - auc: 0.9988
## 152/157 [============================>.] - ETA: 0s - loss: 0.0493 - auc: 0.9988
## 153/157 [============================>.] - ETA: 0s - loss: 0.0492 - auc: 0.9988
## 154/157 [============================>.] - ETA: 0s - loss: 0.0490 - auc: 0.9988
## 155/157 [============================>.] - ETA: 0s - loss: 0.0487 - auc: 0.9988
## 156/157 [============================>.] - ETA: 0s - loss: 0.0485 - auc: 0.9988
## 157/157 [==============================] - 9s 57ms/step - loss: 0.0485 - auc: 0.9988 - val
_loss: 0.0396 - val_auc: 0.9992
## Epoch 7/10
## 
##   1/157 [..............................] - ETA: 8s - loss: 0.0066 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0167 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0366 - auc: 1.0000
##   4/157 [..............................] - ETA: 8s - loss: 0.0464 - auc: 0.9990
##   5/157 [..............................] - ETA: 8s - loss: 0.0412 - auc: 0.9994
##   6/157 [>.............................] - ETA: 8s - loss: 0.0360 - auc: 0.9996
##   7/157 [>.............................] - ETA: 8s - loss: 0.0318 - auc: 0.9997
##   8/157 [>.............................] - ETA: 8s - loss: 0.0288 - auc: 0.9998
##   9/157 [>.............................] - ETA: 8s - loss: 0.0272 - auc: 0.9998
##  10/157 [>.............................] - ETA: 7s - loss: 0.0305 - auc: 0.9997
##  11/157 [=>............................] - ETA: 7s - loss: 0.0306 - auc: 0.9997
##  12/157 [=>............................] - ETA: 7s - loss: 0.0290 - auc: 0.9998
##  13/157 [=>............................] - ETA: 7s - loss: 0.0274 - auc: 0.9998
##  14/157 [=>............................] - ETA: 7s - loss: 0.0258 - auc: 0.9998
##  15/157 [=>............................] - ETA: 7s - loss: 0.0249 - auc: 0.9999
##  16/157 [==>...........................] - ETA: 7s - loss: 0.0241 - auc: 0.9999
##  17/157 [==>...........................] - ETA: 7s - loss: 0.0243 - auc: 0.9999
##  18/157 [==>...........................] - ETA: 7s - loss: 0.0239 - auc: 0.9999
##  19/157 [==>...........................] - ETA: 7s - loss: 0.0231 - auc: 0.9999
##  20/157 [==>...........................] - ETA: 7s - loss: 0.0229 - auc: 0.9999
##  21/157 [===>..........................] - ETA: 7s - loss: 0.0222 - auc: 0.9999
##  22/157 [===>..........................] - ETA: 7s - loss: 0.0216 - auc: 0.9999
##  23/157 [===>..........................] - ETA: 7s - loss: 0.0219 - auc: 0.9999
##  24/157 [===>..........................] - ETA: 7s - loss: 0.0217 - auc: 0.9999
##  25/157 [===>..........................] - ETA: 7s - loss: 0.0218 - auc: 0.9999
##  26/157 [===>..........................] - ETA: 7s - loss: 0.0227 - auc: 1.0000
##  27/157 [====>.........................] - ETA: 7s - loss: 0.0219 - auc: 1.0000
##  28/157 [====>.........................] - ETA: 7s - loss: 0.0228 - auc: 1.0000
##  29/157 [====>.........................] - ETA: 6s - loss: 0.0228 - auc: 1.0000
##  30/157 [====>.........................] - ETA: 6s - loss: 0.0230 - auc: 1.0000
##  31/157 [====>.........................] - ETA: 6s - loss: 0.0316 - auc: 0.9991
##  32/157 [=====>........................] - ETA: 6s - loss: 0.0318 - auc: 0.9991
##  33/157 [=====>........................] - ETA: 6s - loss: 0.0313 - auc: 0.9991
##  34/157 [=====>........................] - ETA: 6s - loss: 0.0309 - auc: 0.9992
##  35/157 [=====>........................] - ETA: 6s - loss: 0.0308 - auc: 0.9992
##  36/157 [=====>........................] - ETA: 6s - loss: 0.0303 - auc: 0.9991
##  37/157 [======>.......................] - ETA: 6s - loss: 0.0301 - auc: 0.9992
##  38/157 [======>.......................] - ETA: 6s - loss: 0.0296 - auc: 0.9992
##  39/157 [======>.......................] - ETA: 6s - loss: 0.0290 - auc: 0.9992
##  40/157 [======>.......................] - ETA: 6s - loss: 0.0285 - auc: 0.9993
##  41/157 [======>.......................] - ETA: 6s - loss: 0.0279 - auc: 0.9993
##  42/157 [=======>......................] - ETA: 6s - loss: 0.0293 - auc: 0.9993
##  43/157 [=======>......................] - ETA: 6s - loss: 0.0288 - auc: 0.9993
```

```
## 44/157 [=======>......................] - ETA: 6s - loss: 0.0287 - auc: 0.9993
## 45/157 [=======>......................] - ETA: 6s - loss: 0.0317 - auc: 0.9993
## 46/157 [=======>......................] - ETA: 6s - loss: 0.0313 - auc: 0.9993
## 47/157 [=======>......................] - ETA: 6s - loss: 0.0310 - auc: 0.9993
## 48/157 [========>.....................] - ETA: 5s - loss: 0.0305 - auc: 0.9993
## 49/157 [========>.....................] - ETA: 5s - loss: 0.0306 - auc: 0.9993
## 50/157 [========>.....................] - ETA: 5s - loss: 0.0302 - auc: 0.9993
## 51/157 [========>.....................] - ETA: 5s - loss: 0.0298 - auc: 0.9993
## 52/157 [========>.....................] - ETA: 5s - loss: 0.0300 - auc: 0.9994
## 53/157 [=========>....................] - ETA: 5s - loss: 0.0343 - auc: 0.9993
## 54/157 [=========>....................] - ETA: 5s - loss: 0.0420 - auc: 0.9990
## 55/157 [=========>....................] - ETA: 5s - loss: 0.0416 - auc: 0.9990
## 56/157 [=========>....................] - ETA: 5s - loss: 0.0410 - auc: 0.9990
## 57/157 [=========>....................] - ETA: 5s - loss: 0.0420 - auc: 0.9989
## 58/157 [==========>...................] - ETA: 5s - loss: 0.0415 - auc: 0.9990
## 59/157 [==========>...................] - ETA: 5s - loss: 0.0410 - auc: 0.9990
## 60/157 [==========>...................] - ETA: 5s - loss: 0.0409 - auc: 0.9990
## 61/157 [==========>...................] - ETA: 5s - loss: 0.0419 - auc: 0.9989
## 62/157 [==========>...................] - ETA: 5s - loss: 0.0415 - auc: 0.9989
## 63/157 [===========>..................] - ETA: 5s - loss: 0.0415 - auc: 0.9989
## 64/157 [===========>..................] - ETA: 5s - loss: 0.0409 - auc: 0.9989
## 65/157 [===========>..................] - ETA: 5s - loss: 0.0420 - auc: 0.9989
## 66/157 [===========>..................] - ETA: 4s - loss: 0.0417 - auc: 0.9989
## 67/157 [===========>..................] - ETA: 4s - loss: 0.0424 - auc: 0.9989
## 68/157 [===========>..................] - ETA: 4s - loss: 0.0428 - auc: 0.9988
## 69/157 [============>.................] - ETA: 4s - loss: 0.0426 - auc: 0.9988
## 70/157 [============>.................] - ETA: 4s - loss: 0.0420 - auc: 0.9989
## 71/157 [============>.................] - ETA: 4s - loss: 0.0417 - auc: 0.9989
## 72/157 [============>.................] - ETA: 4s - loss: 0.0413 - auc: 0.9989
## 73/157 [============>.................] - ETA: 4s - loss: 0.0411 - auc: 0.9989
## 74/157 [=============>................] - ETA: 4s - loss: 0.0417 - auc: 0.9989
## 75/157 [=============>................] - ETA: 4s - loss: 0.0413 - auc: 0.9989
## 76/157 [=============>................] - ETA: 4s - loss: 0.0435 - auc: 0.9989
## 77/157 [=============>................] - ETA: 4s - loss: 0.0430 - auc: 0.9989
## 78/157 [=============>................] - ETA: 4s - loss: 0.0436 - auc: 0.9989
## 79/157 [==============>...............] - ETA: 4s - loss: 0.0432 - auc: 0.9989
## 80/157 [==============>...............] - ETA: 4s - loss: 0.0428 - auc: 0.9990
## 81/157 [==============>...............] - ETA: 4s - loss: 0.0424 - auc: 0.9990
## 82/157 [==============>...............] - ETA: 4s - loss: 0.0420 - auc: 0.9990
## 83/157 [==============>...............] - ETA: 4s - loss: 0.0419 - auc: 0.9990
## 84/157 [===============>..............] - ETA: 3s - loss: 0.0418 - auc: 0.9990
## 85/157 [===============>..............] - ETA: 3s - loss: 0.0416 - auc: 0.9990
## 86/157 [===============>..............] - ETA: 3s - loss: 0.0429 - auc: 0.9990
## 87/157 [===============>..............] - ETA: 3s - loss: 0.0426 - auc: 0.9990
## 88/157 [===============>..............] - ETA: 3s - loss: 0.0429 - auc: 0.9990
## 89/157 [================>.............] - ETA: 3s - loss: 0.0426 - auc: 0.9990
## 90/157 [================>.............] - ETA: 3s - loss: 0.0422 - auc: 0.9990
## 91/157 [================>.............] - ETA: 3s - loss: 0.0421 - auc: 0.9991
## 92/157 [================>.............] - ETA: 3s - loss: 0.0417 - auc: 0.9991
## 93/157 [================>.............] - ETA: 3s - loss: 0.0415 - auc: 0.9991
## 94/157 [================>.............] - ETA: 3s - loss: 0.0411 - auc: 0.9991
## 95/157 [=================>............] - ETA: 3s - loss: 0.0407 - auc: 0.9991
## 96/157 [=================>............] - ETA: 3s - loss: 0.0408 - auc: 0.9991
## 97/157 [=================>............] - ETA: 3s - loss: 0.0405 - auc: 0.9991
## 98/157 [=================>............] - ETA: 3s - loss: 0.0403 - auc: 0.9992
## 99/157 [=================>............] - ETA: 3s - loss: 0.0399 - auc: 0.9992
```

```
## 100/157 [==================>...........] - ETA: 3s - loss: 0.0396 - auc: 0.9992
## 101/157 [==================>...........] - ETA: 3s - loss: 0.0418 - auc: 0.9989
## 102/157 [==================>...........] - ETA: 3s - loss: 0.0415 - auc: 0.9989
## 103/157 [==================>...........] - ETA: 2s - loss: 0.0412 - auc: 0.9990
## 104/157 [==================>...........] - ETA: 2s - loss: 0.0409 - auc: 0.9990
## 105/157 [===================>..........] - ETA: 2s - loss: 0.0409 - auc: 0.9990
## 106/157 [===================>..........] - ETA: 2s - loss: 0.0407 - auc: 0.9990
## 107/157 [===================>..........] - ETA: 2s - loss: 0.0404 - auc: 0.9990
## 108/157 [===================>..........] - ETA: 2s - loss: 0.0401 - auc: 0.9990
## 109/157 [===================>..........] - ETA: 2s - loss: 0.0398 - auc: 0.9990
## 110/157 [====================>.........] - ETA: 2s - loss: 0.0395 - auc: 0.9990
## 111/157 [====================>.........] - ETA: 2s - loss: 0.0392 - auc: 0.9991
## 112/157 [====================>.........] - ETA: 2s - loss: 0.0389 - auc: 0.9991
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0393 - auc: 0.9991
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0391 - auc: 0.9991
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0388 - auc: 0.9991
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0391 - auc: 0.9991
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0389 - auc: 0.9991
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0386 - auc: 0.9991
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0411 - auc: 0.9989
## 120/157 [=====================>........] - ETA: 2s - loss: 0.0408 - auc: 0.9990
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0409 - auc: 0.9990
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0406 - auc: 0.9990
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0403 - auc: 0.9990
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0402 - auc: 0.9990
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0404 - auc: 0.9990
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0401 - auc: 0.9990
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0402 - auc: 0.9990
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0400 - auc: 0.9990
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0398 - auc: 0.9990
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0399 - auc: 0.9990
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0397 - auc: 0.9990
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0400 - auc: 0.9990
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0398 - auc: 0.9990
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0395 - auc: 0.9991
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0401 - auc: 0.9990
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0398 - auc: 0.9991
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0396 - auc: 0.9991
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0397 - auc: 0.9991
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0412 - auc: 0.9989
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0410 - auc: 0.9989
## 141/157 [=========================>....] - ETA: 0s - loss: 0.0408 - auc: 0.9990
## 142/157 [==========================>...] - ETA: 0s - loss: 0.0406 - auc: 0.9990
## 143/157 [==========================>...] - ETA: 0s - loss: 0.0403 - auc: 0.9990
## 144/157 [==========================>...] - ETA: 0s - loss: 0.0402 - auc: 0.9990
## 145/157 [==========================>...] - ETA: 0s - loss: 0.0400 - auc: 0.9990
## 146/157 [==========================>...] - ETA: 0s - loss: 0.0398 - auc: 0.9990
## 147/157 [===========================>..] - ETA: 0s - loss: 0.0396 - auc: 0.9990
## 148/157 [===========================>..] - ETA: 0s - loss: 0.0402 - auc: 0.9990
## 149/157 [===========================>..] - ETA: 0s - loss: 0.0400 - auc: 0.9990
## 150/157 [===========================>..] - ETA: 0s - loss: 0.0399 - auc: 0.9990
## 151/157 [===========================>..] - ETA: 0s - loss: 0.0398 - auc: 0.9990
## 152/157 [============================>.] - ETA: 0s - loss: 0.0399 - auc: 0.9990
## 153/157 [============================>.] - ETA: 0s - loss: 0.0398 - auc: 0.9990
## 154/157 [============================>.] - ETA: 0s - loss: 0.0398 - auc: 0.9990
## 155/157 [============================>.] - ETA: 0s - loss: 0.0397 - auc: 0.9990
```

```
## 156/157 [============================>.] - ETA: 0s - loss: 0.0396 - auc: 0.9990
## 157/157 [==============================] - 9s 57ms/step - loss: 0.0400 - auc: 0.9990 - val
_loss: 0.3231 - val_auc: 0.9991
## Epoch 8/10
## 
##   1/157 [..............................] - ETA: 8s - loss: 0.0865 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0592 - auc: 1.0000
##   3/157 [..............................] - ETA: 8s - loss: 0.0769 - auc: 0.9964
##   4/157 [..............................] - ETA: 8s - loss: 0.0633 - auc: 0.9980
##   5/157 [..............................] - ETA: 8s - loss: 0.0597 - auc: 0.9987
##   6/157 [>.............................] - ETA: 8s - loss: 0.0861 - auc: 0.9976
##   7/157 [>.............................] - ETA: 8s - loss: 0.0752 - auc: 0.9982
##   8/157 [>.............................] - ETA: 8s - loss: 0.0687 - auc: 0.9986
##   9/157 [>.............................] - ETA: 8s - loss: 0.0641 - auc: 0.9989
##  10/157 [>.............................] - ETA: 8s - loss: 0.0590 - auc: 0.9991
##  11/157 [=>............................] - ETA: 8s - loss: 0.0571 - auc: 0.9992
##  12/157 [=>............................] - ETA: 7s - loss: 0.0535 - auc: 0.9993
##  13/157 [=>............................] - ETA: 7s - loss: 0.0584 - auc: 0.9993
##  14/157 [=>............................] - ETA: 7s - loss: 0.0549 - auc: 0.9994
##  15/157 [=>............................] - ETA: 7s - loss: 0.0514 - auc: 0.9994
##  16/157 [==>...........................] - ETA: 7s - loss: 0.0568 - auc: 0.9990
##  17/157 [==>...........................] - ETA: 7s - loss: 0.0544 - auc: 0.9991
##  18/157 [==>...........................] - ETA: 7s - loss: 0.0545 - auc: 0.9991
##  19/157 [==>...........................] - ETA: 7s - loss: 0.0525 - auc: 0.9992
##  20/157 [==>...........................] - ETA: 7s - loss: 0.0503 - auc: 0.9993
##  21/157 [===>..........................] - ETA: 7s - loss: 0.0484 - auc: 0.9993
##  22/157 [===>..........................] - ETA: 7s - loss: 0.0478 - auc: 0.9994
##  23/157 [===>..........................] - ETA: 7s - loss: 0.0459 - auc: 0.9994
##  24/157 [===>..........................] - ETA: 7s - loss: 0.0444 - auc: 0.9995
##  25/157 [===>..........................] - ETA: 7s - loss: 0.0430 - auc: 0.9995
##  26/157 [===>..........................] - ETA: 7s - loss: 0.0551 - auc: 0.9983
##  27/157 [====>.........................] - ETA: 7s - loss: 0.0538 - auc: 0.9984
##  28/157 [====>.........................] - ETA: 7s - loss: 0.0530 - auc: 0.9985
##  29/157 [====>.........................] - ETA: 7s - loss: 0.0529 - auc: 0.9985
##  30/157 [====>.........................] - ETA: 7s - loss: 0.0532 - auc: 0.9985
##  31/157 [====>.........................] - ETA: 6s - loss: 0.0521 - auc: 0.9986
##  32/157 [=====>........................] - ETA: 6s - loss: 0.0519 - auc: 0.9986
##  33/157 [=====>........................] - ETA: 6s - loss: 0.0514 - auc: 0.9986
##  34/157 [=====>........................] - ETA: 6s - loss: 0.0506 - auc: 0.9987
##  35/157 [=====>........................] - ETA: 6s - loss: 0.0494 - auc: 0.9987
##  36/157 [=====>........................] - ETA: 6s - loss: 0.0512 - auc: 0.9987
##  37/157 [======>.......................] - ETA: 6s - loss: 0.0501 - auc: 0.9987
##  38/157 [======>.......................] - ETA: 6s - loss: 0.0490 - auc: 0.9988
##  39/157 [======>.......................] - ETA: 6s - loss: 0.0485 - auc: 0.9988
##  40/157 [======>.......................] - ETA: 6s - loss: 0.0480 - auc: 0.9988
##  41/157 [======>.......................] - ETA: 6s - loss: 0.0470 - auc: 0.9989
##  42/157 [=======>......................] - ETA: 6s - loss: 0.0467 - auc: 0.9989
##  43/157 [=======>......................] - ETA: 6s - loss: 0.0462 - auc: 0.9990
##  44/157 [=======>......................] - ETA: 6s - loss: 0.0463 - auc: 0.9990
##  45/157 [=======>......................] - ETA: 6s - loss: 0.0470 - auc: 0.9990
##  46/157 [=======>......................] - ETA: 6s - loss: 0.0462 - auc: 0.9990
##  47/157 [=======>......................] - ETA: 6s - loss: 0.0455 - auc: 0.9991
##  48/157 [========>.....................] - ETA: 6s - loss: 0.0451 - auc: 0.9991
##  49/157 [========>.....................] - ETA: 5s - loss: 0.0444 - auc: 0.9991
##  50/157 [========>.....................] - ETA: 5s - loss: 0.0436 - auc: 0.9992
##  51/157 [========>.....................] - ETA: 5s - loss: 0.0428 - auc: 0.9992
```

```
##  52/157 [========>.....................] - ETA: 5s - loss: 0.0422 - auc: 0.9992
##  53/157 [=========>....................] - ETA: 5s - loss: 0.0432 - auc: 0.9992
##  54/157 [=========>....................] - ETA: 5s - loss: 0.0437 - auc: 0.9992
##  55/157 [=========>....................] - ETA: 5s - loss: 0.0431 - auc: 0.9992
##  56/157 [=========>....................] - ETA: 5s - loss: 0.0428 - auc: 0.9992
##  57/157 [=========>....................] - ETA: 5s - loss: 0.0425 - auc: 0.9992
##  58/157 [==========>...................] - ETA: 5s - loss: 0.0431 - auc: 0.9992
##  59/157 [==========>...................] - ETA: 5s - loss: 0.0426 - auc: 0.9992
##  60/157 [==========>...................] - ETA: 5s - loss: 0.0425 - auc: 0.9993
##  61/157 [==========>...................] - ETA: 5s - loss: 0.0422 - auc: 0.9993
##  62/157 [==========>...................] - ETA: 5s - loss: 0.0443 - auc: 0.9992
##  63/157 [===========>..................] - ETA: 5s - loss: 0.0437 - auc: 0.9992
##  64/157 [===========>..................] - ETA: 5s - loss: 0.0431 - auc: 0.9992
##  65/157 [===========>..................] - ETA: 5s - loss: 0.0427 - auc: 0.9992
##  66/157 [===========>..................] - ETA: 5s - loss: 0.0422 - auc: 0.9992
##  67/157 [===========>..................] - ETA: 4s - loss: 0.0420 - auc: 0.9993
##  68/157 [===========>..................] - ETA: 4s - loss: 0.0417 - auc: 0.9993
##  69/157 [============>.................] - ETA: 4s - loss: 0.0421 - auc: 0.9992
##  70/157 [============>.................] - ETA: 4s - loss: 0.0418 - auc: 0.9993
##  71/157 [============>.................] - ETA: 4s - loss: 0.0415 - auc: 0.9993
##  72/157 [============>.................] - ETA: 4s - loss: 0.0414 - auc: 0.9993
##  73/157 [============>.................] - ETA: 4s - loss: 0.0409 - auc: 0.9993
##  74/157 [=============>................] - ETA: 4s - loss: 0.0409 - auc: 0.9993
##  75/157 [=============>................] - ETA: 4s - loss: 0.0414 - auc: 0.9993
##  76/157 [=============>................] - ETA: 4s - loss: 0.0420 - auc: 0.9993
##  77/157 [=============>................] - ETA: 4s - loss: 0.0416 - auc: 0.9993
##  78/157 [=============>................] - ETA: 4s - loss: 0.0415 - auc: 0.9993
##  79/157 [==============>...............] - ETA: 4s - loss: 0.0412 - auc: 0.9993
##  80/157 [==============>...............] - ETA: 4s - loss: 0.0408 - auc: 0.9993
##  81/157 [==============>...............] - ETA: 4s - loss: 0.0404 - auc: 0.9993
##  82/157 [==============>...............] - ETA: 4s - loss: 0.0412 - auc: 0.9993
##  83/157 [==============>...............] - ETA: 4s - loss: 0.0415 - auc: 0.9993
##  84/157 [===============>..............] - ETA: 4s - loss: 0.0430 - auc: 0.9992
##  85/157 [===============>..............] - ETA: 3s - loss: 0.0426 - auc: 0.9992
##  86/157 [===============>..............] - ETA: 3s - loss: 0.0422 - auc: 0.9992
##  87/157 [===============>..............] - ETA: 3s - loss: 0.0423 - auc: 0.9992
##  88/157 [===============>..............] - ETA: 3s - loss: 0.0421 - auc: 0.9992
##  89/157 [================>.............] - ETA: 3s - loss: 0.0417 - auc: 0.9992
##  90/157 [================>.............] - ETA: 3s - loss: 0.0413 - auc: 0.9993
##  91/157 [================>.............] - ETA: 3s - loss: 0.0410 - auc: 0.9993
##  92/157 [================>.............] - ETA: 3s - loss: 0.0406 - auc: 0.9993
##  93/157 [================>.............] - ETA: 3s - loss: 0.0407 - auc: 0.9993
##  94/157 [================>.............] - ETA: 3s - loss: 0.0403 - auc: 0.9993
##  95/157 [=================>............] - ETA: 3s - loss: 0.0408 - auc: 0.9993
##  96/157 [=================>............] - ETA: 3s - loss: 0.0404 - auc: 0.9993
##  97/157 [=================>............] - ETA: 3s - loss: 0.0429 - auc: 0.9991
##  98/157 [=================>............] - ETA: 3s - loss: 0.0427 - auc: 0.9991
##  99/157 [=================>............] - ETA: 3s - loss: 0.0424 - auc: 0.9991
## 100/157 [==================>...........] - ETA: 3s - loss: 0.0421 - auc: 0.9991
## 101/157 [==================>...........] - ETA: 3s - loss: 0.0418 - auc: 0.9991
## 102/157 [==================>...........] - ETA: 3s - loss: 0.0416 - auc: 0.9991
## 103/157 [==================>...........] - ETA: 2s - loss: 0.0429 - auc: 0.9990
## 104/157 [==================>...........] - ETA: 2s - loss: 0.0427 - auc: 0.9991
## 105/157 [===================>..........] - ETA: 2s - loss: 0.0435 - auc: 0.9990
## 106/157 [===================>..........] - ETA: 2s - loss: 0.0434 - auc: 0.9990
## 107/157 [===================>..........] - ETA: 2s - loss: 0.0432 - auc: 0.9990
```

```
## 108/157 [===================>..........] - ETA: 2s - loss: 0.0430 - auc: 0.9990
## 109/157 [===================>..........] - ETA: 2s - loss: 0.0426 - auc: 0.9991
## 110/157 [====================>.........] - ETA: 2s - loss: 0.0423 - auc: 0.9991
## 111/157 [====================>.........] - ETA: 2s - loss: 0.0420 - auc: 0.9991
## 112/157 [====================>.........] - ETA: 2s - loss: 0.0493 - auc: 0.9986
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0490 - auc: 0.9986
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0492 - auc: 0.9986
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0489 - auc: 0.9986
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0486 - auc: 0.9987
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0488 - auc: 0.9986
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0505 - auc: 0.9985
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0503 - auc: 0.9985
## 120/157 [=====================>........] - ETA: 2s - loss: 0.0501 - auc: 0.9985
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0497 - auc: 0.9986
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0507 - auc: 0.9985
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0506 - auc: 0.9985
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0503 - auc: 0.9985
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0502 - auc: 0.9985
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0499 - auc: 0.9986
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0496 - auc: 0.9986
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0532 - auc: 0.9979
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0537 - auc: 0.9979
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0536 - auc: 0.9979
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0535 - auc: 0.9979
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0532 - auc: 0.9980
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0530 - auc: 0.9980
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0529 - auc: 0.9980
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0532 - auc: 0.9980
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0533 - auc: 0.9980
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0532 - auc: 0.9980
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0529 - auc: 0.9980
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0527 - auc: 0.9980
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0525 - auc: 0.9980
## 141/157 [=========================>....] - ETA: 0s - loss: 0.0525 - auc: 0.9980
## 142/157 [==========================>...] - ETA: 0s - loss: 0.0522 - auc: 0.9980
## 143/157 [==========================>...] - ETA: 0s - loss: 0.0522 - auc: 0.9980
## 144/157 [==========================>...] - ETA: 0s - loss: 0.0519 - auc: 0.9981
## 145/157 [==========================>...] - ETA: 0s - loss: 0.0516 - auc: 0.9981
## 146/157 [==========================>...] - ETA: 0s - loss: 0.0514 - auc: 0.9981
## 147/157 [===========================>..] - ETA: 0s - loss: 0.0512 - auc: 0.9981
## 148/157 [===========================>..] - ETA: 0s - loss: 0.0509 - auc: 0.9981
## 149/157 [===========================>..] - ETA: 0s - loss: 0.0507 - auc: 0.9982
## 150/157 [===========================>..] - ETA: 0s - loss: 0.0505 - auc: 0.9982
## 151/157 [===========================>..] - ETA: 0s - loss: 0.0503 - auc: 0.9982
## 152/157 [============================>.] - ETA: 0s - loss: 0.0501 - auc: 0.9982
## 153/157 [============================>.] - ETA: 0s - loss: 0.0528 - auc: 0.9979
## 154/157 [============================>.] - ETA: 0s - loss: 0.0525 - auc: 0.9980
## 155/157 [============================>.] - ETA: 0s - loss: 0.0525 - auc: 0.9980
## 156/157 [============================>.] - ETA: 0s - loss: 0.0524 - auc: 0.9980
## 157/157 [==============================] - 9s 57ms/step - loss: 0.0530 - auc: 0.9979 - val
_loss: 0.1232 - val_auc: 0.9994
## Epoch 9/10
##
##   1/157 [..............................] - ETA: 8s - loss: 0.1268 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.0883 - auc: 0.9960
##   3/157 [..............................] - ETA: 8s - loss: 0.1093 - auc: 0.9946
```

```
##   4/157 [..............................] - ETA: 8s - loss: 0.1061 - auc: 0.9921
##   5/157 [..............................] - ETA: 8s - loss: 0.1424 - auc: 0.9864
##   6/157 [>.............................] - ETA: 8s - loss: 0.1483 - auc: 0.9863
##   7/157 [>.............................] - ETA: 8s - loss: 0.1498 - auc: 0.9871
##   8/157 [>.............................] - ETA: 8s - loss: 0.1858 - auc: 0.9825
##   9/157 [>.............................] - ETA: 8s - loss: 0.1659 - auc: 0.9862
##  10/157 [>.............................] - ETA: 8s - loss: 0.2089 - auc: 0.9780
##  11/157 [=>............................] - ETA: 8s - loss: 0.1934 - auc: 0.9804
##  12/157 [=>............................] - ETA: 7s - loss: 0.1788 - auc: 0.9831
##  13/157 [=>............................] - ETA: 7s - loss: 0.1664 - auc: 0.9852
##  14/157 [=>............................] - ETA: 7s - loss: 0.1570 - auc: 0.9867
##  15/157 [=>............................] - ETA: 7s - loss: 0.1489 - auc: 0.9879
##  16/157 [==>...........................] - ETA: 7s - loss: 0.1510 - auc: 0.9876
##  17/157 [==>...........................] - ETA: 7s - loss: 0.1438 - auc: 0.9885
##  18/157 [==>...........................] - ETA: 7s - loss: 0.1419 - auc: 0.9886
##  19/157 [==>...........................] - ETA: 7s - loss: 0.1348 - auc: 0.9898
##  20/157 [==>...........................] - ETA: 7s - loss: 0.1302 - auc: 0.9903
##  21/157 [===>..........................] - ETA: 7s - loss: 0.1251 - auc: 0.9910
##  22/157 [===>..........................] - ETA: 7s - loss: 0.1330 - auc: 0.9901
##  23/157 [===>..........................] - ETA: 7s - loss: 0.1277 - auc: 0.9908
##  24/157 [===>..........................] - ETA: 7s - loss: 0.1286 - auc: 0.9909
##  25/157 [===>..........................] - ETA: 7s - loss: 0.1271 - auc: 0.9912
##  26/157 [===>..........................] - ETA: 7s - loss: 0.1255 - auc: 0.9915
##  27/157 [====>.........................] - ETA: 7s - loss: 0.1213 - auc: 0.9920
##  28/157 [====>.........................] - ETA: 7s - loss: 0.1178 - auc: 0.9924
##  29/157 [====>.........................] - ETA: 7s - loss: 0.1168 - auc: 0.9925
##  30/157 [====>.........................] - ETA: 6s - loss: 0.1182 - auc: 0.9921
##  31/157 [====>.........................] - ETA: 6s - loss: 0.1153 - auc: 0.9924
##  32/157 [=====>........................] - ETA: 6s - loss: 0.1121 - auc: 0.9929
##  33/157 [=====>........................] - ETA: 6s - loss: 0.1103 - auc: 0.9931
##  34/157 [=====>........................] - ETA: 6s - loss: 0.1073 - auc: 0.9935
##  35/157 [=====>........................] - ETA: 6s - loss: 0.1105 - auc: 0.9931
##  36/157 [=====>........................] - ETA: 6s - loss: 0.1084 - auc: 0.9933
##  37/157 [======>.......................] - ETA: 6s - loss: 0.1075 - auc: 0.9935
##  38/157 [======>.......................] - ETA: 6s - loss: 0.1050 - auc: 0.9938
##  39/157 [======>.......................] - ETA: 6s - loss: 0.1036 - auc: 0.9940
##  40/157 [======>.......................] - ETA: 6s - loss: 0.1018 - auc: 0.9942
##  41/157 [======>.......................] - ETA: 6s - loss: 0.1051 - auc: 0.9939
##  42/157 [=======>......................] - ETA: 6s - loss: 0.1030 - auc: 0.9941
##  43/157 [=======>......................] - ETA: 6s - loss: 0.1015 - auc: 0.9942
##  44/157 [=======>......................] - ETA: 6s - loss: 0.0999 - auc: 0.9944
##  45/157 [=======>......................] - ETA: 6s - loss: 0.0989 - auc: 0.9946
##  46/157 [=======>......................] - ETA: 6s - loss: 0.0971 - auc: 0.9948
##  47/157 [=======>......................] - ETA: 6s - loss: 0.0960 - auc: 0.9949
##  48/157 [========>.....................] - ETA: 5s - loss: 0.0944 - auc: 0.9950
##  49/157 [========>.....................] - ETA: 5s - loss: 0.0935 - auc: 0.9951
##  50/157 [========>.....................] - ETA: 5s - loss: 0.0918 - auc: 0.9952
##  51/157 [========>.....................] - ETA: 5s - loss: 0.0904 - auc: 0.9954
##  52/157 [========>.....................] - ETA: 5s - loss: 0.0889 - auc: 0.9955
##  53/157 [=========>....................] - ETA: 5s - loss: 0.0879 - auc: 0.9957
##  54/157 [=========>....................] - ETA: 5s - loss: 0.0866 - auc: 0.9958
##  55/157 [=========>....................] - ETA: 5s - loss: 0.0859 - auc: 0.9958
##  56/157 [=========>....................] - ETA: 5s - loss: 0.0847 - auc: 0.9959
##  57/157 [=========>....................] - ETA: 5s - loss: 0.0836 - auc: 0.9960
##  58/157 [==========>...................] - ETA: 5s - loss: 0.0822 - auc: 0.9962
##  59/157 [==========>...................] - ETA: 5s - loss: 0.0813 - auc: 0.9963
```

```
##  60/157 [==========>...................] - ETA: 5s - loss: 0.0802 - auc: 0.9964
##  61/157 [==========>...................] - ETA: 5s - loss: 0.0795 - auc: 0.9964
##  62/157 [==========>...................] - ETA: 5s - loss: 0.0784 - auc: 0.9965
##  63/157 [===========>..................] - ETA: 5s - loss: 0.0786 - auc: 0.9965
##  64/157 [===========>..................] - ETA: 5s - loss: 0.0778 - auc: 0.9966
##  65/157 [===========>..................] - ETA: 5s - loss: 0.0771 - auc: 0.9966
##  66/157 [===========>..................] - ETA: 4s - loss: 0.0762 - auc: 0.9967
##  67/157 [===========>..................] - ETA: 4s - loss: 0.0752 - auc: 0.9968
##  68/157 [===========>..................] - ETA: 4s - loss: 0.0741 - auc: 0.9969
##  69/157 [============>.................] - ETA: 4s - loss: 0.0751 - auc: 0.9968
##  70/157 [============>.................] - ETA: 4s - loss: 0.0745 - auc: 0.9968
##  71/157 [============>.................] - ETA: 4s - loss: 0.0735 - auc: 0.9969
##  72/157 [============>.................] - ETA: 4s - loss: 0.0726 - auc: 0.9970
##  73/157 [============>.................] - ETA: 4s - loss: 0.0728 - auc: 0.9970
##  74/157 [=============>................] - ETA: 4s - loss: 0.0721 - auc: 0.9971
##  75/157 [=============>................] - ETA: 4s - loss: 0.0713 - auc: 0.9971
##  76/157 [=============>................] - ETA: 4s - loss: 0.0708 - auc: 0.9972
##  77/157 [=============>................] - ETA: 4s - loss: 0.0699 - auc: 0.9972
##  78/157 [=============>................] - ETA: 4s - loss: 0.0691 - auc: 0.9973
##  79/157 [==============>...............] - ETA: 4s - loss: 0.0684 - auc: 0.9973
##  80/157 [==============>...............] - ETA: 4s - loss: 0.0684 - auc: 0.9973
##  81/157 [==============>...............] - ETA: 4s - loss: 0.0677 - auc: 0.9974
##  82/157 [==============>...............] - ETA: 4s - loss: 0.0671 - auc: 0.9974
##  83/157 [==============>...............] - ETA: 4s - loss: 0.0668 - auc: 0.9975
##  84/157 [===============>..............] - ETA: 4s - loss: 0.0660 - auc: 0.9975
##  85/157 [===============>..............] - ETA: 3s - loss: 0.0656 - auc: 0.9975
##  86/157 [===============>..............] - ETA: 3s - loss: 0.0653 - auc: 0.9976
##  87/157 [===============>..............] - ETA: 3s - loss: 0.0648 - auc: 0.9976
##  88/157 [===============>..............] - ETA: 3s - loss: 0.0657 - auc: 0.9976
##  89/157 [================>.............] - ETA: 3s - loss: 0.0656 - auc: 0.9976
##  90/157 [================>.............] - ETA: 3s - loss: 0.0676 - auc: 0.9974
##  91/157 [================>.............] - ETA: 3s - loss: 0.0672 - auc: 0.9975
##  92/157 [================>.............] - ETA: 3s - loss: 0.0667 - auc: 0.9975
##  93/157 [================>.............] - ETA: 3s - loss: 0.0661 - auc: 0.9976
##  94/157 [================>.............] - ETA: 3s - loss: 0.0660 - auc: 0.9976
##  95/157 [=================>............] - ETA: 3s - loss: 0.0656 - auc: 0.9976
##  96/157 [=================>............] - ETA: 3s - loss: 0.0650 - auc: 0.9976
##  97/157 [=================>............] - ETA: 3s - loss: 0.0644 - auc: 0.9977
##  98/157 [=================>............] - ETA: 3s - loss: 0.0644 - auc: 0.9977
##  99/157 [=================>............] - ETA: 3s - loss: 0.0638 - auc: 0.9977
## 100/157 [==================>...........] - ETA: 3s - loss: 0.0632 - auc: 0.9978
## 101/157 [==================>...........] - ETA: 3s - loss: 0.0626 - auc: 0.9978
## 102/157 [==================>...........] - ETA: 3s - loss: 0.0621 - auc: 0.9979
## 103/157 [==================>...........] - ETA: 2s - loss: 0.0616 - auc: 0.9979
## 104/157 [==================>...........] - ETA: 2s - loss: 0.0632 - auc: 0.9977
## 105/157 [===================>..........] - ETA: 2s - loss: 0.0627 - auc: 0.9978
## 106/157 [===================>..........] - ETA: 2s - loss: 0.0625 - auc: 0.9978
## 107/157 [===================>..........] - ETA: 2s - loss: 0.0625 - auc: 0.9978
## 108/157 [===================>..........] - ETA: 2s - loss: 0.0622 - auc: 0.9978
## 109/157 [===================>..........] - ETA: 2s - loss: 0.0623 - auc: 0.9978
## 110/157 [====================>.........] - ETA: 2s - loss: 0.0623 - auc: 0.9978
## 111/157 [====================>.........] - ETA: 2s - loss: 0.0622 - auc: 0.9978
## 112/157 [====================>.........] - ETA: 2s - loss: 0.0618 - auc: 0.9979
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0613 - auc: 0.9979
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0612 - auc: 0.9979
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0608 - auc: 0.9979
```

```
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0605 - auc: 0.9979
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0602 - auc: 0.9980
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0598 - auc: 0.9980
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0598 - auc: 0.9980
## 120/157 [=====================>........] - ETA: 2s - loss: 0.0595 - auc: 0.9980
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0592 - auc: 0.9980
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0591 - auc: 0.9980
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0596 - auc: 0.9980
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0595 - auc: 0.9980
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0591 - auc: 0.9980
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0591 - auc: 0.9980
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0586 - auc: 0.9981
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0583 - auc: 0.9981
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0581 - auc: 0.9981
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0587 - auc: 0.9980
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0584 - auc: 0.9981
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0581 - auc: 0.9981
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0579 - auc: 0.9981
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0577 - auc: 0.9981
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0579 - auc: 0.9981
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0578 - auc: 0.9981
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0575 - auc: 0.9981
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0571 - auc: 0.9982
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0568 - auc: 0.9982
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0567 - auc: 0.9982
## 141/157 [=========================>....] - ETA: 0s - loss: 0.0564 - auc: 0.9982
## 142/157 [==========================>...] - ETA: 0s - loss: 0.0561 - auc: 0.9982
## 143/157 [==========================>...] - ETA: 0s - loss: 0.0559 - auc: 0.9983
## 144/157 [==========================>...] - ETA: 0s - loss: 0.0556 - auc: 0.9983
## 145/157 [==========================>...] - ETA: 0s - loss: 0.0552 - auc: 0.9983
## 146/157 [==========================>...] - ETA: 0s - loss: 0.0552 - auc: 0.9983
## 147/157 [===========================>..] - ETA: 0s - loss: 0.0549 - auc: 0.9983
## 148/157 [===========================>..] - ETA: 0s - loss: 0.0546 - auc: 0.9983
## 149/157 [===========================>..] - ETA: 0s - loss: 0.0550 - auc: 0.9983
## 150/157 [===========================>..] - ETA: 0s - loss: 0.0549 - auc: 0.9983
## 151/157 [===========================>..] - ETA: 0s - loss: 0.0558 - auc: 0.9983
## 152/157 [============================>.] - ETA: 0s - loss: 0.0555 - auc: 0.9983
## 153/157 [============================>.] - ETA: 0s - loss: 0.0552 - auc: 0.9983
## 154/157 [============================>.] - ETA: 0s - loss: 0.0549 - auc: 0.9983
## 155/157 [============================>.] - ETA: 0s - loss: 0.0547 - auc: 0.9983
## 156/157 [============================>.] - ETA: 0s - loss: 0.0548 - auc: 0.9983
## 157/157 [==============================] - 9s 57ms/step - loss: 0.0569 - auc: 0.9981 - val
_loss: 0.5957 - val_auc: 0.9593
## Epoch 10/10
##
##   1/157 [..............................] - ETA: 8s - loss: 0.0858 - auc: 1.0000
##   2/157 [..............................] - ETA: 8s - loss: 0.1490 - auc: 0.9958
##   3/157 [..............................] - ETA: 8s - loss: 0.1276 - auc: 0.9894
##   4/157 [..............................] - ETA: 8s - loss: 0.1623 - auc: 0.9863
##   5/157 [..............................] - ETA: 8s - loss: 0.1593 - auc: 0.9868
##   6/157 [>.............................] - ETA: 8s - loss: 0.1778 - auc: 0.9836
##   7/157 [>.............................] - ETA: 8s - loss: 0.1923 - auc: 0.9805
##   8/157 [>.............................] - ETA: 8s - loss: 0.2145 - auc: 0.9776
##   9/157 [>.............................] - ETA: 8s - loss: 0.1962 - auc: 0.9808
##  10/157 [>.............................] - ETA: 8s - loss: 0.1855 - auc: 0.9819
##  11/157 [=>............................] - ETA: 8s - loss: 0.1767 - auc: 0.9831
```

```
## 12/157 [=>............................] - ETA: 8s - loss: 0.1700 - auc: 0.9839
## 13/157 [=>............................] - ETA: 7s - loss: 0.1687 - auc: 0.9845
## 14/157 [=>............................] - ETA: 7s - loss: 0.1683 - auc: 0.9846
## 15/157 [=>............................] - ETA: 7s - loss: 0.1603 - auc: 0.9855
## 16/157 [==>...........................] - ETA: 7s - loss: 0.1635 - auc: 0.9852
## 17/157 [==>...........................] - ETA: 7s - loss: 0.1571 - auc: 0.9862
## 18/157 [==>...........................] - ETA: 7s - loss: 0.1495 - auc: 0.9873
## 19/157 [==>...........................] - ETA: 7s - loss: 0.1471 - auc: 0.9878
## 20/157 [==>...........................] - ETA: 7s - loss: 0.1601 - auc: 0.9856
## 21/157 [===>..........................] - ETA: 7s - loss: 0.1565 - auc: 0.9860
## 22/157 [===>..........................] - ETA: 7s - loss: 0.1518 - auc: 0.9866
## 23/157 [===>..........................] - ETA: 7s - loss: 0.1605 - auc: 0.9848
## 24/157 [===>..........................] - ETA: 7s - loss: 0.1558 - auc: 0.9854
## 25/157 [===>..........................] - ETA: 7s - loss: 0.1505 - auc: 0.9863
## 26/157 [===>..........................] - ETA: 7s - loss: 0.1475 - auc: 0.9868
## 27/157 [====>.........................] - ETA: 7s - loss: 0.1426 - auc: 0.9875
## 28/157 [====>.........................] - ETA: 7s - loss: 0.1392 - auc: 0.9878
## 29/157 [====>.........................] - ETA: 7s - loss: 0.1387 - auc: 0.9880
## 30/157 [====>.........................] - ETA: 6s - loss: 0.1378 - auc: 0.9881
## 31/157 [====>.........................] - ETA: 6s - loss: 0.1397 - auc: 0.9878
## 32/157 [=====>........................] - ETA: 6s - loss: 0.1393 - auc: 0.9878
## 33/157 [=====>........................] - ETA: 6s - loss: 0.1400 - auc: 0.9878
## 34/157 [=====>........................] - ETA: 6s - loss: 0.1370 - auc: 0.9883
## 35/157 [=====>........................] - ETA: 6s - loss: 0.1337 - auc: 0.9887
## 36/157 [=====>........................] - ETA: 6s - loss: 0.1349 - auc: 0.9886
## 37/157 [======>.......................] - ETA: 6s - loss: 0.1328 - auc: 0.9889
## 38/157 [======>.......................] - ETA: 6s - loss: 0.1328 - auc: 0.9890
## 39/157 [======>.......................] - ETA: 6s - loss: 0.1339 - auc: 0.9888
## 40/157 [======>.......................] - ETA: 6s - loss: 0.1316 - auc: 0.9891
## 41/157 [======>.......................] - ETA: 6s - loss: 0.1297 - auc: 0.9894
## 42/157 [=======>......................] - ETA: 6s - loss: 0.1283 - auc: 0.9896
## 43/157 [=======>......................] - ETA: 6s - loss: 0.1266 - auc: 0.9898
## 44/157 [=======>......................] - ETA: 6s - loss: 0.1258 - auc: 0.9900
## 45/157 [=======>......................] - ETA: 6s - loss: 0.1328 - auc: 0.9890
## 46/157 [=======>......................] - ETA: 6s - loss: 0.1335 - auc: 0.9889
## 47/157 [=======>......................] - ETA: 6s - loss: 0.1342 - auc: 0.9889
## 48/157 [========>.....................] - ETA: 5s - loss: 0.1320 - auc: 0.9892
## 49/157 [========>.....................] - ETA: 5s - loss: 0.1297 - auc: 0.9895
## 50/157 [========>.....................] - ETA: 5s - loss: 0.1276 - auc: 0.9899
## 51/157 [========>.....................] - ETA: 5s - loss: 0.1272 - auc: 0.9899
## 52/157 [========>.....................] - ETA: 5s - loss: 0.1249 - auc: 0.9902
## 53/157 [=========>....................] - ETA: 5s - loss: 0.1241 - auc: 0.9903
## 54/157 [=========>....................] - ETA: 5s - loss: 0.1222 - auc: 0.9906
## 55/157 [=========>....................] - ETA: 5s - loss: 0.1201 - auc: 0.9909
## 56/157 [=========>....................] - ETA: 5s - loss: 0.1181 - auc: 0.9912
## 57/157 [=========>....................] - ETA: 5s - loss: 0.1188 - auc: 0.9912
## 58/157 [==========>...................] - ETA: 5s - loss: 0.1181 - auc: 0.9913
## 59/157 [==========>...................] - ETA: 5s - loss: 0.1204 - auc: 0.9909
## 60/157 [==========>...................] - ETA: 5s - loss: 0.1197 - auc: 0.9910
## 61/157 [==========>...................] - ETA: 5s - loss: 0.1199 - auc: 0.9910
## 62/157 [==========>...................] - ETA: 5s - loss: 0.1182 - auc: 0.9913
## 63/157 [===========>..................] - ETA: 5s - loss: 0.1192 - auc: 0.9912
## 64/157 [===========>..................] - ETA: 5s - loss: 0.1198 - auc: 0.9911
## 65/157 [===========>..................] - ETA: 5s - loss: 0.1183 - auc: 0.9913
## 66/157 [===========>..................] - ETA: 4s - loss: 0.1167 - auc: 0.9915
## 67/157 [===========>..................] - ETA: 4s - loss: 0.1153 - auc: 0.9917
```

```
##  68/157 [===========>..................] - ETA: 4s - loss: 0.1153 - auc: 0.9918
##  69/157 [============>.................] - ETA: 4s - loss: 0.1146 - auc: 0.9919
##  70/157 [============>.................] - ETA: 4s - loss: 0.1160 - auc: 0.9916
##  71/157 [============>.................] - ETA: 4s - loss: 0.1149 - auc: 0.9918
##  72/157 [============>.................] - ETA: 4s - loss: 0.1137 - auc: 0.9920
##  73/157 [============>.................] - ETA: 4s - loss: 0.1123 - auc: 0.9922
##  74/157 [=============>................] - ETA: 4s - loss: 0.1111 - auc: 0.9923
##  75/157 [=============>................] - ETA: 4s - loss: 0.1103 - auc: 0.9924
##  76/157 [=============>................] - ETA: 4s - loss: 0.1167 - auc: 0.9910
##  77/157 [=============>................] - ETA: 4s - loss: 0.1153 - auc: 0.9912
##  78/157 [=============>................] - ETA: 4s - loss: 0.1155 - auc: 0.9913
##  79/157 [==============>...............] - ETA: 4s - loss: 0.1144 - auc: 0.9914
##  80/157 [==============>...............] - ETA: 4s - loss: 0.1137 - auc: 0.9915
##  81/157 [==============>...............] - ETA: 4s - loss: 0.1125 - auc: 0.9917
##  82/157 [==============>...............] - ETA: 4s - loss: 0.1127 - auc: 0.9917
##  83/157 [==============>...............] - ETA: 4s - loss: 0.1117 - auc: 0.9918
##  84/157 [===============>..............] - ETA: 4s - loss: 0.1110 - auc: 0.9920
##  85/157 [===============>..............] - ETA: 3s - loss: 0.1133 - auc: 0.9915
##  86/157 [===============>..............] - ETA: 3s - loss: 0.1120 - auc: 0.9917
##  87/157 [===============>..............] - ETA: 3s - loss: 0.1110 - auc: 0.9918
##  88/157 [===============>..............] - ETA: 3s - loss: 0.1099 - auc: 0.9920
##  89/157 [================>.............] - ETA: 3s - loss: 0.1135 - auc: 0.9914
##  90/157 [================>.............] - ETA: 3s - loss: 0.1134 - auc: 0.9915
##  91/157 [================>.............] - ETA: 3s - loss: 0.1124 - auc: 0.9916
##  92/157 [================>.............] - ETA: 3s - loss: 0.1113 - auc: 0.9917
##  93/157 [================>.............] - ETA: 3s - loss: 0.1105 - auc: 0.9918
##  94/157 [================>.............] - ETA: 3s - loss: 0.1100 - auc: 0.9919
##  95/157 [=================>............] - ETA: 3s - loss: 0.1092 - auc: 0.9920
##  96/157 [=================>............] - ETA: 3s - loss: 0.1091 - auc: 0.9920
##  97/157 [=================>............] - ETA: 3s - loss: 0.1082 - auc: 0.9921
##  98/157 [=================>............] - ETA: 3s - loss: 0.1080 - auc: 0.9922
##  99/157 [=================>............] - ETA: 3s - loss: 0.1078 - auc: 0.9922
## 100/157 [==================>...........] - ETA: 3s - loss: 0.1069 - auc: 0.9923
## 101/157 [==================>...........] - ETA: 3s - loss: 0.1070 - auc: 0.9923
## 102/157 [==================>...........] - ETA: 3s - loss: 0.1066 - auc: 0.9924
## 103/157 [==================>...........] - ETA: 2s - loss: 0.1060 - auc: 0.9924
## 104/157 [==================>...........] - ETA: 2s - loss: 0.1052 - auc: 0.9925
## 105/157 [===================>..........] - ETA: 2s - loss: 0.1045 - auc: 0.9926
## 106/157 [===================>..........] - ETA: 2s - loss: 0.1047 - auc: 0.9926
## 107/157 [===================>..........] - ETA: 2s - loss: 0.1038 - auc: 0.9927
## 108/157 [===================>..........] - ETA: 2s - loss: 0.1030 - auc: 0.9928
## 109/157 [===================>..........] - ETA: 2s - loss: 0.1022 - auc: 0.9929
## 110/157 [====================>.........] - ETA: 2s - loss: 0.1013 - auc: 0.9930
## 111/157 [====================>.........] - ETA: 2s - loss: 0.1005 - auc: 0.9931
## 112/157 [====================>.........] - ETA: 2s - loss: 0.1005 - auc: 0.9931
## 113/157 [====================>.........] - ETA: 2s - loss: 0.0997 - auc: 0.9932
## 114/157 [====================>.........] - ETA: 2s - loss: 0.0992 - auc: 0.9933
## 115/157 [====================>.........] - ETA: 2s - loss: 0.0987 - auc: 0.9934
## 116/157 [=====================>........] - ETA: 2s - loss: 0.0981 - auc: 0.9934
## 117/157 [=====================>........] - ETA: 2s - loss: 0.0972 - auc: 0.9935
## 118/157 [=====================>........] - ETA: 2s - loss: 0.0977 - auc: 0.9935
## 119/157 [=====================>........] - ETA: 2s - loss: 0.0971 - auc: 0.9936
## 120/157 [=====================>........] - ETA: 2s - loss: 0.0967 - auc: 0.9936
## 121/157 [======================>.......] - ETA: 1s - loss: 0.0961 - auc: 0.9937
## 122/157 [======================>.......] - ETA: 1s - loss: 0.0955 - auc: 0.9938
## 123/157 [======================>.......] - ETA: 1s - loss: 0.0956 - auc: 0.9938
```

```
## 124/157 [======================>.......] - ETA: 1s - loss: 0.0953 - auc: 0.9938
## 125/157 [======================>.......] - ETA: 1s - loss: 0.0949 - auc: 0.9938
## 126/157 [=======================>......] - ETA: 1s - loss: 0.0951 - auc: 0.9938
## 127/157 [=======================>......] - ETA: 1s - loss: 0.0944 - auc: 0.9939
## 128/157 [=======================>......] - ETA: 1s - loss: 0.0953 - auc: 0.9938
## 129/157 [=======================>......] - ETA: 1s - loss: 0.0949 - auc: 0.9939
## 130/157 [=======================>......] - ETA: 1s - loss: 0.0943 - auc: 0.9940
## 131/157 [========================>.....] - ETA: 1s - loss: 0.0947 - auc: 0.9939
## 132/157 [========================>.....] - ETA: 1s - loss: 0.0940 - auc: 0.9940
## 133/157 [========================>.....] - ETA: 1s - loss: 0.0933 - auc: 0.9941
## 134/157 [========================>.....] - ETA: 1s - loss: 0.0928 - auc: 0.9942
## 135/157 [========================>.....] - ETA: 1s - loss: 0.0947 - auc: 0.9939
## 136/157 [========================>.....] - ETA: 1s - loss: 0.0944 - auc: 0.9939
## 137/157 [=========================>....] - ETA: 1s - loss: 0.0939 - auc: 0.9940
## 138/157 [=========================>....] - ETA: 1s - loss: 0.0943 - auc: 0.9940
## 139/157 [=========================>....] - ETA: 0s - loss: 0.0940 - auc: 0.9940
## 140/157 [=========================>....] - ETA: 0s - loss: 0.0935 - auc: 0.9941
## 141/157 [=========================>....] - ETA: 0s - loss: 0.0931 - auc: 0.9941
## 142/157 [==========================>...] - ETA: 0s - loss: 0.0927 - auc: 0.9942
## 143/157 [==========================>...] - ETA: 0s - loss: 0.0921 - auc: 0.9942
## 144/157 [==========================>...] - ETA: 0s - loss: 0.0915 - auc: 0.9943
## 145/157 [==========================>...] - ETA: 0s - loss: 0.0910 - auc: 0.9943
## 146/157 [==========================>...] - ETA: 0s - loss: 0.0906 - auc: 0.9944
## 147/157 [===========================>..] - ETA: 0s - loss: 0.0904 - auc: 0.9944
## 148/157 [===========================>..] - ETA: 0s - loss: 0.0899 - auc: 0.9945
## 149/157 [===========================>..] - ETA: 0s - loss: 0.0900 - auc: 0.9945
## 150/157 [===========================>..] - ETA: 0s - loss: 0.0897 - auc: 0.9945
## 151/157 [===========================>..] - ETA: 0s - loss: 0.0897 - auc: 0.9946
## 152/157 [============================>.] - ETA: 0s - loss: 0.0891 - auc: 0.9946
## 153/157 [============================>.] - ETA: 0s - loss: 0.0888 - auc: 0.9946
## 154/157 [============================>.] - ETA: 0s - loss: 0.0883 - auc: 0.9947
## 155/157 [============================>.] - ETA: 0s - loss: 0.0877 - auc: 0.9948
## 156/157 [============================>.] - ETA: 0s - loss: 0.0874 - auc: 0.9948
## 157/157 [==============================] - 9s 57ms/step - loss: 0.0873 - auc: 0.9948 - val
_loss: 0.0267 - val_auc: 0.9994
##
## Adjusted OvR model Gl vs (Mn, Pt):
## Epoch 1/20
##
##  1/75 [..............................] - ETA: 48s - loss: 0.6777 - auc: 0.4792
##  2/75 [..............................] - ETA: 4s - loss: 0.6680 - auc: 0.7013
##  3/75 [>.............................] - ETA: 3s - loss: 0.6288 - auc: 0.7359
##  4/75 [>.............................] - ETA: 3s - loss: 0.6021 - auc: 0.7720
##  5/75 [=>............................] - ETA: 3s - loss: 0.5367 - auc: 0.8266
##  6/75 [=>............................] - ETA: 3s - loss: 0.5049 - auc: 0.8432
##  7/75 [=>............................] - ETA: 3s - loss: 0.4896 - auc: 0.8576
##  8/75 [==>...........................] - ETA: 3s - loss: 0.4733 - auc: 0.8685
##  9/75 [==>...........................] - ETA: 3s - loss: 0.4867 - auc: 0.8604
## 10/75 [===>..........................] - ETA: 3s - loss: 0.4740 - auc: 0.8718
## 11/75 [===>..........................] - ETA: 3s - loss: 0.4694 - auc: 0.8766
## 12/75 [===>..........................] - ETA: 3s - loss: 0.4549 - auc: 0.8823
## 13/75 [====>.........................] - ETA: 3s - loss: 0.4422 - auc: 0.8872
## 14/75 [====>.........................] - ETA: 3s - loss: 0.4526 - auc: 0.8839
## 15/75 [=====>........................] - ETA: 3s - loss: 0.4621 - auc: 0.8809
## 16/75 [=====>........................] - ETA: 3s - loss: 0.4500 - auc: 0.8840
## 17/75 [=====>........................] - ETA: 3s - loss: 0.4491 - auc: 0.8828
```

```
## 18/75 [======>.......................] - ETA: 3s - loss: 0.4442 - auc: 0.8860
## 19/75 [======>.......................] - ETA: 3s - loss: 0.4332 - auc: 0.8936
## 20/75 [=======>......................] - ETA: 3s - loss: 0.4228 - auc: 0.8982
## 21/75 [=======>......................] - ETA: 2s - loss: 0.4225 - auc: 0.8972
## 22/75 [=======>......................] - ETA: 2s - loss: 0.4110 - auc: 0.9043
## 23/75 [========>.....................] - ETA: 2s - loss: 0.4009 - auc: 0.9098
## 24/75 [========>.....................] - ETA: 2s - loss: 0.3932 - auc: 0.9140
## 25/75 [=========>....................] - ETA: 2s - loss: 0.3923 - auc: 0.9159
## 26/75 [=========>....................] - ETA: 2s - loss: 0.3866 - auc: 0.9170
## 27/75 [=========>....................] - ETA: 2s - loss: 0.3807 - auc: 0.9193
## 28/75 [==========>...................] - ETA: 2s - loss: 0.3781 - auc: 0.9190
## 29/75 [==========>...................] - ETA: 2s - loss: 0.3711 - auc: 0.9231
## 30/75 [===========>..................] - ETA: 2s - loss: 0.3697 - auc: 0.9231
## 31/75 [===========>..................] - ETA: 2s - loss: 0.3756 - auc: 0.9175
## 32/75 [===========>..................] - ETA: 2s - loss: 0.3696 - auc: 0.9197
## 33/75 [============>.................] - ETA: 2s - loss: 0.3681 - auc: 0.9210
## 34/75 [============>.................] - ETA: 2s - loss: 0.3625 - auc: 0.9240
## 35/75 [=============>................] - ETA: 2s - loss: 0.3598 - auc: 0.9243
## 36/75 [=============>................] - ETA: 2s - loss: 0.3556 - auc: 0.9264
## 37/75 [=============>................] - ETA: 2s - loss: 0.3494 - auc: 0.9293
## 38/75 [==============>...............] - ETA: 2s - loss: 0.3472 - auc: 0.9306
## 39/75 [==============>...............] - ETA: 1s - loss: 0.3422 - auc: 0.9321
## 40/75 [===============>..............] - ETA: 1s - loss: 0.3396 - auc: 0.9327
## 41/75 [===============>..............] - ETA: 1s - loss: 0.3414 - auc: 0.9308
## 42/75 [===============>..............] - ETA: 1s - loss: 0.3399 - auc: 0.9317
## 43/75 [================>.............] - ETA: 1s - loss: 0.3396 - auc: 0.9314
## 44/75 [================>.............] - ETA: 1s - loss: 0.3356 - auc: 0.9334
## 45/75 [=================>............] - ETA: 1s - loss: 0.3308 - auc: 0.9355
## 46/75 [=================>............] - ETA: 1s - loss: 0.3287 - auc: 0.9361
## 47/75 [=================>............] - ETA: 1s - loss: 0.3296 - auc: 0.9361
## 48/75 [==================>...........] - ETA: 1s - loss: 0.3249 - auc: 0.9381
## 49/75 [==================>...........] - ETA: 1s - loss: 0.3236 - auc: 0.9379
## 50/75 [===================>..........] - ETA: 1s - loss: 0.3222 - auc: 0.9385
## 51/75 [===================>..........] - ETA: 1s - loss: 0.3184 - auc: 0.9397
## 52/75 [===================>..........] - ETA: 1s - loss: 0.3170 - auc: 0.9404
## 53/75 [====================>.........] - ETA: 1s - loss: 0.3152 - auc: 0.9413
## 54/75 [====================>.........] - ETA: 1s - loss: 0.3152 - auc: 0.9410
## 55/75 [=====================>........] - ETA: 1s - loss: 0.3154 - auc: 0.9410
## 56/75 [=====================>........] - ETA: 1s - loss: 0.3114 - auc: 0.9427
## 57/75 [=====================>........] - ETA: 0s - loss: 0.3100 - auc: 0.9434
## 58/75 [======================>.......] - ETA: 0s - loss: 0.3086 - auc: 0.9440
## 59/75 [======================>.......] - ETA: 0s - loss: 0.3089 - auc: 0.9438
## 60/75 [=======================>......] - ETA: 0s - loss: 0.3121 - auc: 0.9412
## 61/75 [=======================>......] - ETA: 0s - loss: 0.3087 - auc: 0.9427
## 62/75 [=======================>......] - ETA: 0s - loss: 0.3070 - auc: 0.9432
## 63/75 [========================>.....] - ETA: 0s - loss: 0.3055 - auc: 0.9440
## 64/75 [========================>.....] - ETA: 0s - loss: 0.3038 - auc: 0.9444
## 65/75 [=========================>....] - ETA: 0s - loss: 0.3036 - auc: 0.9446
## 66/75 [=========================>....] - ETA: 0s - loss: 0.3034 - auc: 0.9448
## 67/75 [=========================>....] - ETA: 0s - loss: 0.3035 - auc: 0.9446
## 68/75 [==========================>...] - ETA: 0s - loss: 0.3013 - auc: 0.9454
## 69/75 [==========================>...] - ETA: 0s - loss: 0.2995 - auc: 0.9462
## 70/75 [===========================>..] - ETA: 0s - loss: 0.2987 - auc: 0.9467
## 71/75 [===========================>..] - ETA: 0s - loss: 0.2969 - auc: 0.9471
## 72/75 [===========================>..] - ETA: 0s - loss: 0.2951 - auc: 0.9479
## 73/75 [============================>.] - ETA: 0s - loss: 0.2945 - auc: 0.9482
```

```
## 74/75 [============================>.] - ETA: 0s - loss: 0.2924 - auc: 0.9491
## 75/75 [==============================] - ETA: 0s - loss: 0.2906 - auc: 0.9499
## 75/75 [==============================] - 6s 69ms/step - loss: 0.2906 - auc: 0.9499 - val_l
oss: 1.6503 - val_auc: 0.8727
## Epoch 2/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.1434 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.2528 - auc: 0.9531
##  3/75 [>.............................] - ETA: 3s - loss: 0.2439 - auc: 0.9653
##  4/75 [>.............................] - ETA: 3s - loss: 0.2260 - auc: 0.9734
##  5/75 [=>............................] - ETA: 3s - loss: 0.2200 - auc: 0.9758
##  6/75 [=>............................] - ETA: 3s - loss: 0.1957 - auc: 0.9826
##  7/75 [=>............................] - ETA: 3s - loss: 0.1969 - auc: 0.9803
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1998 - auc: 0.9802
##  9/75 [==>...........................] - ETA: 3s - loss: 0.2067 - auc: 0.9793
## 10/75 [===>..........................] - ETA: 3s - loss: 0.2018 - auc: 0.9800
## 11/75 [===>..........................] - ETA: 3s - loss: 0.2020 - auc: 0.9797
## 12/75 [===>..........................] - ETA: 3s - loss: 0.1987 - auc: 0.9810
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1981 - auc: 0.9812
## 14/75 [====>.........................] - ETA: 3s - loss: 0.1977 - auc: 0.9816
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1893 - auc: 0.9838
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1972 - auc: 0.9808
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1962 - auc: 0.9809
## 18/75 [======>.......................] - ETA: 3s - loss: 0.2006 - auc: 0.9786
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1998 - auc: 0.9789
## 20/75 [=======>......................] - ETA: 3s - loss: 0.2017 - auc: 0.9775
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1976 - auc: 0.9791
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1965 - auc: 0.9797
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1923 - auc: 0.9806
## 24/75 [========>.....................] - ETA: 2s - loss: 0.2096 - auc: 0.9747
## 25/75 [=========>....................] - ETA: 2s - loss: 0.2076 - auc: 0.9754
## 26/75 [=========>....................] - ETA: 2s - loss: 0.2136 - auc: 0.9729
## 27/75 [=========>....................] - ETA: 2s - loss: 0.2078 - auc: 0.9746
## 28/75 [==========>...................] - ETA: 2s - loss: 0.2053 - auc: 0.9755
## 29/75 [==========>...................] - ETA: 2s - loss: 0.2004 - auc: 0.9769
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1957 - auc: 0.9781
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1963 - auc: 0.9780
## 32/75 [===========>..................] - ETA: 2s - loss: 0.1925 - auc: 0.9789
## 33/75 [============>.................] - ETA: 2s - loss: 0.1955 - auc: 0.9785
## 34/75 [============>.................] - ETA: 2s - loss: 0.1955 - auc: 0.9783
## 35/75 [=============>................] - ETA: 2s - loss: 0.1950 - auc: 0.9786
## 36/75 [=============>................] - ETA: 2s - loss: 0.1966 - auc: 0.9779
## 37/75 [=============>................] - ETA: 2s - loss: 0.1985 - auc: 0.9771
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1953 - auc: 0.9778
## 39/75 [==============>...............] - ETA: 1s - loss: 0.1943 - auc: 0.9780
## 40/75 [===============>..............] - ETA: 1s - loss: 0.1913 - auc: 0.9789
## 41/75 [===============>..............] - ETA: 1s - loss: 0.1911 - auc: 0.9788
## 42/75 [===============>..............] - ETA: 1s - loss: 0.1912 - auc: 0.9787
## 43/75 [================>.............] - ETA: 1s - loss: 0.1971 - auc: 0.9768
## 44/75 [================>.............] - ETA: 1s - loss: 0.1952 - auc: 0.9773
## 45/75 [=================>............] - ETA: 1s - loss: 0.1949 - auc: 0.9773
## 46/75 [=================>............] - ETA: 1s - loss: 0.1929 - auc: 0.9779
## 47/75 [=================>............] - ETA: 1s - loss: 0.1906 - auc: 0.9785
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1892 - auc: 0.9790
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1883 - auc: 0.9791
## 50/75 [===================>..........] - ETA: 1s - loss: 0.1877 - auc: 0.9792
```

```
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1873 - auc: 0.9794
## 52/75 [===================>..........] - ETA: 1s - loss: 0.1863 - auc: 0.9797
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1882 - auc: 0.9792
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1944 - auc: 0.9775
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1926 - auc: 0.9781
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1915 - auc: 0.9784
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1918 - auc: 0.9784
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1907 - auc: 0.9787
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1910 - auc: 0.9787
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1903 - auc: 0.9788
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1928 - auc: 0.9780
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1916 - auc: 0.9783
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1908 - auc: 0.9786
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1896 - auc: 0.9789
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1910 - auc: 0.9784
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1903 - auc: 0.9786
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1884 - auc: 0.9790
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1895 - auc: 0.9787
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1899 - auc: 0.9786
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1941 - auc: 0.9769
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1937 - auc: 0.9771
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1920 - auc: 0.9776
## 73/75 [============================>.] - ETA: 0s - loss: 0.1918 - auc: 0.9778
## 74/75 [============================>.] - ETA: 0s - loss: 0.1905 - auc: 0.9782
## 75/75 [==============================] - 4s 57ms/step - loss: 0.1918 - auc: 0.9777 - val_l
oss: 2.2372 - val_auc: 0.9006
## Epoch 3/20
## 
##  1/75 [..............................] - ETA: 3s - loss: 0.1065 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.1492 - auc: 0.9917
##  3/75 [>.............................] - ETA: 3s - loss: 0.1307 - auc: 0.9941
##  4/75 [>.............................] - ETA: 3s - loss: 0.1943 - auc: 0.9727
##  5/75 [=>............................] - ETA: 3s - loss: 0.1767 - auc: 0.9772
##  6/75 [=>............................] - ETA: 3s - loss: 0.1591 - auc: 0.9821
##  7/75 [=>............................] - ETA: 3s - loss: 0.1694 - auc: 0.9824
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1852 - auc: 0.9801
##  9/75 [==>...........................] - ETA: 3s - loss: 0.2034 - auc: 0.9718
## 10/75 [===>..........................] - ETA: 3s - loss: 0.2590 - auc: 0.9561
## 11/75 [===>..........................] - ETA: 3s - loss: 0.2398 - auc: 0.9618
## 12/75 [===>..........................] - ETA: 3s - loss: 0.2325 - auc: 0.9620
## 13/75 [====>.........................] - ETA: 3s - loss: 0.2209 - auc: 0.9648
## 14/75 [====>.........................] - ETA: 3s - loss: 0.2131 - auc: 0.9679
## 15/75 [=====>........................] - ETA: 3s - loss: 0.2240 - auc: 0.9667
## 16/75 [=====>........................] - ETA: 3s - loss: 0.2173 - auc: 0.9670
## 17/75 [=====>........................] - ETA: 3s - loss: 0.2157 - auc: 0.9682
## 18/75 [======>.......................] - ETA: 3s - loss: 0.2247 - auc: 0.9668
## 19/75 [======>.......................] - ETA: 3s - loss: 0.2354 - auc: 0.9624
## 20/75 [=======>......................] - ETA: 3s - loss: 0.2281 - auc: 0.9643
## 21/75 [=======>......................] - ETA: 2s - loss: 0.2672 - auc: 0.9544
## 22/75 [=======>......................] - ETA: 2s - loss: 0.2612 - auc: 0.9562
## 23/75 [========>.....................] - ETA: 2s - loss: 0.2600 - auc: 0.9550
## 24/75 [========>.....................] - ETA: 2s - loss: 0.2549 - auc: 0.9566
## 25/75 [=========>....................] - ETA: 2s - loss: 0.2483 - auc: 0.9581
## 26/75 [=========>....................] - ETA: 2s - loss: 0.2457 - auc: 0.9600
## 27/75 [=========>....................] - ETA: 2s - loss: 0.2397 - auc: 0.9623
## 28/75 [==========>...................] - ETA: 2s - loss: 0.2384 - auc: 0.9627
```

```
## 29/75 [==========>...................] - ETA: 2s - loss: 0.2446 - auc: 0.9608
## 30/75 [===========>..................] - ETA: 2s - loss: 0.2454 - auc: 0.9599
## 31/75 [===========>..................] - ETA: 2s - loss: 0.2400 - auc: 0.9618
## 32/75 [===========>..................] - ETA: 2s - loss: 0.2388 - auc: 0.9624
## 33/75 [============>.................] - ETA: 2s - loss: 0.2362 - auc: 0.9632
## 34/75 [============>.................] - ETA: 2s - loss: 0.2322 - auc: 0.9648
## 35/75 [=============>................] - ETA: 2s - loss: 0.2310 - auc: 0.9656
## 36/75 [=============>................] - ETA: 2s - loss: 0.2274 - auc: 0.9666
## 37/75 [=============>................] - ETA: 2s - loss: 0.2272 - auc: 0.9667
## 38/75 [==============>...............] - ETA: 2s - loss: 0.2242 - auc: 0.9675
## 39/75 [==============>...............] - ETA: 1s - loss: 0.2267 - auc: 0.9665
## 40/75 [===============>..............] - ETA: 1s - loss: 0.2231 - auc: 0.9676
## 41/75 [===============>..............] - ETA: 1s - loss: 0.2197 - auc: 0.9683
## 42/75 [===============>..............] - ETA: 1s - loss: 0.2180 - auc: 0.9687
## 43/75 [================>.............] - ETA: 1s - loss: 0.2206 - auc: 0.9677
## 44/75 [================>.............] - ETA: 1s - loss: 0.2213 - auc: 0.9680
## 45/75 [=================>............] - ETA: 1s - loss: 0.2196 - auc: 0.9685
## 46/75 [=================>............] - ETA: 1s - loss: 0.2191 - auc: 0.9686
## 47/75 [=================>............] - ETA: 1s - loss: 0.2227 - auc: 0.9672
## 48/75 [==================>...........] - ETA: 1s - loss: 0.2211 - auc: 0.9675
## 49/75 [==================>...........] - ETA: 1s - loss: 0.2186 - auc: 0.9682
## 50/75 [===================>..........] - ETA: 1s - loss: 0.2161 - auc: 0.9688
## 51/75 [===================>..........] - ETA: 1s - loss: 0.2204 - auc: 0.9676
## 52/75 [===================>..........] - ETA: 1s - loss: 0.2177 - auc: 0.9683
## 53/75 [====================>.........] - ETA: 1s - loss: 0.2171 - auc: 0.9683
## 54/75 [====================>.........] - ETA: 1s - loss: 0.2145 - auc: 0.9692
## 55/75 [=====================>........] - ETA: 1s - loss: 0.2168 - auc: 0.9687
## 56/75 [=====================>........] - ETA: 1s - loss: 0.2163 - auc: 0.9689
## 57/75 [=====================>........] - ETA: 0s - loss: 0.2171 - auc: 0.9685
## 58/75 [======================>.......] - ETA: 0s - loss: 0.2218 - auc: 0.9671
## 59/75 [======================>.......] - ETA: 0s - loss: 0.2199 - auc: 0.9675
## 60/75 [=======================>......] - ETA: 0s - loss: 0.2193 - auc: 0.9678
## 61/75 [=======================>......] - ETA: 0s - loss: 0.2208 - auc: 0.9672
## 62/75 [=======================>......] - ETA: 0s - loss: 0.2205 - auc: 0.9672
## 63/75 [========================>.....] - ETA: 0s - loss: 0.2282 - auc: 0.9653
## 64/75 [========================>.....] - ETA: 0s - loss: 0.2256 - auc: 0.9662
## 65/75 [=========================>....] - ETA: 0s - loss: 0.2228 - auc: 0.9670
## 66/75 [=========================>....] - ETA: 0s - loss: 0.2220 - auc: 0.9671
## 67/75 [=========================>....] - ETA: 0s - loss: 0.2249 - auc: 0.9662
## 68/75 [==========================>...] - ETA: 0s - loss: 0.2235 - auc: 0.9665
## 69/75 [==========================>...] - ETA: 0s - loss: 0.2235 - auc: 0.9662
## 70/75 [===========================>..] - ETA: 0s - loss: 0.2215 - auc: 0.9667
## 71/75 [===========================>..] - ETA: 0s - loss: 0.2211 - auc: 0.9672
## 72/75 [===========================>..] - ETA: 0s - loss: 0.2209 - auc: 0.9671
## 73/75 [============================>.] - ETA: 0s - loss: 0.2216 - auc: 0.9670
## 74/75 [============================>.] - ETA: 0s - loss: 0.2200 - auc: 0.9676
## 75/75 [==============================] - 4s 57ms/step - loss: 0.2192 - auc: 0.9678 - val_l
oss: 2.3331 - val_auc: 0.9044
## Epoch 4/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.1979 - auc: 0.9833
##  2/75 [..............................] - ETA: 4s - loss: 0.1201 - auc: 0.9958
##  3/75 [>.............................] - ETA: 3s - loss: 0.1147 - auc: 0.9980
##  4/75 [>.............................] - ETA: 3s - loss: 0.1131 - auc: 0.9978
##  5/75 [=>............................] - ETA: 3s - loss: 0.1107 - auc: 0.9986
##  6/75 [=>............................] - ETA: 3s - loss: 0.1088 - auc: 0.9986
```

```
##  7/75 [=>............................] - ETA: 3s - loss: 0.1179 - auc: 0.9965
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1130 - auc: 0.9970
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1097 - auc: 0.9974
## 10/75 [===>..........................] - ETA: 3s - loss: 0.1088 - auc: 0.9974
## 11/75 [===>..........................] - ETA: 3s - loss: 0.1135 - auc: 0.9972
## 12/75 [===>..........................] - ETA: 3s - loss: 0.1149 - auc: 0.9971
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1265 - auc: 0.9942
## 14/75 [====>.........................] - ETA: 3s - loss: 0.1259 - auc: 0.9943
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1192 - auc: 0.9951
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1231 - auc: 0.9944
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1222 - auc: 0.9946
## 18/75 [======>.......................] - ETA: 3s - loss: 0.1344 - auc: 0.9919
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1339 - auc: 0.9919
## 20/75 [=======>......................] - ETA: 3s - loss: 0.1326 - auc: 0.9924
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1366 - auc: 0.9908
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1424 - auc: 0.9889
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1400 - auc: 0.9892
## 24/75 [========>.....................] - ETA: 2s - loss: 0.1423 - auc: 0.9886
## 25/75 [=========>....................] - ETA: 2s - loss: 0.1428 - auc: 0.9887
## 26/75 [=========>....................] - ETA: 2s - loss: 0.1422 - auc: 0.9890
## 27/75 [=========>....................] - ETA: 2s - loss: 0.1408 - auc: 0.9892
## 28/75 [==========>...................] - ETA: 2s - loss: 0.1406 - auc: 0.9891
## 29/75 [==========>...................] - ETA: 2s - loss: 0.1426 - auc: 0.9889
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1424 - auc: 0.9891
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1413 - auc: 0.9894
## 32/75 [===========>..................] - ETA: 2s - loss: 0.1397 - auc: 0.9898
## 33/75 [============>.................] - ETA: 2s - loss: 0.1380 - auc: 0.9902
## 34/75 [============>.................] - ETA: 2s - loss: 0.1364 - auc: 0.9906
## 35/75 [=============>................] - ETA: 2s - loss: 0.1367 - auc: 0.9907
## 36/75 [=============>................] - ETA: 2s - loss: 0.1361 - auc: 0.9909
## 37/75 [=============>................] - ETA: 2s - loss: 0.1340 - auc: 0.9913
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1458 - auc: 0.9870
## 39/75 [==============>...............] - ETA: 1s - loss: 0.1438 - auc: 0.9875
## 40/75 [===============>..............] - ETA: 1s - loss: 0.1441 - auc: 0.9877
## 41/75 [===============>..............] - ETA: 1s - loss: 0.1439 - auc: 0.9880
## 42/75 [===============>..............] - ETA: 1s - loss: 0.1429 - auc: 0.9883
## 43/75 [================>.............] - ETA: 1s - loss: 0.1416 - auc: 0.9886
## 44/75 [================>.............] - ETA: 1s - loss: 0.1407 - auc: 0.9889
## 45/75 [=================>............] - ETA: 1s - loss: 0.1401 - auc: 0.9889
## 46/75 [=================>............] - ETA: 1s - loss: 0.1405 - auc: 0.9888
## 47/75 [=================>............] - ETA: 1s - loss: 0.1387 - auc: 0.9892
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1407 - auc: 0.9884
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1441 - auc: 0.9878
## 50/75 [===================>..........] - ETA: 1s - loss: 0.1461 - auc: 0.9874
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1530 - auc: 0.9860
## 52/75 [===================>..........] - ETA: 1s - loss: 0.1537 - auc: 0.9857
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1518 - auc: 0.9860
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1566 - auc: 0.9846
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1561 - auc: 0.9848
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1556 - auc: 0.9848
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1548 - auc: 0.9850
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1545 - auc: 0.9852
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1553 - auc: 0.9850
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1555 - auc: 0.9849
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1549 - auc: 0.9850
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1533 - auc: 0.9854
```

```
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1520 - auc: 0.9857
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1524 - auc: 0.9857
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1547 - auc: 0.9847
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1577 - auc: 0.9839
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1562 - auc: 0.9843
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1547 - auc: 0.9847
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1543 - auc: 0.9848
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1553 - auc: 0.9847
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1548 - auc: 0.9849
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1535 - auc: 0.9852
## 73/75 [============================>.] - ETA: 0s - loss: 0.1524 - auc: 0.9854
## 74/75 [============================>.] - ETA: 0s - loss: 0.1511 - auc: 0.9857
## 75/75 [==============================] - 4s 57ms/step - loss: 0.1507 - auc: 0.9859 - val_l
oss: 2.0941 - val_auc: 0.9261
## Epoch 5/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.3092 - auc: 0.9273
##  2/75 [..............................] - ETA: 4s - loss: 0.1918 - auc: 0.9750
##  3/75 [>.............................] - ETA: 3s - loss: 0.1350 - auc: 0.9863
##  4/75 [>.............................] - ETA: 3s - loss: 0.1316 - auc: 0.9871
##  5/75 [=>............................] - ETA: 3s - loss: 0.1216 - auc: 0.9900
##  6/75 [=>............................] - ETA: 3s - loss: 0.1103 - auc: 0.9927
##  7/75 [=>............................] - ETA: 3s - loss: 0.1029 - auc: 0.9940
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1159 - auc: 0.9922
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1281 - auc: 0.9903
## 10/75 [===>..........................] - ETA: 3s - loss: 0.1357 - auc: 0.9886
## 11/75 [===>..........................] - ETA: 3s - loss: 0.1266 - auc: 0.9902
## 12/75 [===>..........................] - ETA: 3s - loss: 0.1213 - auc: 0.9906
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1174 - auc: 0.9910
## 14/75 [====>.........................] - ETA: 3s - loss: 0.1182 - auc: 0.9911
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1165 - auc: 0.9916
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1136 - auc: 0.9921
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1146 - auc: 0.9920
## 18/75 [======>.......................] - ETA: 3s - loss: 0.1232 - auc: 0.9907
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1290 - auc: 0.9896
## 20/75 [=======>......................] - ETA: 3s - loss: 0.1349 - auc: 0.9881
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1356 - auc: 0.9881
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1324 - auc: 0.9889
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1297 - auc: 0.9894
## 24/75 [========>.....................] - ETA: 2s - loss: 0.1279 - auc: 0.9898
## 25/75 [=========>....................] - ETA: 2s - loss: 0.1293 - auc: 0.9893
## 26/75 [=========>....................] - ETA: 2s - loss: 0.1296 - auc: 0.9892
## 27/75 [=========>....................] - ETA: 2s - loss: 0.1308 - auc: 0.9893
## 28/75 [==========>...................] - ETA: 2s - loss: 0.1295 - auc: 0.9895
## 29/75 [==========>...................] - ETA: 2s - loss: 0.1281 - auc: 0.9897
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1275 - auc: 0.9897
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1276 - auc: 0.9900
## 32/75 [===========>..................] - ETA: 2s - loss: 0.1327 - auc: 0.9891
## 33/75 [============>.................] - ETA: 2s - loss: 0.1393 - auc: 0.9876
## 34/75 [============>.................] - ETA: 2s - loss: 0.1388 - auc: 0.9879
## 35/75 [=============>................] - ETA: 2s - loss: 0.1374 - auc: 0.9882
## 36/75 [=============>................] - ETA: 2s - loss: 0.1398 - auc: 0.9878
## 37/75 [=============>................] - ETA: 2s - loss: 0.1390 - auc: 0.9881
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1379 - auc: 0.9884
## 39/75 [==============>...............] - ETA: 1s - loss: 0.1358 - auc: 0.9888
## 40/75 [===============>..............] - ETA: 1s - loss: 0.1341 - auc: 0.9892
```

```
## 41/75 [===============>..............] - ETA: 1s - loss: 0.1332 - auc: 0.9895
## 42/75 [===============>..............] - ETA: 1s - loss: 0.1324 - auc: 0.9895
## 43/75 [================>.............] - ETA: 1s - loss: 0.1306 - auc: 0.9898
## 44/75 [================>.............] - ETA: 1s - loss: 0.1281 - auc: 0.9902
## 45/75 [=================>............] - ETA: 1s - loss: 0.1270 - auc: 0.9905
## 46/75 [=================>............] - ETA: 1s - loss: 0.1262 - auc: 0.9907
## 47/75 [=================>............] - ETA: 1s - loss: 0.1309 - auc: 0.9896
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1335 - auc: 0.9892
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1326 - auc: 0.9893
## 50/75 [===================>..........] - ETA: 1s - loss: 0.1324 - auc: 0.9894
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1306 - auc: 0.9898
## 52/75 [===================>..........] - ETA: 1s - loss: 0.1299 - auc: 0.9899
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1292 - auc: 0.9901
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1273 - auc: 0.9904
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1265 - auc: 0.9906
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1257 - auc: 0.9908
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1324 - auc: 0.9881
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1312 - auc: 0.9883
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1306 - auc: 0.9884
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1295 - auc: 0.9887
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1281 - auc: 0.9889
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1303 - auc: 0.9884
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1308 - auc: 0.9884
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1295 - auc: 0.9886
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1280 - auc: 0.9889
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1284 - auc: 0.9890
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1314 - auc: 0.9884
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1337 - auc: 0.9881
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1351 - auc: 0.9879
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1362 - auc: 0.9878
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1355 - auc: 0.9880
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1364 - auc: 0.9879
## 73/75 [============================>.] - ETA: 0s - loss: 0.1375 - auc: 0.9876
## 74/75 [============================>.] - ETA: 0s - loss: 0.1381 - auc: 0.9875
## 75/75 [==============================] - 4s 57ms/step - loss: 0.1370 - auc: 0.9877 - val_l
oss: 1.7069 - val_auc: 0.9602
## Epoch 6/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.1421 - auc: 0.9841
##  2/75 [..............................] - ETA: 3s - loss: 0.1105 - auc: 0.9909
##  3/75 [>.............................] - ETA: 4s - loss: 0.0912 - auc: 0.9937
##  4/75 [>.............................] - ETA: 3s - loss: 0.1003 - auc: 0.9955
##  5/75 [=>............................] - ETA: 3s - loss: 0.1009 - auc: 0.9959
##  6/75 [=>............................] - ETA: 3s - loss: 0.1042 - auc: 0.9955
##  7/75 [=>............................] - ETA: 3s - loss: 0.0987 - auc: 0.9964
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1041 - auc: 0.9966
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1021 - auc: 0.9970
## 10/75 [===>..........................] - ETA: 3s - loss: 0.1041 - auc: 0.9970
## 11/75 [===>..........................] - ETA: 3s - loss: 0.1022 - auc: 0.9973
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0973 - auc: 0.9978
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1001 - auc: 0.9972
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0955 - auc: 0.9975
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1001 - auc: 0.9973
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1092 - auc: 0.9951
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1181 - auc: 0.9932
## 18/75 [======>.......................] - ETA: 3s - loss: 0.1141 - auc: 0.9938
```

```
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1113 - auc: 0.9942
## 20/75 [=======>......................] - ETA: 3s - loss: 0.1221 - auc: 0.9930
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1216 - auc: 0.9932
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1194 - auc: 0.9936
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1163 - auc: 0.9940
## 24/75 [========>.....................] - ETA: 2s - loss: 0.1128 - auc: 0.9945
## 25/75 [=========>....................] - ETA: 2s - loss: 0.1167 - auc: 0.9923
## 26/75 [=========>....................] - ETA: 2s - loss: 0.1142 - auc: 0.9927
## 27/75 [=========>....................] - ETA: 2s - loss: 0.1159 - auc: 0.9926
## 28/75 [==========>...................] - ETA: 2s - loss: 0.1303 - auc: 0.9894
## 29/75 [==========>...................] - ETA: 2s - loss: 0.1269 - auc: 0.9901
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1250 - auc: 0.9906
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1268 - auc: 0.9902
## 32/75 [===========>..................] - ETA: 2s - loss: 0.1300 - auc: 0.9898
## 33/75 [============>.................] - ETA: 2s - loss: 0.1287 - auc: 0.9901
## 34/75 [============>.................] - ETA: 2s - loss: 0.1269 - auc: 0.9905
## 35/75 [=============>................] - ETA: 2s - loss: 0.1256 - auc: 0.9907
## 36/75 [=============>................] - ETA: 2s - loss: 0.1240 - auc: 0.9909
## 37/75 [=============>................] - ETA: 2s - loss: 0.1242 - auc: 0.9909
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1238 - auc: 0.9910
## 39/75 [==============>...............] - ETA: 1s - loss: 0.1232 - auc: 0.9911
## 40/75 [===============>..............] - ETA: 1s - loss: 0.1215 - auc: 0.9913
## 41/75 [===============>..............] - ETA: 1s - loss: 0.1209 - auc: 0.9914
## 42/75 [===============>..............] - ETA: 1s - loss: 0.1190 - auc: 0.9917
## 43/75 [================>.............] - ETA: 1s - loss: 0.1188 - auc: 0.9917
## 44/75 [================>.............] - ETA: 1s - loss: 0.1167 - auc: 0.9921
## 45/75 [=================>............] - ETA: 1s - loss: 0.1198 - auc: 0.9911
## 46/75 [=================>............] - ETA: 1s - loss: 0.1188 - auc: 0.9913
## 47/75 [=================>............] - ETA: 1s - loss: 0.1215 - auc: 0.9909
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1252 - auc: 0.9897
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1238 - auc: 0.9900
## 50/75 [===================>..........] - ETA: 1s - loss: 0.1221 - auc: 0.9904
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1217 - auc: 0.9905
## 52/75 [===================>..........] - ETA: 1s - loss: 0.1212 - auc: 0.9906
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1193 - auc: 0.9909
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1193 - auc: 0.9909
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1188 - auc: 0.9911
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1189 - auc: 0.9912
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1180 - auc: 0.9914
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1206 - auc: 0.9905
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1202 - auc: 0.9906
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1204 - auc: 0.9907
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1198 - auc: 0.9907
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1196 - auc: 0.9908
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1197 - auc: 0.9910
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1205 - auc: 0.9908
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1191 - auc: 0.9911
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1182 - auc: 0.9912
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1183 - auc: 0.9913
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1170 - auc: 0.9915
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1160 - auc: 0.9917
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1158 - auc: 0.9919
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1147 - auc: 0.9921
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1150 - auc: 0.9921
## 73/75 [============================>.] - ETA: 0s - loss: 0.1146 - auc: 0.9922
## 74/75 [============================>.] - ETA: 0s - loss: 0.1141 - auc: 0.9923
```

```
## 75/75 [==============================] - 4s 58ms/step - loss: 0.1138 - auc: 0.9923 - val_l
oss: 1.0336 - val_auc: 0.9865
## Epoch 7/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0573 - auc: 1.0000
##  2/75 [..............................] - ETA: 4s - loss: 0.0461 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.1020 - auc: 0.9951
##  4/75 [>.............................] - ETA: 3s - loss: 0.1136 - auc: 0.9940
##  5/75 [=>............................] - ETA: 3s - loss: 0.1516 - auc: 0.9931
##  6/75 [=>............................] - ETA: 3s - loss: 0.1357 - auc: 0.9909
##  7/75 [=>............................] - ETA: 3s - loss: 0.1578 - auc: 0.9908
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1482 - auc: 0.9926
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1360 - auc: 0.9925
## 10/75 [===>..........................] - ETA: 3s - loss: 0.1254 - auc: 0.9932
## 11/75 [===>..........................] - ETA: 3s - loss: 0.1209 - auc: 0.9931
## 12/75 [===>..........................] - ETA: 3s - loss: 0.1181 - auc: 0.9939
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1115 - auc: 0.9948
## 14/75 [====>.........................] - ETA: 3s - loss: 0.1202 - auc: 0.9912
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1153 - auc: 0.9921
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1111 - auc: 0.9928
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1127 - auc: 0.9920
## 18/75 [======>.......................] - ETA: 3s - loss: 0.1073 - auc: 0.9927
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1054 - auc: 0.9929
## 20/75 [=======>......................] - ETA: 3s - loss: 0.1063 - auc: 0.9927
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1078 - auc: 0.9925
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1079 - auc: 0.9925
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1052 - auc: 0.9929
## 24/75 [========>.....................] - ETA: 2s - loss: 0.1039 - auc: 0.9932
## 25/75 [=========>....................] - ETA: 2s - loss: 0.1016 - auc: 0.9934
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0995 - auc: 0.9938
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0977 - auc: 0.9940
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0965 - auc: 0.9942
## 29/75 [==========>...................] - ETA: 2s - loss: 0.1057 - auc: 0.9933
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1062 - auc: 0.9933
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1233 - auc: 0.9898
## 32/75 [===========>..................] - ETA: 2s - loss: 0.1210 - auc: 0.9902
## 33/75 [============>.................] - ETA: 2s - loss: 0.1191 - auc: 0.9906
## 34/75 [============>.................] - ETA: 2s - loss: 0.1163 - auc: 0.9910
## 35/75 [=============>................] - ETA: 2s - loss: 0.1143 - auc: 0.9914
## 36/75 [=============>................] - ETA: 2s - loss: 0.1114 - auc: 0.9919
## 37/75 [=============>................] - ETA: 2s - loss: 0.1143 - auc: 0.9915
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1146 - auc: 0.9915
## 39/75 [==============>...............] - ETA: 1s - loss: 0.1130 - auc: 0.9918
## 40/75 [===============>..............] - ETA: 1s - loss: 0.1128 - auc: 0.9919
## 41/75 [===============>..............] - ETA: 1s - loss: 0.1110 - auc: 0.9921
## 42/75 [===============>..............] - ETA: 1s - loss: 0.1117 - auc: 0.9920
## 43/75 [================>.............] - ETA: 1s - loss: 0.1096 - auc: 0.9923
## 44/75 [================>.............] - ETA: 1s - loss: 0.1075 - auc: 0.9926
## 45/75 [=================>............] - ETA: 1s - loss: 0.1106 - auc: 0.9922
## 46/75 [=================>............] - ETA: 1s - loss: 0.1086 - auc: 0.9925
## 47/75 [=================>............] - ETA: 1s - loss: 0.1069 - auc: 0.9927
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1051 - auc: 0.9930
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1086 - auc: 0.9922
## 50/75 [===================>..........] - ETA: 1s - loss: 0.1101 - auc: 0.9919
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1113 - auc: 0.9917
## 52/75 [===================>..........] - ETA: 1s - loss: 0.1097 - auc: 0.9920
```

```
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1102 - auc: 0.9919
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1089 - auc: 0.9921
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1088 - auc: 0.9922
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1101 - auc: 0.9920
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1096 - auc: 0.9921
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1085 - auc: 0.9923
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1122 - auc: 0.9915
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1109 - auc: 0.9918
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1125 - auc: 0.9916
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1113 - auc: 0.9918
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1127 - auc: 0.9917
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1127 - auc: 0.9918
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1121 - auc: 0.9919
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1115 - auc: 0.9920
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1106 - auc: 0.9921
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1091 - auc: 0.9923
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1079 - auc: 0.9925
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1073 - auc: 0.9926
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1062 - auc: 0.9928
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1063 - auc: 0.9928
## 73/75 [============================>.] - ETA: 0s - loss: 0.1053 - auc: 0.9930
## 74/75 [============================>.] - ETA: 0s - loss: 0.1057 - auc: 0.9930
## 75/75 [==============================] - 4s 58ms/step - loss: 0.1054 - auc: 0.9930 - val_l
oss: 0.6300 - val_auc: 0.9836
## Epoch 8/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.2025 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.1155 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.1832 - auc: 0.9766
##  4/75 [>.............................] - ETA: 3s - loss: 0.1418 - auc: 0.9864
##  5/75 [=>............................] - ETA: 3s - loss: 0.1290 - auc: 0.9891
##  6/75 [=>............................] - ETA: 3s - loss: 0.1178 - auc: 0.9902
##  7/75 [=>............................] - ETA: 3s - loss: 0.1069 - auc: 0.9918
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1160 - auc: 0.9914
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1107 - auc: 0.9923
## 10/75 [===>..........................] - ETA: 3s - loss: 0.1128 - auc: 0.9921
## 11/75 [===>..........................] - ETA: 3s - loss: 0.1097 - auc: 0.9927
## 12/75 [===>..........................] - ETA: 3s - loss: 0.1074 - auc: 0.9934
## 13/75 [====>.........................] - ETA: 3s - loss: 0.1123 - auc: 0.9929
## 14/75 [====>.........................] - ETA: 3s - loss: 0.1080 - auc: 0.9935
## 15/75 [=====>........................] - ETA: 3s - loss: 0.1083 - auc: 0.9937
## 16/75 [=====>........................] - ETA: 3s - loss: 0.1112 - auc: 0.9932
## 17/75 [=====>........................] - ETA: 3s - loss: 0.1162 - auc: 0.9915
## 18/75 [======>.......................] - ETA: 3s - loss: 0.1118 - auc: 0.9925
## 19/75 [======>.......................] - ETA: 3s - loss: 0.1109 - auc: 0.9928
## 20/75 [=======>......................] - ETA: 3s - loss: 0.1062 - auc: 0.9935
## 21/75 [=======>......................] - ETA: 2s - loss: 0.1094 - auc: 0.9933
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1067 - auc: 0.9937
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1069 - auc: 0.9938
## 24/75 [========>.....................] - ETA: 2s - loss: 0.1073 - auc: 0.9938
## 25/75 [=========>....................] - ETA: 2s - loss: 0.1074 - auc: 0.9937
## 26/75 [=========>....................] - ETA: 2s - loss: 0.1049 - auc: 0.9941
## 27/75 [=========>....................] - ETA: 2s - loss: 0.1026 - auc: 0.9944
## 28/75 [==========>...................] - ETA: 2s - loss: 0.1023 - auc: 0.9943
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0993 - auc: 0.9947
## 30/75 [===========>..................] - ETA: 2s - loss: 0.1021 - auc: 0.9943
```

```
## 31/75 [===========>..................] - ETA: 2s - loss: 0.1014 - auc: 0.9944
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0998 - auc: 0.9946
## 33/75 [============>.................] - ETA: 2s - loss: 0.0997 - auc: 0.9947
## 34/75 [============>.................] - ETA: 2s - loss: 0.0980 - auc: 0.9949
## 35/75 [=============>................] - ETA: 2s - loss: 0.0979 - auc: 0.9950
## 36/75 [=============>................] - ETA: 2s - loss: 0.0965 - auc: 0.9952
## 37/75 [=============>................] - ETA: 2s - loss: 0.1016 - auc: 0.9945
## 38/75 [==============>...............] - ETA: 2s - loss: 0.1002 - auc: 0.9947
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0988 - auc: 0.9949
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0968 - auc: 0.9951
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0962 - auc: 0.9952
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0944 - auc: 0.9954
## 43/75 [================>.............] - ETA: 1s - loss: 0.0936 - auc: 0.9956
## 44/75 [================>.............] - ETA: 1s - loss: 0.0940 - auc: 0.9956
## 45/75 [=================>............] - ETA: 1s - loss: 0.0932 - auc: 0.9957
## 46/75 [=================>............] - ETA: 1s - loss: 0.1046 - auc: 0.9931
## 47/75 [=================>............] - ETA: 1s - loss: 0.1035 - auc: 0.9932
## 48/75 [==================>...........] - ETA: 1s - loss: 0.1019 - auc: 0.9935
## 49/75 [==================>...........] - ETA: 1s - loss: 0.1008 - auc: 0.9937
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0998 - auc: 0.9938
## 51/75 [===================>..........] - ETA: 1s - loss: 0.1000 - auc: 0.9938
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0985 - auc: 0.9940
## 53/75 [====================>.........] - ETA: 1s - loss: 0.1010 - auc: 0.9934
## 54/75 [====================>.........] - ETA: 1s - loss: 0.1008 - auc: 0.9935
## 55/75 [=====================>........] - ETA: 1s - loss: 0.1025 - auc: 0.9935
## 56/75 [=====================>........] - ETA: 1s - loss: 0.1012 - auc: 0.9936
## 57/75 [=====================>........] - ETA: 0s - loss: 0.1003 - auc: 0.9937
## 58/75 [======================>.......] - ETA: 0s - loss: 0.1000 - auc: 0.9938
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0991 - auc: 0.9939
## 60/75 [=======================>......] - ETA: 0s - loss: 0.1002 - auc: 0.9937
## 61/75 [=======================>......] - ETA: 0s - loss: 0.1003 - auc: 0.9938
## 62/75 [=======================>......] - ETA: 0s - loss: 0.1000 - auc: 0.9939
## 63/75 [========================>.....] - ETA: 0s - loss: 0.1012 - auc: 0.9938
## 64/75 [========================>.....] - ETA: 0s - loss: 0.1007 - auc: 0.9939
## 65/75 [=========================>....] - ETA: 0s - loss: 0.1012 - auc: 0.9938
## 66/75 [=========================>....] - ETA: 0s - loss: 0.1003 - auc: 0.9939
## 67/75 [=========================>....] - ETA: 0s - loss: 0.1036 - auc: 0.9935
## 68/75 [==========================>...] - ETA: 0s - loss: 0.1038 - auc: 0.9934
## 69/75 [==========================>...] - ETA: 0s - loss: 0.1034 - auc: 0.9935
## 70/75 [===========================>..] - ETA: 0s - loss: 0.1040 - auc: 0.9935
## 71/75 [===========================>..] - ETA: 0s - loss: 0.1029 - auc: 0.9936
## 72/75 [===========================>..] - ETA: 0s - loss: 0.1023 - auc: 0.9937
## 73/75 [============================>.] - ETA: 0s - loss: 0.1016 - auc: 0.9938
## 74/75 [============================>.] - ETA: 0s - loss: 0.1008 - auc: 0.9940
## 75/75 [==============================] - 4s 58ms/step - loss: 0.1002 - auc: 0.9940 - val_l
oss: 0.5057 - val_auc: 0.9958
## Epoch 9/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.1398 - auc: 0.9818
##  2/75 [..............................] - ETA: 4s - loss: 0.0763 - auc: 0.9955
##  3/75 [>.............................] - ETA: 3s - loss: 0.0595 - auc: 0.9980
##  4/75 [>.............................] - ETA: 3s - loss: 0.0597 - auc: 0.9988
##  5/75 [=>............................] - ETA: 3s - loss: 0.0668 - auc: 0.9986
##  6/75 [=>............................] - ETA: 3s - loss: 0.0641 - auc: 0.9986
##  7/75 [=>............................] - ETA: 3s - loss: 0.0718 - auc: 0.9986
##  8/75 [==>...........................] - ETA: 3s - loss: 0.1114 - auc: 0.9837
```

```
##  9/75 [==>...........................] - ETA: 3s - loss: 0.1002 - auc: 0.9853
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0964 - auc: 0.9867
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0915 - auc: 0.9874
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0943 - auc: 0.9886
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0953 - auc: 0.9897
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0904 - auc: 0.9906
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0978 - auc: 0.9907
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0959 - auc: 0.9911
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0923 - auc: 0.9916
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0884 - auc: 0.9921
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0907 - auc: 0.9917
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0867 - auc: 0.9923
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0996 - auc: 0.9904
## 22/75 [=======>......................] - ETA: 2s - loss: 0.1039 - auc: 0.9902
## 23/75 [========>.....................] - ETA: 2s - loss: 0.1019 - auc: 0.9910
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0985 - auc: 0.9916
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0957 - auc: 0.9919
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0947 - auc: 0.9919
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0921 - auc: 0.9923
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0897 - auc: 0.9925
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0873 - auc: 0.9929
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0913 - auc: 0.9924
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0903 - auc: 0.9929
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0889 - auc: 0.9930
## 33/75 [============>.................] - ETA: 2s - loss: 0.0913 - auc: 0.9928
## 34/75 [============>.................] - ETA: 2s - loss: 0.0905 - auc: 0.9930
## 35/75 [=============>................] - ETA: 2s - loss: 0.0895 - auc: 0.9932
## 36/75 [=============>................] - ETA: 2s - loss: 0.0901 - auc: 0.9935
## 37/75 [=============>................] - ETA: 2s - loss: 0.0923 - auc: 0.9933
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0915 - auc: 0.9935
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0914 - auc: 0.9936
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0905 - auc: 0.9937
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0970 - auc: 0.9927
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0958 - auc: 0.9930
## 43/75 [================>.............] - ETA: 1s - loss: 0.0946 - auc: 0.9931
## 44/75 [================>.............] - ETA: 1s - loss: 0.0939 - auc: 0.9933
## 45/75 [=================>............] - ETA: 1s - loss: 0.0926 - auc: 0.9935
## 46/75 [=================>............] - ETA: 1s - loss: 0.0920 - auc: 0.9935
## 47/75 [=================>............] - ETA: 1s - loss: 0.0909 - auc: 0.9937
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0910 - auc: 0.9937
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0899 - auc: 0.9938
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0889 - auc: 0.9939
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0920 - auc: 0.9936
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0911 - auc: 0.9937
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0917 - auc: 0.9937
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0925 - auc: 0.9937
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0913 - auc: 0.9938
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0993 - auc: 0.9924
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0998 - auc: 0.9925
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0985 - auc: 0.9927
## 59/75 [======================>.......] - ETA: 0s - loss: 0.1004 - auc: 0.9925
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0995 - auc: 0.9926
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0990 - auc: 0.9927
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0993 - auc: 0.9927
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0998 - auc: 0.9927
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0991 - auc: 0.9928
```

```
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0983 - auc: 0.9929
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0976 - auc: 0.9930
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0992 - auc: 0.9929
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0988 - auc: 0.9929
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0976 - auc: 0.9931
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0965 - auc: 0.9932
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0961 - auc: 0.9933
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0958 - auc: 0.9933
## 73/75 [============================>.] - ETA: 0s - loss: 0.0960 - auc: 0.9933
## 74/75 [============================>.] - ETA: 0s - loss: 0.0954 - auc: 0.9934
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0971 - auc: 0.9931 - val_l
oss: 0.5577 - val_auc: 0.9943
## Epoch 10/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0317 - auc: 1.0000
##  2/75 [..............................] - ETA: 4s - loss: 0.0232 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0300 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0299 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0487 - auc: 0.9993
##  6/75 [=>............................] - ETA: 3s - loss: 0.0461 - auc: 0.9995
##  7/75 [=>............................] - ETA: 3s - loss: 0.0460 - auc: 0.9992
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0541 - auc: 0.9992
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0502 - auc: 0.9993
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0566 - auc: 0.9991
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0531 - auc: 0.9993
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0580 - auc: 0.9988
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0633 - auc: 0.9983
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0612 - auc: 0.9986
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0782 - auc: 0.9947
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0818 - auc: 0.9950
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0850 - auc: 0.9951
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0846 - auc: 0.9952
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0853 - auc: 0.9949
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0849 - auc: 0.9948
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0911 - auc: 0.9939
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0909 - auc: 0.9942
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0912 - auc: 0.9943
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0895 - auc: 0.9945
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0888 - auc: 0.9947
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0873 - auc: 0.9951
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0852 - auc: 0.9954
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0852 - auc: 0.9955
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0986 - auc: 0.9930
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0963 - auc: 0.9934
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0955 - auc: 0.9935
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0962 - auc: 0.9933
## 33/75 [============>.................] - ETA: 2s - loss: 0.0940 - auc: 0.9937
## 34/75 [============>.................] - ETA: 2s - loss: 0.0936 - auc: 0.9938
## 35/75 [=============>................] - ETA: 2s - loss: 0.0919 - auc: 0.9940
## 36/75 [=============>................] - ETA: 2s - loss: 0.0906 - auc: 0.9943
## 37/75 [=============>................] - ETA: 2s - loss: 0.0900 - auc: 0.9944
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0917 - auc: 0.9942
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0928 - auc: 0.9942
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0917 - auc: 0.9944
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0903 - auc: 0.9946
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0924 - auc: 0.9943
```

```
## 43/75 [================>.............] - ETA: 1s - loss: 0.0933 - auc: 0.9943
## 44/75 [================>.............] - ETA: 1s - loss: 0.0938 - auc: 0.9942
## 45/75 [=================>............] - ETA: 1s - loss: 0.0923 - auc: 0.9944
## 46/75 [=================>............] - ETA: 1s - loss: 0.0936 - auc: 0.9944
## 47/75 [=================>............] - ETA: 1s - loss: 0.0920 - auc: 0.9946
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0919 - auc: 0.9946
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0908 - auc: 0.9948
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0894 - auc: 0.9950
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0904 - auc: 0.9949
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0897 - auc: 0.9950
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0903 - auc: 0.9951
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0893 - auc: 0.9952
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0887 - auc: 0.9953
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0880 - auc: 0.9954
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0875 - auc: 0.9954
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0863 - auc: 0.9955
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0852 - auc: 0.9957
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0842 - auc: 0.9958
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0876 - auc: 0.9954
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0866 - auc: 0.9955
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0869 - auc: 0.9955
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0860 - auc: 0.9956
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0850 - auc: 0.9957
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0844 - auc: 0.9958
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0839 - auc: 0.9958
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0851 - auc: 0.9957
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0853 - auc: 0.9957
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0845 - auc: 0.9958
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0838 - auc: 0.9959
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0833 - auc: 0.9960
## 73/75 [============================>.] - ETA: 0s - loss: 0.0828 - auc: 0.9961
## 74/75 [============================>.] - ETA: 0s - loss: 0.0834 - auc: 0.9960
## 75/75 [==============================] - 4s 58ms/step - loss: 0.0827 - auc: 0.9961 - val_l
oss: 0.2005 - val_auc: 0.9955
## Epoch 11/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.1632 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0963 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0734 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0680 - auc: 0.9988
##  5/75 [=>............................] - ETA: 3s - loss: 0.0574 - auc: 0.9992
##  6/75 [=>............................] - ETA: 3s - loss: 0.0522 - auc: 0.9995
##  7/75 [=>............................] - ETA: 3s - loss: 0.0543 - auc: 0.9989
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0535 - auc: 0.9989
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0894 - auc: 0.9954
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0905 - auc: 0.9950
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0888 - auc: 0.9953
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0883 - auc: 0.9955
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0826 - auc: 0.9961
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0835 - auc: 0.9960
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0825 - auc: 0.9961
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0837 - auc: 0.9959
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0851 - auc: 0.9963
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0817 - auc: 0.9967
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0786 - auc: 0.9970
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0780 - auc: 0.9972
```

```
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0747 - auc: 0.9975
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0725 - auc: 0.9976
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0708 - auc: 0.9978
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0694 - auc: 0.9979
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0679 - auc: 0.9981
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0673 - auc: 0.9982
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0688 - auc: 0.9981
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0668 - auc: 0.9983
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0656 - auc: 0.9984
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0643 - auc: 0.9985
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0668 - auc: 0.9982
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0658 - auc: 0.9984
## 33/75 [============>.................] - ETA: 2s - loss: 0.0646 - auc: 0.9985
## 34/75 [============>.................] - ETA: 2s - loss: 0.0634 - auc: 0.9985
## 35/75 [=============>................] - ETA: 2s - loss: 0.0620 - auc: 0.9986
## 36/75 [=============>................] - ETA: 2s - loss: 0.0607 - auc: 0.9987
## 37/75 [=============>................] - ETA: 2s - loss: 0.0613 - auc: 0.9987
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0614 - auc: 0.9987
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0604 - auc: 0.9987
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0602 - auc: 0.9987
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0617 - auc: 0.9985
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0606 - auc: 0.9986
## 43/75 [================>.............] - ETA: 1s - loss: 0.0663 - auc: 0.9980
## 44/75 [================>.............] - ETA: 1s - loss: 0.0651 - auc: 0.9981
## 45/75 [=================>............] - ETA: 1s - loss: 0.0650 - auc: 0.9981
## 46/75 [=================>............] - ETA: 1s - loss: 0.0683 - auc: 0.9979
## 47/75 [=================>............] - ETA: 1s - loss: 0.0670 - auc: 0.9980
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0720 - auc: 0.9969
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0710 - auc: 0.9970
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0728 - auc: 0.9969
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0717 - auc: 0.9970
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0721 - auc: 0.9970
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0713 - auc: 0.9971
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0702 - auc: 0.9972
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0710 - auc: 0.9971
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0745 - auc: 0.9965
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0766 - auc: 0.9962
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0761 - auc: 0.9963
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0751 - auc: 0.9963
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0743 - auc: 0.9964
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0741 - auc: 0.9965
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0736 - auc: 0.9965
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0727 - auc: 0.9966
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0724 - auc: 0.9967
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0725 - auc: 0.9967
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0723 - auc: 0.9968
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0730 - auc: 0.9968
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0729 - auc: 0.9968
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0730 - auc: 0.9968
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0722 - auc: 0.9968
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0714 - auc: 0.9969
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0708 - auc: 0.9970
## 73/75 [============================>.] - ETA: 0s - loss: 0.0710 - auc: 0.9970
## 74/75 [============================>.] - ETA: 0s - loss: 0.0736 - auc: 0.9966
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0745 - auc: 0.9965 - val_l
oss: 0.7702 - val_auc: 0.9965
```

```
## Epoch 12/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0750 - auc: 1.0000
##  2/75 [..............................] - ETA: 4s - loss: 0.0569 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0523 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0573 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0489 - auc: 1.0000
##  6/75 [=>............................] - ETA: 3s - loss: 0.0503 - auc: 1.0000
##  7/75 [=>............................] - ETA: 3s - loss: 0.0502 - auc: 1.0000
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0486 - auc: 1.0000
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0494 - auc: 1.0000
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0496 - auc: 1.0000
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0485 - auc: 1.0000
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0519 - auc: 0.9998
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0495 - auc: 0.9998
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0468 - auc: 0.9998
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0451 - auc: 0.9998
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0450 - auc: 0.9999
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0439 - auc: 0.9999
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0426 - auc: 0.9999
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0502 - auc: 0.9989
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0528 - auc: 0.9986
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0518 - auc: 0.9987
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0519 - auc: 0.9988
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0517 - auc: 0.9989
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0535 - auc: 0.9989
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0558 - auc: 0.9988
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0554 - auc: 0.9989
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0541 - auc: 0.9989
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0528 - auc: 0.9990
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0534 - auc: 0.9990
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0665 - auc: 0.9974
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0647 - auc: 0.9975
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0638 - auc: 0.9977
## 33/75 [============>.................] - ETA: 2s - loss: 0.0629 - auc: 0.9979
## 34/75 [============>.................] - ETA: 2s - loss: 0.0636 - auc: 0.9978
## 35/75 [=============>................] - ETA: 2s - loss: 0.0621 - auc: 0.9979
## 36/75 [=============>................] - ETA: 2s - loss: 0.0629 - auc: 0.9980
## 37/75 [=============>................] - ETA: 2s - loss: 0.0684 - auc: 0.9979
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0674 - auc: 0.9980
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0665 - auc: 0.9980
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0652 - auc: 0.9981
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0650 - auc: 0.9981
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0637 - auc: 0.9982
## 43/75 [================>.............] - ETA: 1s - loss: 0.0627 - auc: 0.9983
## 44/75 [================>.............] - ETA: 1s - loss: 0.0621 - auc: 0.9983
## 45/75 [=================>............] - ETA: 1s - loss: 0.0645 - auc: 0.9979
## 46/75 [=================>............] - ETA: 1s - loss: 0.0649 - auc: 0.9979
## 47/75 [=================>............] - ETA: 1s - loss: 0.0641 - auc: 0.9979
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0634 - auc: 0.9980
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0639 - auc: 0.9979
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0632 - auc: 0.9980
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0631 - auc: 0.9980
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0622 - auc: 0.9980
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0618 - auc: 0.9980
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0612 - auc: 0.9981
```

```
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0613 - auc: 0.9982
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0606 - auc: 0.9982
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0630 - auc: 0.9977
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0650 - auc: 0.9976
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0645 - auc: 0.9977
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0637 - auc: 0.9978
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0632 - auc: 0.9978
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0623 - auc: 0.9979
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0615 - auc: 0.9979
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0639 - auc: 0.9976
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0637 - auc: 0.9976
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0631 - auc: 0.9977
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0623 - auc: 0.9977
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0618 - auc: 0.9978
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0628 - auc: 0.9978
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0661 - auc: 0.9974
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0657 - auc: 0.9974
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0653 - auc: 0.9975
## 73/75 [============================>.] - ETA: 0s - loss: 0.0649 - auc: 0.9975
## 74/75 [============================>.] - ETA: 0s - loss: 0.0652 - auc: 0.9975
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0648 - auc: 0.9975 - val_l
oss: 0.4534 - val_auc: 0.9961
## Epoch 13/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0572 - auc: 1.0000
##  2/75 [..............................] - ETA: 4s - loss: 0.0527 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0743 - auc: 0.9980
##  4/75 [>.............................] - ETA: 3s - loss: 0.0571 - auc: 0.9989
##  5/75 [=>............................] - ETA: 3s - loss: 0.0556 - auc: 0.9986
##  6/75 [=>............................] - ETA: 3s - loss: 0.0513 - auc: 0.9990
##  7/75 [=>............................] - ETA: 3s - loss: 0.0550 - auc: 0.9993
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0588 - auc: 0.9994
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0542 - auc: 0.9996
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0522 - auc: 0.9996
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0501 - auc: 0.9997
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0473 - auc: 0.9998
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0481 - auc: 0.9997
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0536 - auc: 0.9991
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0528 - auc: 0.9991
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0611 - auc: 0.9978
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0585 - auc: 0.9980
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0582 - auc: 0.9982
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0601 - auc: 0.9983
## 20/75 [=======>......................] - ETA: 2s - loss: 0.0580 - auc: 0.9984
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0605 - auc: 0.9982
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0582 - auc: 0.9984
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0687 - auc: 0.9974
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0670 - auc: 0.9976
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0647 - auc: 0.9978
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0681 - auc: 0.9975
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0676 - auc: 0.9975
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0659 - auc: 0.9977
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0657 - auc: 0.9978
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0644 - auc: 0.9979
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0637 - auc: 0.9980
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0626 - auc: 0.9980
```

```
## 33/75 [============>.................] - ETA: 2s - loss: 0.0616 - auc: 0.9981
## 34/75 [============>.................] - ETA: 2s - loss: 0.0602 - auc: 0.9982
## 35/75 [=============>................] - ETA: 2s - loss: 0.0591 - auc: 0.9983
## 36/75 [=============>................] - ETA: 2s - loss: 0.0618 - auc: 0.9980
## 37/75 [=============>................] - ETA: 2s - loss: 0.0696 - auc: 0.9973
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0682 - auc: 0.9974
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0668 - auc: 0.9976
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0654 - auc: 0.9977
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0640 - auc: 0.9978
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0638 - auc: 0.9978
## 43/75 [================>.............] - ETA: 1s - loss: 0.0657 - auc: 0.9977
## 44/75 [================>.............] - ETA: 1s - loss: 0.0652 - auc: 0.9977
## 45/75 [=================>............] - ETA: 1s - loss: 0.0645 - auc: 0.9978
## 46/75 [=================>............] - ETA: 1s - loss: 0.0637 - auc: 0.9978
## 47/75 [=================>............] - ETA: 1s - loss: 0.0660 - auc: 0.9976
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0652 - auc: 0.9977
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0670 - auc: 0.9976
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0665 - auc: 0.9976
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0659 - auc: 0.9977
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0653 - auc: 0.9978
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0643 - auc: 0.9979
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0636 - auc: 0.9979
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0635 - auc: 0.9979
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0626 - auc: 0.9980
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0625 - auc: 0.9980
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0630 - auc: 0.9980
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0644 - auc: 0.9979
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0636 - auc: 0.9979
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0631 - auc: 0.9980
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0622 - auc: 0.9980
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0619 - auc: 0.9981
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0613 - auc: 0.9981
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0605 - auc: 0.9982
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0598 - auc: 0.9982
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0594 - auc: 0.9982
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0586 - auc: 0.9983
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0578 - auc: 0.9983
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0579 - auc: 0.9983
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0584 - auc: 0.9983
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0579 - auc: 0.9984
## 73/75 [============================>.] - ETA: 0s - loss: 0.0573 - auc: 0.9984
## 74/75 [============================>.] - ETA: 0s - loss: 0.0567 - auc: 0.9984
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0570 - auc: 0.9984 - val_l
oss: 0.8573 - val_auc: 0.9682
## Epoch 14/20
## 
##  1/75 [..............................] - ETA: 3s - loss: 0.1055 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0654 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0511 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0441 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0782 - auc: 0.9974
##  6/75 [=>............................] - ETA: 3s - loss: 0.0737 - auc: 0.9977
##  7/75 [=>............................] - ETA: 3s - loss: 0.0718 - auc: 0.9980
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0707 - auc: 0.9982
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0643 - auc: 0.9986
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0622 - auc: 0.9987
```

```
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0666 - auc: 0.9986
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0633 - auc: 0.9988
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0595 - auc: 0.9990
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0563 - auc: 0.9991
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0600 - auc: 0.9991
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0581 - auc: 0.9992
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0571 - auc: 0.9992
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0549 - auc: 0.9993
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0532 - auc: 0.9994
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0520 - auc: 0.9995
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0566 - auc: 0.9989
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0585 - auc: 0.9987
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0566 - auc: 0.9988
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0572 - auc: 0.9988
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0559 - auc: 0.9989
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0568 - auc: 0.9989
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0568 - auc: 0.9989
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0557 - auc: 0.9990
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0571 - auc: 0.9989
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0596 - auc: 0.9988
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0595 - auc: 0.9988
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0591 - auc: 0.9989
## 33/75 [============>.................] - ETA: 2s - loss: 0.0582 - auc: 0.9989
## 34/75 [============>.................] - ETA: 2s - loss: 0.0573 - auc: 0.9990
## 35/75 [=============>................] - ETA: 2s - loss: 0.0570 - auc: 0.9990
## 36/75 [=============>................] - ETA: 2s - loss: 0.0567 - auc: 0.9990
## 37/75 [=============>................] - ETA: 2s - loss: 0.0561 - auc: 0.9991
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0555 - auc: 0.9991
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0543 - auc: 0.9991
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0600 - auc: 0.9989
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0589 - auc: 0.9989
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0626 - auc: 0.9983
## 43/75 [================>.............] - ETA: 1s - loss: 0.0617 - auc: 0.9983
## 44/75 [================>.............] - ETA: 1s - loss: 0.0612 - auc: 0.9984
## 45/75 [=================>............] - ETA: 1s - loss: 0.0601 - auc: 0.9984
## 46/75 [=================>............] - ETA: 1s - loss: 0.0597 - auc: 0.9985
## 47/75 [=================>............] - ETA: 1s - loss: 0.0597 - auc: 0.9985
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0587 - auc: 0.9985
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0582 - auc: 0.9986
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0573 - auc: 0.9986
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0574 - auc: 0.9986
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0573 - auc: 0.9986
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0567 - auc: 0.9987
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0559 - auc: 0.9987
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0560 - auc: 0.9987
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0554 - auc: 0.9988
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0545 - auc: 0.9988
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0538 - auc: 0.9988
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0531 - auc: 0.9989
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0523 - auc: 0.9989
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0517 - auc: 0.9989
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0513 - auc: 0.9990
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0529 - auc: 0.9989
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0533 - auc: 0.9989
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0529 - auc: 0.9989
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0525 - auc: 0.9989
```

```
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0523 - auc: 0.9989
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0521 - auc: 0.9990
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0519 - auc: 0.9990
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0538 - auc: 0.9988
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0536 - auc: 0.9988
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0532 - auc: 0.9988
## 73/75 [============================>.] - ETA: 0s - loss: 0.0527 - auc: 0.9988
## 74/75 [============================>.] - ETA: 0s - loss: 0.0520 - auc: 0.9989
## 75/75 [==============================] - 4s 58ms/step - loss: 0.0520 - auc: 0.9989 - val_l
oss: 0.1609 - val_auc: 0.9978
## Epoch 15/20
## 
##  1/75 [..............................] - ETA: 3s - loss: 0.1464 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0834 - auc: 0.9952
##  3/75 [>.............................] - ETA: 3s - loss: 0.0736 - auc: 0.9958
##  4/75 [>.............................] - ETA: 3s - loss: 0.0578 - auc: 0.9977
##  5/75 [=>............................] - ETA: 3s - loss: 0.0534 - auc: 0.9979
##  6/75 [=>............................] - ETA: 3s - loss: 0.0716 - auc: 0.9974
##  7/75 [=>............................] - ETA: 3s - loss: 0.0643 - auc: 0.9981
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0638 - auc: 0.9983
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0616 - auc: 0.9985
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0569 - auc: 0.9988
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0536 - auc: 0.9988
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0531 - auc: 0.9988
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0527 - auc: 0.9988
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0503 - auc: 0.9989
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0481 - auc: 0.9991
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0471 - auc: 0.9991
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0462 - auc: 0.9991
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0453 - auc: 0.9992
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0444 - auc: 0.9992
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0431 - auc: 0.9993
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0415 - auc: 0.9994
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0410 - auc: 0.9994
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0398 - auc: 0.9994
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0385 - auc: 0.9995
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0381 - auc: 0.9995
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0400 - auc: 0.9995
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0391 - auc: 0.9995
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0421 - auc: 0.9995
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0561 - auc: 0.9984
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0556 - auc: 0.9984
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0540 - auc: 0.9985
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0542 - auc: 0.9986
## 33/75 [============>.................] - ETA: 2s - loss: 0.0538 - auc: 0.9986
## 34/75 [============>.................] - ETA: 2s - loss: 0.0529 - auc: 0.9987
## 35/75 [=============>................] - ETA: 2s - loss: 0.0517 - auc: 0.9987
## 36/75 [=============>................] - ETA: 2s - loss: 0.0512 - auc: 0.9987
## 37/75 [=============>................] - ETA: 2s - loss: 0.0501 - auc: 0.9988
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0489 - auc: 0.9989
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0492 - auc: 0.9989
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0482 - auc: 0.9989
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0475 - auc: 0.9990
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0470 - auc: 0.9990
## 43/75 [================>.............] - ETA: 1s - loss: 0.0473 - auc: 0.9990
## 44/75 [================>.............] - ETA: 1s - loss: 0.0464 - auc: 0.9990
```

```
## 45/75 [=================>............] - ETA: 1s - loss: 0.0488 - auc: 0.9990
## 46/75 [=================>............] - ETA: 1s - loss: 0.0491 - auc: 0.9990
## 47/75 [=================>............] - ETA: 1s - loss: 0.0485 - auc: 0.9990
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0478 - auc: 0.9990
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0476 - auc: 0.9990
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0468 - auc: 0.9991
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0467 - auc: 0.9991
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0462 - auc: 0.9991
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0455 - auc: 0.9991
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0450 - auc: 0.9992
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0472 - auc: 0.9990
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0486 - auc: 0.9989
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0518 - auc: 0.9987
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0515 - auc: 0.9987
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0513 - auc: 0.9987
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0549 - auc: 0.9984
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0544 - auc: 0.9985
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0538 - auc: 0.9985
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0532 - auc: 0.9986
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0531 - auc: 0.9986
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0541 - auc: 0.9985
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0560 - auc: 0.9984
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0594 - auc: 0.9981
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0590 - auc: 0.9982
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0600 - auc: 0.9981
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0597 - auc: 0.9981
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0590 - auc: 0.9982
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0595 - auc: 0.9981
## 73/75 [============================>.] - ETA: 0s - loss: 0.0590 - auc: 0.9982
## 74/75 [============================>.] - ETA: 0s - loss: 0.0586 - auc: 0.9982
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0580 - auc: 0.9982 - val_l
oss: 0.1596 - val_auc: 0.9975
## Epoch 16/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0071 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0164 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0353 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0475 - auc: 0.9990
##  5/75 [=>............................] - ETA: 3s - loss: 0.0418 - auc: 0.9994
##  6/75 [=>............................] - ETA: 3s - loss: 0.0368 - auc: 0.9996
##  7/75 [=>............................] - ETA: 3s - loss: 0.0531 - auc: 0.9990
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0557 - auc: 0.9992
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0510 - auc: 0.9994
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0612 - auc: 0.9982
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0572 - auc: 0.9985
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0596 - auc: 0.9988
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0557 - auc: 0.9989
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0560 - auc: 0.9991
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0537 - auc: 0.9992
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0521 - auc: 0.9993
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0502 - auc: 0.9994
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0480 - auc: 0.9994
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0494 - auc: 0.9993
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0486 - auc: 0.9993
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0471 - auc: 0.9994
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0458 - auc: 0.9994
```

```
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0482 - auc: 0.9993
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0468 - auc: 0.9994
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0455 - auc: 0.9994
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0478 - auc: 0.9993
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0469 - auc: 0.9993
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0466 - auc: 0.9994
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0454 - auc: 0.9994
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0450 - auc: 0.9995
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0440 - auc: 0.9995
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0428 - auc: 0.9995
## 33/75 [============>.................] - ETA: 2s - loss: 0.0417 - auc: 0.9996
## 34/75 [============>.................] - ETA: 2s - loss: 0.0412 - auc: 0.9996
## 35/75 [=============>................] - ETA: 2s - loss: 0.0401 - auc: 0.9996
## 36/75 [=============>................] - ETA: 2s - loss: 0.0393 - auc: 0.9996
## 37/75 [=============>................] - ETA: 2s - loss: 0.0385 - auc: 0.9996
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0414 - auc: 0.9995
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0409 - auc: 0.9995
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0404 - auc: 0.9995
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0399 - auc: 0.9995
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0394 - auc: 0.9995
## 43/75 [================>.............] - ETA: 1s - loss: 0.0392 - auc: 0.9996
## 44/75 [================>.............] - ETA: 1s - loss: 0.0395 - auc: 0.9996
## 45/75 [=================>............] - ETA: 1s - loss: 0.0436 - auc: 0.9992
## 46/75 [=================>............] - ETA: 1s - loss: 0.0429 - auc: 0.9992
## 47/75 [=================>............] - ETA: 1s - loss: 0.0422 - auc: 0.9993
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0416 - auc: 0.9993
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0409 - auc: 0.9993
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0414 - auc: 0.9993
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0413 - auc: 0.9993
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0420 - auc: 0.9993
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0414 - auc: 0.9993
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0424 - auc: 0.9992
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0419 - auc: 0.9993
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0461 - auc: 0.9989
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0454 - auc: 0.9990
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0448 - auc: 0.9990
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0446 - auc: 0.9990
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0441 - auc: 0.9990
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0442 - auc: 0.9990
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0441 - auc: 0.9991
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0435 - auc: 0.9991
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0506 - auc: 0.9971
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0500 - auc: 0.9972
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0498 - auc: 0.9973
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0492 - auc: 0.9973
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0498 - auc: 0.9973
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0499 - auc: 0.9973
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0494 - auc: 0.9974
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0490 - auc: 0.9975
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0486 - auc: 0.9975
## 73/75 [============================>.] - ETA: 0s - loss: 0.0485 - auc: 0.9975
## 74/75 [============================>.] - ETA: 0s - loss: 0.0480 - auc: 0.9976
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0477 - auc: 0.9976 - val_l
oss: 0.7687 - val_auc: 0.9891
## Epoch 17/20
##
```

```
##  1/75 [..............................] - ETA: 4s - loss: 0.0177 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0193 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0347 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0271 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0280 - auc: 1.0000
##  6/75 [=>............................] - ETA: 3s - loss: 0.0291 - auc: 1.0000
##  7/75 [=>............................] - ETA: 3s - loss: 0.0272 - auc: 1.0000
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0250 - auc: 1.0000
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0368 - auc: 1.0000
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0363 - auc: 1.0000
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0354 - auc: 1.0000
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0371 - auc: 1.0000
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0354 - auc: 1.0000
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0374 - auc: 1.0000
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0369 - auc: 1.0000
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0355 - auc: 1.0000
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0340 - auc: 1.0000
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0325 - auc: 1.0000
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0324 - auc: 1.0000
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0321 - auc: 1.0000
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0314 - auc: 1.0000
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0312 - auc: 1.0000
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0455 - auc: 0.9993
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0445 - auc: 0.9993
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0431 - auc: 0.9994
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0418 - auc: 0.9994
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0411 - auc: 0.9994
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0405 - auc: 0.9995
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0419 - auc: 0.9994
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0415 - auc: 0.9995
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0419 - auc: 0.9995
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0429 - auc: 0.9995
## 33/75 [============>.................] - ETA: 2s - loss: 0.0433 - auc: 0.9994
## 34/75 [============>.................] - ETA: 2s - loss: 0.0436 - auc: 0.9994
## 35/75 [=============>................] - ETA: 2s - loss: 0.0430 - auc: 0.9994
## 36/75 [=============>................] - ETA: 2s - loss: 0.0427 - auc: 0.9995
## 37/75 [=============>................] - ETA: 2s - loss: 0.0433 - auc: 0.9994
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0462 - auc: 0.9993
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0452 - auc: 0.9994
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0446 - auc: 0.9994
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0443 - auc: 0.9994
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0449 - auc: 0.9994
## 43/75 [================>.............] - ETA: 1s - loss: 0.0440 - auc: 0.9994
## 44/75 [================>.............] - ETA: 1s - loss: 0.0432 - auc: 0.9994
## 45/75 [=================>............] - ETA: 1s - loss: 0.0428 - auc: 0.9994
## 46/75 [=================>............] - ETA: 1s - loss: 0.0423 - auc: 0.9995
## 47/75 [=================>............] - ETA: 1s - loss: 0.0417 - auc: 0.9995
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0416 - auc: 0.9995
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0418 - auc: 0.9995
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0412 - auc: 0.9995
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0415 - auc: 0.9995
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0424 - auc: 0.9995
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0420 - auc: 0.9995
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0414 - auc: 0.9995
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0411 - auc: 0.9995
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0438 - auc: 0.9993
```

```
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0432 - auc: 0.9993
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0434 - auc: 0.9993
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0429 - auc: 0.9993
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0427 - auc: 0.9994
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0426 - auc: 0.9994
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0421 - auc: 0.9994
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0428 - auc: 0.9994
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0423 - auc: 0.9994
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0442 - auc: 0.9992
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0441 - auc: 0.9992
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0444 - auc: 0.9992
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0440 - auc: 0.9993
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0468 - auc: 0.9991
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0462 - auc: 0.9991
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0462 - auc: 0.9991
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0457 - auc: 0.9991
## 73/75 [============================>.] - ETA: 0s - loss: 0.0453 - auc: 0.9992
## 74/75 [============================>.] - ETA: 0s - loss: 0.0448 - auc: 0.9992
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0443 - auc: 0.9992 - val_l
oss: 0.4663 - val_auc: 0.9978
## Epoch 18/20
## 
##  1/75 [..............................] - ETA: 3s - loss: 0.0138 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0338 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0285 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0247 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0339 - auc: 1.0000
##  6/75 [=>............................] - ETA: 3s - loss: 0.0445 - auc: 0.9993
##  7/75 [=>............................] - ETA: 3s - loss: 0.0397 - auc: 0.9995
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0423 - auc: 0.9996
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0414 - auc: 0.9997
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0386 - auc: 0.9997
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0384 - auc: 0.9998
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0411 - auc: 0.9998
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0386 - auc: 0.9998
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0367 - auc: 0.9999
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0379 - auc: 0.9999
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0372 - auc: 0.9999
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0354 - auc: 0.9999
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0341 - auc: 0.9999
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0355 - auc: 0.9999
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0384 - auc: 0.9998
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0411 - auc: 0.9998
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0395 - auc: 0.9998
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0407 - auc: 0.9998
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0396 - auc: 0.9998
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0389 - auc: 0.9998
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0382 - auc: 0.9998
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0369 - auc: 0.9998
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0358 - auc: 0.9999
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0357 - auc: 0.9999
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0357 - auc: 0.9999
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0361 - auc: 0.9998
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0354 - auc: 0.9999
## 33/75 [============>.................] - ETA: 2s - loss: 0.0359 - auc: 0.9999
## 34/75 [============>.................] - ETA: 2s - loss: 0.0354 - auc: 0.9999
```

```
## 35/75 [=============>................] - ETA: 2s - loss: 0.0350 - auc: 0.9999
## 36/75 [=============>................] - ETA: 2s - loss: 0.0342 - auc: 0.9999
## 37/75 [=============>................] - ETA: 2s - loss: 0.0339 - auc: 0.9999
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0335 - auc: 0.9999
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0328 - auc: 0.9999
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0348 - auc: 0.9998
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0380 - auc: 0.9996
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0375 - auc: 0.9996
## 43/75 [================>.............] - ETA: 1s - loss: 0.0368 - auc: 0.9996
## 44/75 [================>.............] - ETA: 1s - loss: 0.0374 - auc: 0.9996
## 45/75 [=================>............] - ETA: 1s - loss: 0.0368 - auc: 0.9996
## 46/75 [=================>............] - ETA: 1s - loss: 0.0364 - auc: 0.9997
## 47/75 [=================>............] - ETA: 1s - loss: 0.0357 - auc: 0.9997
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0357 - auc: 0.9997
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0381 - auc: 0.9995
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0390 - auc: 0.9995
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0386 - auc: 0.9995
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0380 - auc: 0.9995
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0380 - auc: 0.9995
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0374 - auc: 0.9996
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0372 - auc: 0.9996
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0371 - auc: 0.9996
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0365 - auc: 0.9996
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0361 - auc: 0.9996
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0365 - auc: 0.9996
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0368 - auc: 0.9996
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0385 - auc: 0.9996
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0393 - auc: 0.9996
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0388 - auc: 0.9996
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0409 - auc: 0.9994
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0403 - auc: 0.9994
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0398 - auc: 0.9995
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0407 - auc: 0.9994
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0402 - auc: 0.9994
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0398 - auc: 0.9994
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0409 - auc: 0.9994
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0405 - auc: 0.9994
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0402 - auc: 0.9994
## 73/75 [============================>.] - ETA: 0s - loss: 0.0407 - auc: 0.9994
## 74/75 [============================>.] - ETA: 0s - loss: 0.0404 - auc: 0.9994
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0413 - auc: 0.9993 - val_l
oss: 0.1943 - val_auc: 0.9964
## Epoch 19/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0305 - auc: 1.0000
##  2/75 [..............................] - ETA: 4s - loss: 0.0469 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0328 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0286 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0312 - auc: 1.0000
##  6/75 [=>............................] - ETA: 3s - loss: 0.0275 - auc: 1.0000
##  7/75 [=>............................] - ETA: 3s - loss: 0.0476 - auc: 0.9993
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0429 - auc: 0.9994
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0394 - auc: 0.9996
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0396 - auc: 0.9996
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0388 - auc: 0.9997
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0388 - auc: 0.9996
```

```
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0417 - auc: 0.9997
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0394 - auc: 0.9997
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0568 - auc: 0.9974
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0548 - auc: 0.9977
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0540 - auc: 0.9977
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0529 - auc: 0.9978
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0514 - auc: 0.9979
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0509 - auc: 0.9980
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0491 - auc: 0.9981
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0472 - auc: 0.9983
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0460 - auc: 0.9984
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0443 - auc: 0.9985
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0433 - auc: 0.9986
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0424 - auc: 0.9987
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0426 - auc: 0.9988
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0426 - auc: 0.9989
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0426 - auc: 0.9988
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0416 - auc: 0.9989
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0405 - auc: 0.9989
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0393 - auc: 0.9990
## 33/75 [============>.................] - ETA: 2s - loss: 0.0395 - auc: 0.9990
## 34/75 [============>.................] - ETA: 2s - loss: 0.0390 - auc: 0.9991
## 35/75 [=============>................] - ETA: 2s - loss: 0.0392 - auc: 0.9991
## 36/75 [=============>................] - ETA: 2s - loss: 0.0384 - auc: 0.9991
## 37/75 [=============>................] - ETA: 2s - loss: 0.0383 - auc: 0.9991
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0377 - auc: 0.9991
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0419 - auc: 0.9988
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0410 - auc: 0.9988
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0410 - auc: 0.9989
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0404 - auc: 0.9989
## 43/75 [================>.............] - ETA: 1s - loss: 0.0395 - auc: 0.9990
## 44/75 [================>.............] - ETA: 1s - loss: 0.0391 - auc: 0.9990
## 45/75 [=================>............] - ETA: 1s - loss: 0.0385 - auc: 0.9991
## 46/75 [=================>............] - ETA: 1s - loss: 0.0379 - auc: 0.9991
## 47/75 [=================>............] - ETA: 1s - loss: 0.0380 - auc: 0.9991
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0376 - auc: 0.9991
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0370 - auc: 0.9991
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0365 - auc: 0.9992
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0365 - auc: 0.9992
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0373 - auc: 0.9992
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0367 - auc: 0.9992
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0385 - auc: 0.9992
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0380 - auc: 0.9992
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0376 - auc: 0.9992
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0422 - auc: 0.9990
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0417 - auc: 0.9990
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0413 - auc: 0.9990
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0407 - auc: 0.9991
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0401 - auc: 0.9991
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0397 - auc: 0.9991
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0393 - auc: 0.9991
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0387 - auc: 0.9991
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0385 - auc: 0.9991
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0383 - auc: 0.9992
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0395 - auc: 0.9991
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0396 - auc: 0.9991
```

```
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0408 - auc: 0.9990
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0408 - auc: 0.9990
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0409 - auc: 0.9990
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0409 - auc: 0.9990
## 73/75 [============================>.] - ETA: 0s - loss: 0.0407 - auc: 0.9990
## 74/75 [============================>.] - ETA: 0s - loss: 0.0408 - auc: 0.9990
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0404 - auc: 0.9991 - val_l
oss: 0.5158 - val_auc: 0.9965
## Epoch 20/20
##
##  1/75 [..............................] - ETA: 3s - loss: 0.0197 - auc: 1.0000
##  2/75 [..............................] - ETA: 3s - loss: 0.0118 - auc: 1.0000
##  3/75 [>.............................] - ETA: 3s - loss: 0.0104 - auc: 1.0000
##  4/75 [>.............................] - ETA: 3s - loss: 0.0227 - auc: 1.0000
##  5/75 [=>............................] - ETA: 3s - loss: 0.0197 - auc: 1.0000
##  6/75 [=>............................] - ETA: 3s - loss: 0.0174 - auc: 1.0000
##  7/75 [=>............................] - ETA: 3s - loss: 0.0158 - auc: 1.0000
##  8/75 [==>...........................] - ETA: 3s - loss: 0.0153 - auc: 1.0000
##  9/75 [==>...........................] - ETA: 3s - loss: 0.0197 - auc: 1.0000
## 10/75 [===>..........................] - ETA: 3s - loss: 0.0228 - auc: 1.0000
## 11/75 [===>..........................] - ETA: 3s - loss: 0.0211 - auc: 1.0000
## 12/75 [===>..........................] - ETA: 3s - loss: 0.0198 - auc: 1.0000
## 13/75 [====>.........................] - ETA: 3s - loss: 0.0186 - auc: 1.0000
## 14/75 [====>.........................] - ETA: 3s - loss: 0.0287 - auc: 0.9999
## 15/75 [=====>........................] - ETA: 3s - loss: 0.0271 - auc: 0.9999
## 16/75 [=====>........................] - ETA: 3s - loss: 0.0255 - auc: 0.9999
## 17/75 [=====>........................] - ETA: 3s - loss: 0.0264 - auc: 0.9999
## 18/75 [======>.......................] - ETA: 3s - loss: 0.0264 - auc: 0.9999
## 19/75 [======>.......................] - ETA: 3s - loss: 0.0259 - auc: 0.9999
## 20/75 [=======>......................] - ETA: 3s - loss: 0.0261 - auc: 0.9999
## 21/75 [=======>......................] - ETA: 2s - loss: 0.0255 - auc: 0.9999
## 22/75 [=======>......................] - ETA: 2s - loss: 0.0247 - auc: 0.9999
## 23/75 [========>.....................] - ETA: 2s - loss: 0.0309 - auc: 0.9997
## 24/75 [========>.....................] - ETA: 2s - loss: 0.0301 - auc: 0.9997
## 25/75 [=========>....................] - ETA: 2s - loss: 0.0295 - auc: 0.9997
## 26/75 [=========>....................] - ETA: 2s - loss: 0.0314 - auc: 0.9997
## 27/75 [=========>....................] - ETA: 2s - loss: 0.0309 - auc: 0.9997
## 28/75 [==========>...................] - ETA: 2s - loss: 0.0317 - auc: 0.9997
## 29/75 [==========>...................] - ETA: 2s - loss: 0.0327 - auc: 0.9996
## 30/75 [===========>..................] - ETA: 2s - loss: 0.0326 - auc: 0.9997
## 31/75 [===========>..................] - ETA: 2s - loss: 0.0317 - auc: 0.9997
## 32/75 [===========>..................] - ETA: 2s - loss: 0.0321 - auc: 0.9996
## 33/75 [============>.................] - ETA: 2s - loss: 0.0312 - auc: 0.9996
## 34/75 [============>.................] - ETA: 2s - loss: 0.0306 - auc: 0.9997
## 35/75 [=============>................] - ETA: 2s - loss: 0.0301 - auc: 0.9997
## 36/75 [=============>................] - ETA: 2s - loss: 0.0296 - auc: 0.9997
## 37/75 [=============>................] - ETA: 2s - loss: 0.0295 - auc: 0.9997
## 38/75 [==============>...............] - ETA: 2s - loss: 0.0289 - auc: 0.9997
## 39/75 [==============>...............] - ETA: 1s - loss: 0.0290 - auc: 0.9997
## 40/75 [===============>..............] - ETA: 1s - loss: 0.0299 - auc: 0.9997
## 41/75 [===============>..............] - ETA: 1s - loss: 0.0295 - auc: 0.9997
## 42/75 [===============>..............] - ETA: 1s - loss: 0.0291 - auc: 0.9997
## 43/75 [================>.............] - ETA: 1s - loss: 0.0286 - auc: 0.9997
## 44/75 [================>.............] - ETA: 1s - loss: 0.0280 - auc: 0.9997
## 45/75 [=================>............] - ETA: 1s - loss: 0.0324 - auc: 0.9995
## 46/75 [=================>............] - ETA: 1s - loss: 0.0319 - auc: 0.9996
```

```
## 47/75 [=================>............] - ETA: 1s - loss: 0.0314 - auc: 0.9996
## 48/75 [==================>...........] - ETA: 1s - loss: 0.0316 - auc: 0.9995
## 49/75 [==================>...........] - ETA: 1s - loss: 0.0313 - auc: 0.9995
## 50/75 [===================>..........] - ETA: 1s - loss: 0.0309 - auc: 0.9996
## 51/75 [===================>..........] - ETA: 1s - loss: 0.0306 - auc: 0.9996
## 52/75 [===================>..........] - ETA: 1s - loss: 0.0308 - auc: 0.9996
## 53/75 [====================>.........] - ETA: 1s - loss: 0.0304 - auc: 0.9996
## 54/75 [====================>.........] - ETA: 1s - loss: 0.0327 - auc: 0.9995
## 55/75 [=====================>........] - ETA: 1s - loss: 0.0327 - auc: 0.9995
## 56/75 [=====================>........] - ETA: 1s - loss: 0.0324 - auc: 0.9995
## 57/75 [=====================>........] - ETA: 0s - loss: 0.0322 - auc: 0.9995
## 58/75 [======================>.......] - ETA: 0s - loss: 0.0323 - auc: 0.9995
## 59/75 [======================>.......] - ETA: 0s - loss: 0.0322 - auc: 0.9995
## 60/75 [=======================>......] - ETA: 0s - loss: 0.0320 - auc: 0.9995
## 61/75 [=======================>......] - ETA: 0s - loss: 0.0317 - auc: 0.9995
## 62/75 [=======================>......] - ETA: 0s - loss: 0.0315 - auc: 0.9995
## 63/75 [========================>.....] - ETA: 0s - loss: 0.0334 - auc: 0.9994
## 64/75 [========================>.....] - ETA: 0s - loss: 0.0334 - auc: 0.9994
## 65/75 [=========================>....] - ETA: 0s - loss: 0.0330 - auc: 0.9994
## 66/75 [=========================>....] - ETA: 0s - loss: 0.0327 - auc: 0.9995
## 67/75 [=========================>....] - ETA: 0s - loss: 0.0324 - auc: 0.9995
## 68/75 [==========================>...] - ETA: 0s - loss: 0.0325 - auc: 0.9995
## 69/75 [==========================>...] - ETA: 0s - loss: 0.0321 - auc: 0.9995
## 70/75 [===========================>..] - ETA: 0s - loss: 0.0317 - auc: 0.9995
## 71/75 [===========================>..] - ETA: 0s - loss: 0.0317 - auc: 0.9995
## 72/75 [===========================>..] - ETA: 0s - loss: 0.0354 - auc: 0.9991
## 73/75 [============================>.] - ETA: 0s - loss: 0.0385 - auc: 0.9988
## 74/75 [============================>.] - ETA: 0s - loss: 0.0382 - auc: 0.9988
## 75/75 [==============================] - 4s 57ms/step - loss: 0.0397 - auc: 0.9987 - val_l
oss: 0.4914 - val_auc: 0.9981
##
## Adjusted OvR model Mn vs Pt:
## Epoch 1/25
##
##  1/49 [..............................] - ETA: 31s - loss: 1.0817 - auc: 0.2857
##  2/49 [>.............................] - ETA: 2s - loss: 0.8681 - auc: 0.5586
##  3/49 [>.............................] - ETA: 2s - loss: 0.6711 - auc: 0.7483
##  4/49 [=>............................] - ETA: 2s - loss: 0.6058 - auc: 0.8079
##  5/49 [==>...........................] - ETA: 2s - loss: 0.6046 - auc: 0.8384
##  6/49 [==>...........................] - ETA: 2s - loss: 0.5743 - auc: 0.8578
##  7/49 [===>..........................] - ETA: 2s - loss: 0.5282 - auc: 0.8721
##  8/49 [===>..........................] - ETA: 2s - loss: 0.5063 - auc: 0.8805
##  9/49 [====>.........................] - ETA: 2s - loss: 0.4675 - auc: 0.8949
## 10/49 [=====>........................] - ETA: 2s - loss: 0.4440 - auc: 0.9034
## 11/49 [=====>........................] - ETA: 2s - loss: 0.4565 - auc: 0.8907
## 12/49 [======>.......................] - ETA: 2s - loss: 0.4394 - auc: 0.8907
## 13/49 [======>.......................] - ETA: 1s - loss: 0.4215 - auc: 0.9013
## 14/49 [=======>......................] - ETA: 1s - loss: 0.4138 - auc: 0.9075
## 15/49 [========>.....................] - ETA: 1s - loss: 0.3986 - auc: 0.9158
## 16/49 [========>.....................] - ETA: 1s - loss: 0.3948 - auc: 0.9182
## 17/49 [=========>....................] - ETA: 1s - loss: 0.3870 - auc: 0.9203
## 18/49 [==========>...................] - ETA: 1s - loss: 0.3870 - auc: 0.9230
## 19/49 [==========>...................] - ETA: 1s - loss: 0.3975 - auc: 0.9209
## 20/49 [===========>..................] - ETA: 1s - loss: 0.3906 - auc: 0.9228
## 21/49 [===========>..................] - ETA: 1s - loss: 0.3767 - auc: 0.9278
## 22/49 [============>.................] - ETA: 1s - loss: 0.3638 - auc: 0.9330
```

```
## 23/49 [=============>................] - ETA: 1s - loss: 0.3570 - auc: 0.9336
## 24/49 [=============>................] - ETA: 1s - loss: 0.3559 - auc: 0.9353
## 25/49 [==============>...............] - ETA: 1s - loss: 0.3466 - auc: 0.9379
## 26/49 [==============>...............] - ETA: 1s - loss: 0.3507 - auc: 0.9369
## 27/49 [===============>..............] - ETA: 1s - loss: 0.3453 - auc: 0.9396
## 28/49 [================>.............] - ETA: 1s - loss: 0.3379 - auc: 0.9416
## 29/49 [================>.............] - ETA: 1s - loss: 0.3326 - auc: 0.9440
## 30/49 [=================>............] - ETA: 1s - loss: 0.3328 - auc: 0.9433
## 31/49 [=================>............] - ETA: 0s - loss: 0.3253 - auc: 0.9459
## 32/49 [==================>...........] - ETA: 0s - loss: 0.3212 - auc: 0.9478
## 33/49 [===================>..........] - ETA: 0s - loss: 0.3168 - auc: 0.9482
## 34/49 [===================>..........] - ETA: 0s - loss: 0.3226 - auc: 0.9440
## 35/49 [====================>.........] - ETA: 0s - loss: 0.3207 - auc: 0.9454
## 36/49 [=====================>........] - ETA: 0s - loss: 0.3199 - auc: 0.9461
## 37/49 [=====================>........] - ETA: 0s - loss: 0.3171 - auc: 0.9471
## 38/49 [======================>.......] - ETA: 0s - loss: 0.3184 - auc: 0.9471
## 39/49 [======================>.......] - ETA: 0s - loss: 0.3203 - auc: 0.9451
## 40/49 [=======================>......] - ETA: 0s - loss: 0.3178 - auc: 0.9448
## 41/49 [========================>.....] - ETA: 0s - loss: 0.3162 - auc: 0.9460
## 42/49 [========================>.....] - ETA: 0s - loss: 0.3117 - auc: 0.9467
## 43/49 [=========================>....] - ETA: 0s - loss: 0.3063 - auc: 0.9484
## 44/49 [=========================>....] - ETA: 0s - loss: 0.3042 - auc: 0.9483
## 45/49 [==========================>...] - ETA: 0s - loss: 0.3011 - auc: 0.9499
## 46/49 [===========================>..] - ETA: 0s - loss: 0.2970 - auc: 0.9512
## 47/49 [===========================>..] - ETA: 0s - loss: 0.2954 - auc: 0.9516
## 48/49 [============================>.] - ETA: 0s - loss: 0.2920 - auc: 0.9529
## 49/49 [==============================] - 4s 61ms/step - loss: 0.2895 - auc: 0.9533 - val_l
oss: 0.9101 - val_auc: 0.8191
## Epoch 2/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0708 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.2277 - auc: 0.9683
##  3/49 [>.............................] - ETA: 2s - loss: 0.2066 - auc: 0.9757
##  4/49 [=>............................] - ETA: 2s - loss: 0.2260 - auc: 0.9702
##  5/49 [==>...........................] - ETA: 2s - loss: 0.3251 - auc: 0.9484
##  6/49 [==>...........................] - ETA: 2s - loss: 0.3296 - auc: 0.9523
##  7/49 [===>..........................] - ETA: 2s - loss: 0.3035 - auc: 0.9607
##  8/49 [===>..........................] - ETA: 2s - loss: 0.2834 - auc: 0.9643
##  9/49 [====>.........................] - ETA: 2s - loss: 0.2690 - auc: 0.9692
## 10/49 [=====>........................] - ETA: 2s - loss: 0.2473 - auc: 0.9734
## 11/49 [=====>........................] - ETA: 2s - loss: 0.2391 - auc: 0.9726
## 12/49 [======>.......................] - ETA: 2s - loss: 0.2312 - auc: 0.9706
## 13/49 [======>.......................] - ETA: 1s - loss: 0.2377 - auc: 0.9712
## 14/49 [=======>......................] - ETA: 1s - loss: 0.2249 - auc: 0.9740
## 15/49 [========>.....................] - ETA: 1s - loss: 0.2202 - auc: 0.9733
## 16/49 [========>.....................] - ETA: 1s - loss: 0.2191 - auc: 0.9739
## 17/49 [=========>....................] - ETA: 1s - loss: 0.2175 - auc: 0.9730
## 18/49 [==========>...................] - ETA: 1s - loss: 0.2290 - auc: 0.9706
## 19/49 [==========>...................] - ETA: 1s - loss: 0.2364 - auc: 0.9696
## 20/49 [===========>..................] - ETA: 1s - loss: 0.2307 - auc: 0.9702
## 21/49 [===========>..................] - ETA: 1s - loss: 0.2314 - auc: 0.9701
## 22/49 [============>.................] - ETA: 1s - loss: 0.2253 - auc: 0.9713
## 23/49 [=============>................] - ETA: 1s - loss: 0.2302 - auc: 0.9693
## 24/49 [=============>................] - ETA: 1s - loss: 0.2270 - auc: 0.9703
## 25/49 [==============>...............] - ETA: 1s - loss: 0.2201 - auc: 0.9723
## 26/49 [==============>...............] - ETA: 1s - loss: 0.2179 - auc: 0.9727
```

```
## 27/49 [===============>..............] - ETA: 1s - loss: 0.2159 - auc: 0.9729
## 28/49 [================>.............] - ETA: 1s - loss: 0.2112 - auc: 0.9737
## 29/49 [================>.............] - ETA: 1s - loss: 0.2056 - auc: 0.9751
## 30/49 [=================>............] - ETA: 1s - loss: 0.2068 - auc: 0.9743
## 31/49 [=================>............] - ETA: 0s - loss: 0.2016 - auc: 0.9756
## 32/49 [==================>...........] - ETA: 0s - loss: 0.2026 - auc: 0.9747
## 33/49 [===================>..........] - ETA: 0s - loss: 0.1995 - auc: 0.9756
## 34/49 [===================>..........] - ETA: 0s - loss: 0.1977 - auc: 0.9760
## 35/49 [====================>.........] - ETA: 0s - loss: 0.1966 - auc: 0.9762
## 36/49 [=====================>........] - ETA: 0s - loss: 0.1931 - auc: 0.9774
## 37/49 [=====================>........] - ETA: 0s - loss: 0.1909 - auc: 0.9781
## 38/49 [======================>.......] - ETA: 0s - loss: 0.1877 - auc: 0.9788
## 39/49 [======================>.......] - ETA: 0s - loss: 0.1903 - auc: 0.9780
## 40/49 [=======================>......] - ETA: 0s - loss: 0.1875 - auc: 0.9788
## 41/49 [========================>.....] - ETA: 0s - loss: 0.1855 - auc: 0.9791
## 42/49 [========================>.....] - ETA: 0s - loss: 0.1863 - auc: 0.9790
## 43/49 [=========================>....] - ETA: 0s - loss: 0.1871 - auc: 0.9787
## 44/49 [=========================>....] - ETA: 0s - loss: 0.1845 - auc: 0.9794
## 45/49 [==========================>...] - ETA: 0s - loss: 0.1811 - auc: 0.9801
## 46/49 [===========================>..] - ETA: 0s - loss: 0.1832 - auc: 0.9797
## 47/49 [===========================>..] - ETA: 0s - loss: 0.1825 - auc: 0.9801
## 48/49 [============================>.] - ETA: 0s - loss: 0.1863 - auc: 0.9794
## 49/49 [==============================] - 3s 57ms/step - loss: 0.1856 - auc: 0.9795 - val_l
oss: 1.0433 - val_auc: 0.8662
## Epoch 3/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0478 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0677 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0776 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0746 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.1147 - auc: 0.9969
##  6/49 [==>...........................] - ETA: 2s - loss: 0.1406 - auc: 0.9974
##  7/49 [===>..........................] - ETA: 2s - loss: 0.1490 - auc: 0.9950
##  8/49 [===>..........................] - ETA: 2s - loss: 0.1506 - auc: 0.9946
##  9/49 [====>.........................] - ETA: 2s - loss: 0.1437 - auc: 0.9955
## 10/49 [=====>........................] - ETA: 2s - loss: 0.1426 - auc: 0.9959
## 11/49 [=====>........................] - ETA: 2s - loss: 0.1708 - auc: 0.9856
## 12/49 [======>.......................] - ETA: 2s - loss: 0.1858 - auc: 0.9815
## 13/49 [======>.......................] - ETA: 1s - loss: 0.1747 - auc: 0.9842
## 14/49 [=======>......................] - ETA: 1s - loss: 0.1760 - auc: 0.9844
## 15/49 [========>.....................] - ETA: 1s - loss: 0.1716 - auc: 0.9857
## 16/49 [========>.....................] - ETA: 1s - loss: 0.1689 - auc: 0.9862
## 17/49 [=========>....................] - ETA: 1s - loss: 0.1666 - auc: 0.9862
## 18/49 [==========>...................] - ETA: 1s - loss: 0.1678 - auc: 0.9864
## 19/49 [==========>...................] - ETA: 1s - loss: 0.1676 - auc: 0.9865
## 20/49 [===========>..................] - ETA: 1s - loss: 0.1639 - auc: 0.9872
## 21/49 [===========>..................] - ETA: 1s - loss: 0.1847 - auc: 0.9818
## 22/49 [============>.................] - ETA: 1s - loss: 0.1931 - auc: 0.9794
## 23/49 [=============>................] - ETA: 1s - loss: 0.1890 - auc: 0.9806
## 24/49 [=============>................] - ETA: 1s - loss: 0.1946 - auc: 0.9796
## 25/49 [==============>...............] - ETA: 1s - loss: 0.1908 - auc: 0.9802
## 26/49 [==============>...............] - ETA: 1s - loss: 0.1932 - auc: 0.9797
## 27/49 [===============>..............] - ETA: 1s - loss: 0.1956 - auc: 0.9796
## 28/49 [================>.............] - ETA: 1s - loss: 0.1966 - auc: 0.9793
## 29/49 [================>.............] - ETA: 1s - loss: 0.1949 - auc: 0.9801
## 30/49 [=================>............] - ETA: 1s - loss: 0.1914 - auc: 0.9807
```

```
## 31/49 [=================>............] - ETA: 0s - loss: 0.1873 - auc: 0.9814
## 32/49 [==================>...........] - ETA: 0s - loss: 0.1909 - auc: 0.9799
## 33/49 [===================>..........] - ETA: 0s - loss: 0.1886 - auc: 0.9803
## 34/49 [===================>..........] - ETA: 0s - loss: 0.1876 - auc: 0.9806
## 35/49 [====================>.........] - ETA: 0s - loss: 0.1862 - auc: 0.9808
## 36/49 [=====================>........] - ETA: 0s - loss: 0.1839 - auc: 0.9812
## 37/49 [=====================>........] - ETA: 0s - loss: 0.1830 - auc: 0.9811
## 38/49 [======================>.......] - ETA: 0s - loss: 0.1799 - auc: 0.9818
## 39/49 [======================>.......] - ETA: 0s - loss: 0.1766 - auc: 0.9825
## 40/49 [=======================>......] - ETA: 0s - loss: 0.1739 - auc: 0.9831
## 41/49 [========================>.....] - ETA: 0s - loss: 0.1727 - auc: 0.9834
## 42/49 [========================>.....] - ETA: 0s - loss: 0.1701 - auc: 0.9840
## 43/49 [=========================>....] - ETA: 0s - loss: 0.1692 - auc: 0.9842
## 44/49 [=========================>....] - ETA: 0s - loss: 0.1671 - auc: 0.9847
## 45/49 [==========================>...] - ETA: 0s - loss: 0.1655 - auc: 0.9851
## 46/49 [===========================>..] - ETA: 0s - loss: 0.1648 - auc: 0.9851
## 47/49 [===========================>..] - ETA: 0s - loss: 0.1660 - auc: 0.9847
## 48/49 [============================>.] - ETA: 0s - loss: 0.1635 - auc: 0.9852
## 49/49 [==============================] - 3s 57ms/step - loss: 0.1627 - auc: 0.9854 - val_l
oss: 1.2780 - val_auc: 0.8515
## Epoch 4/25
## 
##  1/49 [..............................] - ETA: 2s - loss: 0.1772 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.1454 - auc: 0.9913
##  3/49 [>.............................] - ETA: 2s - loss: 0.1337 - auc: 0.9929
##  4/49 [=>............................] - ETA: 2s - loss: 0.1110 - auc: 0.9960
##  5/49 [==>...........................] - ETA: 2s - loss: 0.1207 - auc: 0.9962
##  6/49 [==>...........................] - ETA: 2s - loss: 0.1140 - auc: 0.9974
##  7/49 [===>..........................] - ETA: 2s - loss: 0.1111 - auc: 0.9974
##  8/49 [===>..........................] - ETA: 2s - loss: 0.1069 - auc: 0.9976
##  9/49 [====>.........................] - ETA: 2s - loss: 0.1150 - auc: 0.9956
## 10/49 [=====>........................] - ETA: 2s - loss: 0.1162 - auc: 0.9954
## 11/49 [=====>........................] - ETA: 2s - loss: 0.1206 - auc: 0.9947
## 12/49 [======>.......................] - ETA: 2s - loss: 0.1171 - auc: 0.9954
## 13/49 [======>.......................] - ETA: 1s - loss: 0.1158 - auc: 0.9955
## 14/49 [=======>......................] - ETA: 1s - loss: 0.1333 - auc: 0.9922
## 15/49 [========>.....................] - ETA: 1s - loss: 0.1265 - auc: 0.9932
## 16/49 [========>.....................] - ETA: 1s - loss: 0.1309 - auc: 0.9926
## 17/49 [=========>....................] - ETA: 1s - loss: 0.1362 - auc: 0.9917
## 18/49 [==========>...................] - ETA: 1s - loss: 0.1355 - auc: 0.9918
## 19/49 [==========>...................] - ETA: 1s - loss: 0.1375 - auc: 0.9915
## 20/49 [===========>..................] - ETA: 1s - loss: 0.1341 - auc: 0.9919
## 21/49 [===========>..................] - ETA: 1s - loss: 0.1310 - auc: 0.9925
## 22/49 [============>.................] - ETA: 1s - loss: 0.1312 - auc: 0.9922
## 23/49 [=============>................] - ETA: 1s - loss: 0.1293 - auc: 0.9927
## 24/49 [=============>................] - ETA: 1s - loss: 0.1264 - auc: 0.9932
## 25/49 [==============>...............] - ETA: 1s - loss: 0.1226 - auc: 0.9937
## 26/49 [==============>...............] - ETA: 1s - loss: 0.1208 - auc: 0.9939
## 27/49 [===============>..............] - ETA: 1s - loss: 0.1180 - auc: 0.9943
## 28/49 [================>.............] - ETA: 1s - loss: 0.1190 - auc: 0.9945
## 29/49 [================>.............] - ETA: 1s - loss: 0.1177 - auc: 0.9947
## 30/49 [=================>............] - ETA: 1s - loss: 0.1206 - auc: 0.9937
## 31/49 [=================>............] - ETA: 0s - loss: 0.1189 - auc: 0.9940
## 32/49 [==================>...........] - ETA: 0s - loss: 0.1289 - auc: 0.9929
## 33/49 [===================>..........] - ETA: 0s - loss: 0.1354 - auc: 0.9909
## 34/49 [===================>..........] - ETA: 0s - loss: 0.1346 - auc: 0.9910
```

```
## 35/49 [====================>.........] - ETA: 0s - loss: 0.1329 - auc: 0.9912
## 36/49 [=====================>........] - ETA: 0s - loss: 0.1306 - auc: 0.9915
## 37/49 [=====================>........] - ETA: 0s - loss: 0.1280 - auc: 0.9919
## 38/49 [======================>.......] - ETA: 0s - loss: 0.1264 - auc: 0.9922
## 39/49 [======================>.......] - ETA: 0s - loss: 0.1248 - auc: 0.9924
## 40/49 [=======================>......] - ETA: 0s - loss: 0.1275 - auc: 0.9919
## 41/49 [========================>.....] - ETA: 0s - loss: 0.1274 - auc: 0.9920
## 42/49 [========================>.....] - ETA: 0s - loss: 0.1261 - auc: 0.9922
## 43/49 [=========================>....] - ETA: 0s - loss: 0.1266 - auc: 0.9920
## 44/49 [=========================>....] - ETA: 0s - loss: 0.1246 - auc: 0.9924
## 45/49 [==========================>...] - ETA: 0s - loss: 0.1239 - auc: 0.9924
## 46/49 [===========================>..] - ETA: 0s - loss: 0.1252 - auc: 0.9922
## 47/49 [===========================>..] - ETA: 0s - loss: 0.1268 - auc: 0.9920
## 48/49 [============================>.] - ETA: 0s - loss: 0.1264 - auc: 0.9920
## 49/49 [==============================] - 3s 57ms/step - loss: 0.1263 - auc: 0.9921 - val_l
oss: 1.4049 - val_auc: 0.8959
## Epoch 5/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0927 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0813 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.1006 - auc: 0.9965
##  4/49 [=>............................] - ETA: 2s - loss: 0.1017 - auc: 0.9961
##  5/49 [==>...........................] - ETA: 2s - loss: 0.1284 - auc: 0.9902
##  6/49 [==>...........................] - ETA: 2s - loss: 0.1163 - auc: 0.9921
##  7/49 [===>..........................] - ETA: 2s - loss: 0.1119 - auc: 0.9938
##  8/49 [===>..........................] - ETA: 2s - loss: 0.1041 - auc: 0.9953
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0975 - auc: 0.9957
## 10/49 [=====>........................] - ETA: 2s - loss: 0.1145 - auc: 0.9927
## 11/49 [=====>........................] - ETA: 2s - loss: 0.1069 - auc: 0.9937
## 12/49 [======>.......................] - ETA: 2s - loss: 0.1148 - auc: 0.9930
## 13/49 [======>.......................] - ETA: 1s - loss: 0.1099 - auc: 0.9939
## 14/49 [=======>......................] - ETA: 1s - loss: 0.1072 - auc: 0.9942
## 15/49 [========>.....................] - ETA: 1s - loss: 0.1107 - auc: 0.9936
## 16/49 [========>.....................] - ETA: 1s - loss: 0.1088 - auc: 0.9939
## 17/49 [=========>....................] - ETA: 1s - loss: 0.1089 - auc: 0.9941
## 18/49 [==========>...................] - ETA: 1s - loss: 0.1150 - auc: 0.9934
## 19/49 [==========>...................] - ETA: 1s - loss: 0.1107 - auc: 0.9940
## 20/49 [===========>..................] - ETA: 1s - loss: 0.1121 - auc: 0.9938
## 21/49 [===========>..................] - ETA: 1s - loss: 0.1137 - auc: 0.9938
## 22/49 [============>.................] - ETA: 1s - loss: 0.1136 - auc: 0.9939
## 23/49 [=============>................] - ETA: 1s - loss: 0.1176 - auc: 0.9933
## 24/49 [=============>................] - ETA: 1s - loss: 0.1182 - auc: 0.9933
## 25/49 [==============>...............] - ETA: 1s - loss: 0.1199 - auc: 0.9929
## 26/49 [==============>...............] - ETA: 1s - loss: 0.1198 - auc: 0.9930
## 27/49 [===============>..............] - ETA: 1s - loss: 0.1187 - auc: 0.9931
## 28/49 [================>.............] - ETA: 1s - loss: 0.1157 - auc: 0.9936
## 29/49 [================>.............] - ETA: 1s - loss: 0.1150 - auc: 0.9937
## 30/49 [=================>............] - ETA: 1s - loss: 0.1176 - auc: 0.9934
## 31/49 [=================>............] - ETA: 0s - loss: 0.1165 - auc: 0.9937
## 32/49 [==================>...........] - ETA: 0s - loss: 0.1181 - auc: 0.9935
## 33/49 [===================>..........] - ETA: 0s - loss: 0.1180 - auc: 0.9935
## 34/49 [===================>..........] - ETA: 0s - loss: 0.1197 - auc: 0.9932
## 35/49 [====================>.........] - ETA: 0s - loss: 0.1197 - auc: 0.9932
## 36/49 [=====================>........] - ETA: 0s - loss: 0.1221 - auc: 0.9928
## 37/49 [=====================>........] - ETA: 0s - loss: 0.1199 - auc: 0.9931
## 38/49 [======================>.......] - ETA: 0s - loss: 0.1179 - auc: 0.9934
```

```
## 39/49 [======================>.......] - ETA: 0s - loss: 0.1162 - auc: 0.9936
## 40/49 [=======================>......] - ETA: 0s - loss: 0.1175 - auc: 0.9936
## 41/49 [========================>.....] - ETA: 0s - loss: 0.1164 - auc: 0.9938
## 42/49 [========================>.....] - ETA: 0s - loss: 0.1162 - auc: 0.9938
## 43/49 [=========================>....] - ETA: 0s - loss: 0.1168 - auc: 0.9937
## 44/49 [=========================>....] - ETA: 0s - loss: 0.1167 - auc: 0.9937
## 45/49 [==========================>...] - ETA: 0s - loss: 0.1161 - auc: 0.9939
## 46/49 [===========================>..] - ETA: 0s - loss: 0.1148 - auc: 0.9941
## 47/49 [===========================>..] - ETA: 0s - loss: 0.1173 - auc: 0.9938
## 48/49 [============================>.] - ETA: 0s - loss: 0.1157 - auc: 0.9940
## 49/49 [==============================] - 3s 57ms/step - loss: 0.1164 - auc: 0.9939 - val_l
oss: 1.4811 - val_auc: 0.9075
## Epoch 6/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0556 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0711 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0661 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0841 - auc: 0.9990
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0834 - auc: 0.9994
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0967 - auc: 0.9986
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0935 - auc: 0.9987
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0928 - auc: 0.9984
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0941 - auc: 0.9983
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0925 - auc: 0.9983
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0891 - auc: 0.9984
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0901 - auc: 0.9984
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0956 - auc: 0.9980
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0970 - auc: 0.9980
## 15/49 [========>.....................] - ETA: 1s - loss: 0.1013 - auc: 0.9973
## 16/49 [========>.....................] - ETA: 1s - loss: 0.1036 - auc: 0.9969
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0999 - auc: 0.9971
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0996 - auc: 0.9971
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0969 - auc: 0.9973
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0945 - auc: 0.9975
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0930 - auc: 0.9976
## 22/49 [============>.................] - ETA: 1s - loss: 0.1064 - auc: 0.9955
## 23/49 [=============>................] - ETA: 1s - loss: 0.1047 - auc: 0.9957
## 24/49 [=============>................] - ETA: 1s - loss: 0.1011 - auc: 0.9961
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0997 - auc: 0.9962
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0968 - auc: 0.9965
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0942 - auc: 0.9967
## 28/49 [================>.............] - ETA: 1s - loss: 0.0917 - auc: 0.9970
## 29/49 [================>.............] - ETA: 1s - loss: 0.0929 - auc: 0.9967
## 30/49 [=================>............] - ETA: 1s - loss: 0.0930 - auc: 0.9967
## 31/49 [=================>............] - ETA: 0s - loss: 0.0984 - auc: 0.9962
## 32/49 [==================>...........] - ETA: 0s - loss: 0.1012 - auc: 0.9957
## 33/49 [===================>..........] - ETA: 0s - loss: 0.1032 - auc: 0.9955
## 34/49 [===================>..........] - ETA: 0s - loss: 0.1029 - auc: 0.9956
## 35/49 [====================>.........] - ETA: 0s - loss: 0.1024 - auc: 0.9956
## 36/49 [=====================>........] - ETA: 0s - loss: 0.1014 - auc: 0.9957
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0993 - auc: 0.9959
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0982 - auc: 0.9961
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0962 - auc: 0.9963
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0961 - auc: 0.9963
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0963 - auc: 0.9963
## 42/49 [========================>.....] - ETA: 0s - loss: 0.1001 - auc: 0.9956
```

```
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0983 - auc: 0.9958
## 44/49 [=========================>....] - ETA: 0s - loss: 0.1035 - auc: 0.9953
## 45/49 [==========================>...] - ETA: 0s - loss: 0.1023 - auc: 0.9954
## 46/49 [===========================>..] - ETA: 0s - loss: 0.1037 - auc: 0.9954
## 47/49 [===========================>..] - ETA: 0s - loss: 0.1030 - auc: 0.9955
## 48/49 [============================>.] - ETA: 0s - loss: 0.1017 - auc: 0.9957
## 49/49 [==============================] - 3s 57ms/step - loss: 0.1011 - auc: 0.9957 - val_l
oss: 1.8566 - val_auc: 0.9392
## Epoch 7/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.1125 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.1161 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0999 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0809 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0693 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0751 - auc: 0.9996
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0978 - auc: 0.9957
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0975 - auc: 0.9955
##  9/49 [====>.........................] - ETA: 2s - loss: 0.1011 - auc: 0.9949
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0930 - auc: 0.9959
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0879 - auc: 0.9964
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0854 - auc: 0.9969
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0809 - auc: 0.9974
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0816 - auc: 0.9972
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0771 - auc: 0.9976
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0752 - auc: 0.9979
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0742 - auc: 0.9981
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0746 - auc: 0.9982
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0722 - auc: 0.9984
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0696 - auc: 0.9986
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0729 - auc: 0.9983
## 22/49 [============>.................] - ETA: 1s - loss: 0.0739 - auc: 0.9983
## 23/49 [=============>................] - ETA: 1s - loss: 0.0746 - auc: 0.9982
## 24/49 [=============>................] - ETA: 1s - loss: 0.0761 - auc: 0.9982
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0808 - auc: 0.9977
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0795 - auc: 0.9979
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0797 - auc: 0.9978
## 28/49 [================>.............] - ETA: 1s - loss: 0.0805 - auc: 0.9978
## 29/49 [================>.............] - ETA: 1s - loss: 0.0810 - auc: 0.9978
## 30/49 [=================>............] - ETA: 1s - loss: 0.0814 - auc: 0.9977
## 31/49 [=================>............] - ETA: 0s - loss: 0.0803 - auc: 0.9978
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0787 - auc: 0.9979
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0779 - auc: 0.9980
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0801 - auc: 0.9979
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0817 - auc: 0.9977
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0805 - auc: 0.9978
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0794 - auc: 0.9979
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0791 - auc: 0.9979
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0812 - auc: 0.9978
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0820 - auc: 0.9978
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0810 - auc: 0.9979
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0811 - auc: 0.9979
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0839 - auc: 0.9978
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0831 - auc: 0.9979
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0817 - auc: 0.9980
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0808 - auc: 0.9981
```

```
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0803 - auc: 0.9981
## 48/49 [============================>.] - ETA: 0s - loss: 0.0791 - auc: 0.9982
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0781 - auc: 0.9983 - val_l
oss: 1.6155 - val_auc: 0.9604
## Epoch 8/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0372 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0282 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0403 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0873 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0818 - auc: 0.9981
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0777 - auc: 0.9987
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0691 - auc: 0.9990
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0653 - auc: 0.9993
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0919 - auc: 0.9982
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0848 - auc: 0.9986
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0963 - auc: 0.9961
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0959 - auc: 0.9959
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0966 - auc: 0.9957
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0973 - auc: 0.9953
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0965 - auc: 0.9952
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0923 - auc: 0.9957
## 17/49 [=========>....................] - ETA: 1s - loss: 0.1031 - auc: 0.9943
## 18/49 [==========>...................] - ETA: 1s - loss: 0.1019 - auc: 0.9947
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0999 - auc: 0.9949
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0966 - auc: 0.9954
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0951 - auc: 0.9956
## 22/49 [============>.................] - ETA: 1s - loss: 0.0937 - auc: 0.9958
## 23/49 [=============>................] - ETA: 1s - loss: 0.0977 - auc: 0.9953
## 24/49 [=============>................] - ETA: 1s - loss: 0.1036 - auc: 0.9948
## 25/49 [==============>...............] - ETA: 1s - loss: 0.1035 - auc: 0.9947
## 26/49 [==============>...............] - ETA: 1s - loss: 0.1002 - auc: 0.9951
## 27/49 [===============>..............] - ETA: 1s - loss: 0.1003 - auc: 0.9953
## 28/49 [================>.............] - ETA: 1s - loss: 0.1003 - auc: 0.9953
## 29/49 [================>.............] - ETA: 1s - loss: 0.0983 - auc: 0.9955
## 30/49 [=================>............] - ETA: 1s - loss: 0.0954 - auc: 0.9958
## 31/49 [=================>............] - ETA: 0s - loss: 0.0944 - auc: 0.9959
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0950 - auc: 0.9959
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0935 - auc: 0.9961
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0944 - auc: 0.9959
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0933 - auc: 0.9961
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0922 - auc: 0.9962
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0943 - auc: 0.9957
## 38/49 [======================>.......] - ETA: 0s - loss: 0.1022 - auc: 0.9941
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0999 - auc: 0.9944
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0996 - auc: 0.9945
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0988 - auc: 0.9946
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0984 - auc: 0.9947
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0968 - auc: 0.9949
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0958 - auc: 0.9950
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0948 - auc: 0.9951
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0934 - auc: 0.9953
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0937 - auc: 0.9953
## 48/49 [============================>.] - ETA: 0s - loss: 0.0927 - auc: 0.9955
## 49/49 [==============================] - ETA: 0s - loss: 0.0915 - auc: 0.9956
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0915 - auc: 0.9956 - val_l
```

```
oss: 1.2669 - val_auc: 0.9587
## Epoch 9/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0421 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0480 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0421 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0677 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0694 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0725 - auc: 0.9996
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0789 - auc: 0.9990
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0751 - auc: 0.9993
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0786 - auc: 0.9986
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0731 - auc: 0.9989
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0698 - auc: 0.9991
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0687 - auc: 0.9992
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0728 - auc: 0.9988
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0690 - auc: 0.9990
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0661 - auc: 0.9991
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0638 - auc: 0.9992
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0620 - auc: 0.9993
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0600 - auc: 0.9994
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0747 - auc: 0.9974
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0729 - auc: 0.9975
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0724 - auc: 0.9976
## 22/49 [============>.................] - ETA: 1s - loss: 0.0708 - auc: 0.9978
## 23/49 [=============>................] - ETA: 1s - loss: 0.0684 - auc: 0.9979
## 24/49 [=============>................] - ETA: 1s - loss: 0.0667 - auc: 0.9980
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0695 - auc: 0.9978
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0701 - auc: 0.9978
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0705 - auc: 0.9978
## 28/49 [================>.............] - ETA: 1s - loss: 0.0684 - auc: 0.9980
## 29/49 [================>.............] - ETA: 1s - loss: 0.0668 - auc: 0.9980
## 30/49 [=================>............] - ETA: 1s - loss: 0.0657 - auc: 0.9982
## 31/49 [=================>............] - ETA: 0s - loss: 0.0643 - auc: 0.9982
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0633 - auc: 0.9983
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0625 - auc: 0.9983
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0614 - auc: 0.9984
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0604 - auc: 0.9985
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0590 - auc: 0.9986
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0745 - auc: 0.9957
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0737 - auc: 0.9959
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0727 - auc: 0.9959
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0714 - auc: 0.9961
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0702 - auc: 0.9962
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0729 - auc: 0.9961
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0716 - auc: 0.9962
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0711 - auc: 0.9963
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0717 - auc: 0.9963
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0716 - auc: 0.9964
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0709 - auc: 0.9965
## 48/49 [============================>.] - ETA: 0s - loss: 0.0703 - auc: 0.9965
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0718 - auc: 0.9965 - val_l
oss: 0.8096 - val_auc: 0.9863
## Epoch 10/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0357 - auc: 1.0000
```

```
##  2/49 [>.............................] - ETA: 2s - loss: 0.0428 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0599 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0630 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0582 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0577 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0527 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0506 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0639 - auc: 0.9998
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0646 - auc: 0.9998
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0687 - auc: 0.9997
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0675 - auc: 0.9998
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0633 - auc: 0.9998
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0647 - auc: 0.9998
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0634 - auc: 0.9998
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0606 - auc: 0.9998
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0580 - auc: 0.9998
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0557 - auc: 0.9999
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0635 - auc: 0.9994
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0629 - auc: 0.9993
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0615 - auc: 0.9993
## 22/49 [============>.................] - ETA: 1s - loss: 0.0624 - auc: 0.9993
## 23/49 [=============>................] - ETA: 1s - loss: 0.0635 - auc: 0.9992
## 24/49 [=============>................] - ETA: 1s - loss: 0.0617 - auc: 0.9993
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0598 - auc: 0.9993
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0592 - auc: 0.9994
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0582 - auc: 0.9994
## 28/49 [================>.............] - ETA: 1s - loss: 0.0582 - auc: 0.9994
## 29/49 [================>.............] - ETA: 1s - loss: 0.0566 - auc: 0.9994
## 30/49 [=================>............] - ETA: 1s - loss: 0.0579 - auc: 0.9994
## 31/49 [=================>............] - ETA: 0s - loss: 0.0571 - auc: 0.9994
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0564 - auc: 0.9994
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0555 - auc: 0.9994
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0566 - auc: 0.9994
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0557 - auc: 0.9994
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0547 - auc: 0.9994
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0535 - auc: 0.9995
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0533 - auc: 0.9995
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0536 - auc: 0.9994
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0534 - auc: 0.9995
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0531 - auc: 0.9995
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0522 - auc: 0.9995
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0521 - auc: 0.9995
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0529 - auc: 0.9995
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0522 - auc: 0.9995
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0535 - auc: 0.9995
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0536 - auc: 0.9995
## 48/49 [============================>.] - ETA: 0s - loss: 0.0534 - auc: 0.9995
## 49/49 [==============================] - ETA: 0s - loss: 0.0532 - auc: 0.9995
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0532 - auc: 0.9995 - val_l
oss: 0.6487 - val_auc: 0.9805
## Epoch 11/25
## 
##  1/49 [..............................] - ETA: 2s - loss: 0.0630 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0751 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0625 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0508 - auc: 1.0000
```

```
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0630 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0588 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0531 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0472 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0481 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0560 - auc: 0.9997
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0553 - auc: 0.9996
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0538 - auc: 0.9997
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0515 - auc: 0.9997
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0492 - auc: 0.9998
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0471 - auc: 0.9998
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0467 - auc: 0.9998
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0479 - auc: 0.9997
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0483 - auc: 0.9997
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0463 - auc: 0.9997
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0451 - auc: 0.9998
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0443 - auc: 0.9998
## 22/49 [============>.................] - ETA: 1s - loss: 0.0434 - auc: 0.9998
## 23/49 [=============>................] - ETA: 1s - loss: 0.0453 - auc: 0.9998
## 24/49 [=============>................] - ETA: 1s - loss: 0.0438 - auc: 0.9998
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0441 - auc: 0.9998
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0434 - auc: 0.9998
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0428 - auc: 0.9998
## 28/49 [================>.............] - ETA: 1s - loss: 0.0423 - auc: 0.9998
## 29/49 [================>.............] - ETA: 1s - loss: 0.0429 - auc: 0.9999
## 30/49 [=================>............] - ETA: 1s - loss: 0.0444 - auc: 0.9997
## 31/49 [=================>............] - ETA: 0s - loss: 0.0450 - auc: 0.9998
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0492 - auc: 0.9994
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0493 - auc: 0.9994
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0493 - auc: 0.9995
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0481 - auc: 0.9995
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0481 - auc: 0.9995
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0477 - auc: 0.9995
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0467 - auc: 0.9995
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0467 - auc: 0.9995
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0466 - auc: 0.9995
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0463 - auc: 0.9996
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0455 - auc: 0.9996
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0447 - auc: 0.9996
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0447 - auc: 0.9996
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0442 - auc: 0.9996
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0435 - auc: 0.9996
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0430 - auc: 0.9997
## 48/49 [============================>.] - ETA: 0s - loss: 0.0425 - auc: 0.9997
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0442 - auc: 0.9996 - val_l
oss: 0.4853 - val_auc: 0.9815
## Epoch 12/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0074 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0096 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0109 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0238 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0337 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0317 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0302 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0504 - auc: 0.9988
```

```
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0471 - auc: 0.9990
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0441 - auc: 0.9992
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0415 - auc: 0.9994
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0401 - auc: 0.9993
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0415 - auc: 0.9992
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0408 - auc: 0.9992
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0397 - auc: 0.9993
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0395 - auc: 0.9993
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0407 - auc: 0.9994
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0394 - auc: 0.9994
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0381 - auc: 0.9995
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0367 - auc: 0.9996
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0364 - auc: 0.9996
## 22/49 [============>.................] - ETA: 1s - loss: 0.0372 - auc: 0.9996
## 23/49 [=============>................] - ETA: 1s - loss: 0.0365 - auc: 0.9996
## 24/49 [=============>................] - ETA: 1s - loss: 0.0356 - auc: 0.9997
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0347 - auc: 0.9997
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0343 - auc: 0.9997
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0342 - auc: 0.9997
## 28/49 [================>.............] - ETA: 1s - loss: 0.0344 - auc: 0.9997
## 29/49 [================>.............] - ETA: 1s - loss: 0.0337 - auc: 0.9997
## 30/49 [=================>............] - ETA: 1s - loss: 0.0349 - auc: 0.9997
## 31/49 [=================>............] - ETA: 0s - loss: 0.0340 - auc: 0.9997
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0351 - auc: 0.9997
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0360 - auc: 0.9997
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0353 - auc: 0.9997
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0345 - auc: 0.9998
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0340 - auc: 0.9998
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0340 - auc: 0.9998
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0342 - auc: 0.9997
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0339 - auc: 0.9998
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0345 - auc: 0.9998
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0344 - auc: 0.9998
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0339 - auc: 0.9998
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0335 - auc: 0.9998
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0331 - auc: 0.9998
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0328 - auc: 0.9998
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0323 - auc: 0.9998
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0324 - auc: 0.9998
## 48/49 [============================>.] - ETA: 0s - loss: 0.0325 - auc: 0.9998
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0324 - auc: 0.9998 - val_l
oss: 0.3413 - val_auc: 0.9989
## Epoch 13/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0131 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0519 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0453 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0518 - auc: 0.9990
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0541 - auc: 0.9994
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0488 - auc: 0.9996
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0441 - auc: 0.9997
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0410 - auc: 0.9998
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0377 - auc: 0.9998
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0375 - auc: 0.9998
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0374 - auc: 0.9999
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0348 - auc: 0.9999
```

```
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0335 - auc: 0.9999
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0332 - auc: 0.9999
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0346 - auc: 0.9998
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0329 - auc: 0.9998
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0352 - auc: 0.9998
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0375 - auc: 0.9998
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0361 - auc: 0.9998
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0361 - auc: 0.9998
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0350 - auc: 0.9998
## 22/49 [============>.................] - ETA: 1s - loss: 0.0346 - auc: 0.9998
## 23/49 [=============>................] - ETA: 1s - loss: 0.0334 - auc: 0.9999
## 24/49 [=============>................] - ETA: 1s - loss: 0.0328 - auc: 0.9999
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0326 - auc: 0.9999
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0327 - auc: 0.9999
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0320 - auc: 0.9999
## 28/49 [================>.............] - ETA: 1s - loss: 0.0317 - auc: 0.9999
## 29/49 [================>.............] - ETA: 1s - loss: 0.0309 - auc: 0.9999
## 30/49 [=================>............] - ETA: 1s - loss: 0.0313 - auc: 0.9999
## 31/49 [=================>............] - ETA: 0s - loss: 0.0327 - auc: 0.9999
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0349 - auc: 0.9998
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0344 - auc: 0.9998
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0364 - auc: 0.9997
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0364 - auc: 0.9997
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0356 - auc: 0.9997
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0361 - auc: 0.9997
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0372 - auc: 0.9997
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0386 - auc: 0.9997
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0379 - auc: 0.9997
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0373 - auc: 0.9997
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0394 - auc: 0.9997
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0389 - auc: 0.9997
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0385 - auc: 0.9997
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0387 - auc: 0.9997
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0383 - auc: 0.9997
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0385 - auc: 0.9997
## 48/49 [============================>.] - ETA: 0s - loss: 0.0384 - auc: 0.9997
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0380 - auc: 0.9997 - val_l
oss: 0.2714 - val_auc: 0.9994
## Epoch 14/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0711 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0461 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0853 - auc: 0.9965
##  4/49 [=>............................] - ETA: 2s - loss: 0.0667 - auc: 0.9980
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0642 - auc: 0.9987
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0763 - auc: 0.9969
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0668 - auc: 0.9978
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0610 - auc: 0.9983
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0583 - auc: 0.9984
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0551 - auc: 0.9987
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0564 - auc: 0.9988
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0585 - auc: 0.9988
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0550 - auc: 0.9990
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0517 - auc: 0.9991
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0525 - auc: 0.9990
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0504 - auc: 0.9991
```

```
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0492 - auc: 0.9992
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0486 - auc: 0.9992
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0484 - auc: 0.9993
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0467 - auc: 0.9994
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0470 - auc: 0.9994
## 22/49 [============>.................] - ETA: 1s - loss: 0.0451 - auc: 0.9995
## 23/49 [=============>................] - ETA: 1s - loss: 0.0459 - auc: 0.9995
## 24/49 [=============>................] - ETA: 1s - loss: 0.0467 - auc: 0.9995
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0458 - auc: 0.9995
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0444 - auc: 0.9995
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0434 - auc: 0.9996
## 28/49 [================>.............] - ETA: 1s - loss: 0.0424 - auc: 0.9996
## 29/49 [================>.............] - ETA: 1s - loss: 0.0421 - auc: 0.9996
## 30/49 [=================>............] - ETA: 1s - loss: 0.0415 - auc: 0.9997
## 31/49 [=================>............] - ETA: 0s - loss: 0.0409 - auc: 0.9997
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0399 - auc: 0.9997
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0390 - auc: 0.9997
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0399 - auc: 0.9996
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0396 - auc: 0.9997
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0387 - auc: 0.9997
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0382 - auc: 0.9997
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0376 - auc: 0.9997
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0368 - auc: 0.9997
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0366 - auc: 0.9997
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0359 - auc: 0.9998
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0377 - auc: 0.9997
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0382 - auc: 0.9997
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0378 - auc: 0.9997
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0375 - auc: 0.9997
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0377 - auc: 0.9997
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0384 - auc: 0.9997
## 48/49 [============================>.] - ETA: 0s - loss: 0.0409 - auc: 0.9996
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0403 - auc: 0.9996 - val_l
oss: 0.2431 - val_auc: 0.9994
## Epoch 15/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0137 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0582 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0522 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0433 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0531 - auc: 0.9984
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0557 - auc: 0.9980
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0509 - auc: 0.9986
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0467 - auc: 0.9989
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0492 - auc: 0.9989
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0465 - auc: 0.9991
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0468 - auc: 0.9992
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0457 - auc: 0.9993
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0458 - auc: 0.9994
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0434 - auc: 0.9995
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0408 - auc: 0.9995
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0402 - auc: 0.9996
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0398 - auc: 0.9996
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0392 - auc: 0.9997
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0380 - auc: 0.9997
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0372 - auc: 0.9997
```

```
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0363 - auc: 0.9998
## 22/49 [============>.................] - ETA: 1s - loss: 0.0724 - auc: 0.9949
## 23/49 [=============>................] - ETA: 1s - loss: 0.0704 - auc: 0.9952
## 24/49 [=============>................] - ETA: 1s - loss: 0.0683 - auc: 0.9955
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0666 - auc: 0.9958
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0644 - auc: 0.9960
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0625 - auc: 0.9962
## 28/49 [================>.............] - ETA: 1s - loss: 0.0607 - auc: 0.9963
## 29/49 [================>.............] - ETA: 1s - loss: 0.0599 - auc: 0.9964
## 30/49 [=================>............] - ETA: 1s - loss: 0.0594 - auc: 0.9965
## 31/49 [=================>............] - ETA: 0s - loss: 0.0577 - auc: 0.9967
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0567 - auc: 0.9968
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0556 - auc: 0.9969
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0544 - auc: 0.9970
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0534 - auc: 0.9971
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0528 - auc: 0.9972
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0519 - auc: 0.9972
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0516 - auc: 0.9973
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0510 - auc: 0.9973
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0503 - auc: 0.9974
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0499 - auc: 0.9974
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0489 - auc: 0.9975
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0481 - auc: 0.9975
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0481 - auc: 0.9975
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0498 - auc: 0.9976
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0499 - auc: 0.9976
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0492 - auc: 0.9977
## 48/49 [============================>.] - ETA: 0s - loss: 0.0485 - auc: 0.9977
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0483 - auc: 0.9977 - val_l
oss: 0.2064 - val_auc: 0.9994
## Epoch 16/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0118 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0128 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0466 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0362 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0349 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0371 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0357 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0331 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0310 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0292 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0282 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0381 - auc: 0.9996
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0378 - auc: 0.9996
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0429 - auc: 0.9994
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0422 - auc: 0.9994
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0432 - auc: 0.9993
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0438 - auc: 0.9994
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0418 - auc: 0.9995
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0404 - auc: 0.9995
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0388 - auc: 0.9996
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0378 - auc: 0.9996
## 22/49 [============>.................] - ETA: 1s - loss: 0.0374 - auc: 0.9996
## 23/49 [=============>................] - ETA: 1s - loss: 0.0365 - auc: 0.9997
## 24/49 [=============>................] - ETA: 1s - loss: 0.0353 - auc: 0.9997
```

```
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0343 - auc: 0.9997
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0334 - auc: 0.9997
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0335 - auc: 0.9998
## 28/49 [================>.............] - ETA: 1s - loss: 0.0325 - auc: 0.9998
## 29/49 [================>.............] - ETA: 1s - loss: 0.0323 - auc: 0.9998
## 30/49 [=================>............] - ETA: 1s - loss: 0.0319 - auc: 0.9998
## 31/49 [=================>............] - ETA: 0s - loss: 0.0318 - auc: 0.9998
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0310 - auc: 0.9998
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0308 - auc: 0.9998
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0301 - auc: 0.9999
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0295 - auc: 0.9999
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0290 - auc: 0.9999
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0285 - auc: 0.9999
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0281 - auc: 0.9999
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0276 - auc: 0.9999
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0278 - auc: 0.9999
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0275 - auc: 0.9999
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0273 - auc: 0.9999
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0282 - auc: 0.9999
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0279 - auc: 0.9999
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0290 - auc: 0.9999
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0285 - auc: 0.9999
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0283 - auc: 0.9999
## 48/49 [============================>.] - ETA: 0s - loss: 0.0279 - auc: 0.9999
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0278 - auc: 0.9999 - val_l
oss: 0.1568 - val_auc: 1.0000
## Epoch 17/25
## 
##  1/49 [..............................] - ETA: 2s - loss: 0.0343 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0238 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0262 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0224 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0301 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0270 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0247 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0224 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0226 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0210 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0242 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0226 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0217 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0207 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0205 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0209 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0209 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0200 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0207 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0202 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0199 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0223 - auc: 1.0000
## 23/49 [=============>................] - ETA: 1s - loss: 0.0228 - auc: 1.0000
## 24/49 [=============>................] - ETA: 1s - loss: 0.0223 - auc: 1.0000
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0226 - auc: 1.0000
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0225 - auc: 1.0000
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0223 - auc: 1.0000
## 28/49 [================>.............] - ETA: 1s - loss: 0.0217 - auc: 1.0000
```

```
## 29/49 [================>.............] - ETA: 1s - loss: 0.0213 - auc: 1.0000
## 30/49 [=================>............] - ETA: 1s - loss: 0.0209 - auc: 1.0000
## 31/49 [=================>............] - ETA: 0s - loss: 0.0217 - auc: 1.0000
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0218 - auc: 1.0000
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0216 - auc: 1.0000
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0214 - auc: 1.0000
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0214 - auc: 1.0000
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0217 - auc: 1.0000
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0222 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0218 - auc: 1.0000
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0214 - auc: 1.0000
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0212 - auc: 1.0000
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0211 - auc: 1.0000
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0215 - auc: 1.0000
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0223 - auc: 1.0000
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0220 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0228 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0230 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0226 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0226 - auc: 1.0000
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0224 - auc: 1.0000 - val_l
oss: 0.2429 - val_auc: 1.0000
## Epoch 18/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0051 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0041 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0056 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0208 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0175 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0186 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0168 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0152 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0190 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0203 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0212 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0199 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0186 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0178 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0176 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0185 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0176 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0170 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0162 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0303 - auc: 0.9996
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0305 - auc: 0.9996
## 22/49 [============>.................] - ETA: 1s - loss: 0.0299 - auc: 0.9996
## 23/49 [=============>................] - ETA: 1s - loss: 0.0313 - auc: 0.9996
## 24/49 [=============>................] - ETA: 1s - loss: 0.0312 - auc: 0.9996
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0304 - auc: 0.9996
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0295 - auc: 0.9997
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0289 - auc: 0.9997
## 28/49 [================>.............] - ETA: 1s - loss: 0.0289 - auc: 0.9997
## 29/49 [================>.............] - ETA: 1s - loss: 0.0282 - auc: 0.9997
## 30/49 [=================>............] - ETA: 1s - loss: 0.0310 - auc: 0.9997
## 31/49 [=================>............] - ETA: 0s - loss: 0.0318 - auc: 0.9997
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0333 - auc: 0.9997
```

```
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0327 - auc: 0.9997
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0338 - auc: 0.9996
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0333 - auc: 0.9996
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0327 - auc: 0.9997
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0320 - auc: 0.9997
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0326 - auc: 0.9997
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0331 - auc: 0.9997
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0324 - auc: 0.9997
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0318 - auc: 0.9997
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0311 - auc: 0.9997
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0306 - auc: 0.9997
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0309 - auc: 0.9997
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0303 - auc: 0.9997
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0298 - auc: 0.9997
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0294 - auc: 0.9998
## 48/49 [============================>.] - ETA: 0s - loss: 0.0293 - auc: 0.9998
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0289 - auc: 0.9998 - val_l
oss: 0.1666 - val_auc: 1.0000
## Epoch 19/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0113 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0090 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0154 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0161 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0212 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0237 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0217 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0247 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0233 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0216 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0201 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0190 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0206 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0212 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0254 - auc: 0.9999
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0240 - auc: 0.9999
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0227 - auc: 0.9999
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0217 - auc: 0.9999
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0209 - auc: 0.9999
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0209 - auc: 0.9999
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0217 - auc: 0.9999
## 22/49 [============>.................] - ETA: 1s - loss: 0.0212 - auc: 0.9999
## 23/49 [=============>................] - ETA: 1s - loss: 0.0208 - auc: 0.9999
## 24/49 [=============>................] - ETA: 1s - loss: 0.0209 - auc: 0.9999
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0207 - auc: 0.9999
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0204 - auc: 0.9999
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0199 - auc: 0.9999
## 28/49 [================>.............] - ETA: 1s - loss: 0.0195 - auc: 0.9999
## 29/49 [================>.............] - ETA: 1s - loss: 0.0190 - auc: 0.9999
## 30/49 [=================>............] - ETA: 1s - loss: 0.0186 - auc: 0.9999
## 31/49 [=================>............] - ETA: 0s - loss: 0.0181 - auc: 1.0000
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0188 - auc: 0.9999
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0184 - auc: 0.9999
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0181 - auc: 0.9999
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0181 - auc: 0.9999
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0187 - auc: 1.0000
```

```
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0185 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0184 - auc: 1.0000
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0182 - auc: 1.0000
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0183 - auc: 1.0000
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0182 - auc: 1.0000
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0178 - auc: 1.0000
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0176 - auc: 1.0000
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0178 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0175 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0174 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0171 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0172 - auc: 1.0000
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0171 - auc: 1.0000 - val_l
oss: 0.2874 - val_auc: 1.0000
## Epoch 20/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0074 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0063 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0097 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0111 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0105 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0133 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0128 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0131 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0125 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0125 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0116 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0117 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0124 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0121 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0123 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0147 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0141 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0162 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0172 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0169 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0164 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0163 - auc: 1.0000
## 23/49 [=============>................] - ETA: 1s - loss: 0.0157 - auc: 1.0000
## 24/49 [=============>................] - ETA: 1s - loss: 0.0151 - auc: 1.0000
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0157 - auc: 1.0000
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0178 - auc: 1.0000
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0176 - auc: 1.0000
## 28/49 [================>.............] - ETA: 1s - loss: 0.0172 - auc: 1.0000
## 29/49 [================>.............] - ETA: 1s - loss: 0.0169 - auc: 1.0000
## 30/49 [=================>............] - ETA: 1s - loss: 0.0208 - auc: 0.9999
## 31/49 [=================>............] - ETA: 0s - loss: 0.0203 - auc: 0.9999
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0201 - auc: 0.9999
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0202 - auc: 0.9999
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0200 - auc: 0.9999
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0197 - auc: 0.9999
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0198 - auc: 0.9999
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0196 - auc: 0.9999
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0197 - auc: 0.9999
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0194 - auc: 0.9999
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0191 - auc: 0.9999
```

```
## 41/49 [=======================>.....] - ETA: 0s - loss: 0.0189 - auc: 0.9999
## 42/49 [=======================>.....] - ETA: 0s - loss: 0.0189 - auc: 0.9999
## 43/49 [========================>....] - ETA: 0s - loss: 0.0190 - auc: 0.9999
## 44/49 [========================>....] - ETA: 0s - loss: 0.0192 - auc: 0.9999
## 45/49 [=========================>...] - ETA: 0s - loss: 0.0193 - auc: 0.9999
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0192 - auc: 0.9999
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0191 - auc: 0.9999
## 48/49 [============================>.] - ETA: 0s - loss: 0.0188 - auc: 0.9999
## 49/49 [==============================] - ETA: 0s - loss: 0.0185 - auc: 0.9999
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0185 - auc: 0.9999 - val_l
oss: 0.2140 - val_auc: 0.9997
## Epoch 21/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0093 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0760 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0563 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0434 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0369 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0314 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0282 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0262 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0244 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0230 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0216 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0221 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0216 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0206 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0198 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0191 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0200 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0218 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0217 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0212 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0212 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0209 - auc: 1.0000
## 23/49 [=============>................] - ETA: 1s - loss: 0.0205 - auc: 1.0000
## 24/49 [=============>................] - ETA: 1s - loss: 0.0199 - auc: 1.0000
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0193 - auc: 1.0000
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0187 - auc: 1.0000
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0181 - auc: 1.0000
## 28/49 [================>.............] - ETA: 1s - loss: 0.0176 - auc: 1.0000
## 29/49 [================>.............] - ETA: 1s - loss: 0.0174 - auc: 1.0000
## 30/49 [=================>............] - ETA: 1s - loss: 0.0170 - auc: 1.0000
## 31/49 [=================>............] - ETA: 0s - loss: 0.0169 - auc: 1.0000
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0179 - auc: 1.0000
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0178 - auc: 1.0000
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0185 - auc: 1.0000
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0181 - auc: 1.0000
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0182 - auc: 1.0000
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0178 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0209 - auc: 0.9999
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0205 - auc: 0.9999
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0201 - auc: 1.0000
## 41/49 [=======================>.....] - ETA: 0s - loss: 0.0199 - auc: 1.0000
## 42/49 [=======================>.....] - ETA: 0s - loss: 0.0203 - auc: 0.9999
## 43/49 [========================>....] - ETA: 0s - loss: 0.0202 - auc: 0.9999
```

```
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0201 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0200 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0197 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0195 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0194 - auc: 1.0000
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0191 - auc: 1.0000 - val_l
oss: 0.2998 - val_auc: 1.0000
## Epoch 22/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0090 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0149 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0131 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0113 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0394 - auc: 0.9994
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0338 - auc: 0.9996
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0408 - auc: 0.9997
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0405 - auc: 0.9998
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0389 - auc: 0.9996
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0360 - auc: 0.9997
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0423 - auc: 0.9992
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0397 - auc: 0.9993
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0378 - auc: 0.9994
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0354 - auc: 0.9995
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0336 - auc: 0.9995
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0333 - auc: 0.9996
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0322 - auc: 0.9996
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0309 - auc: 0.9996
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0303 - auc: 0.9997
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0291 - auc: 0.9997
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0288 - auc: 0.9997
## 22/49 [============>.................] - ETA: 1s - loss: 0.0281 - auc: 0.9997
## 23/49 [=============>................] - ETA: 1s - loss: 0.0284 - auc: 0.9997
## 24/49 [=============>................] - ETA: 1s - loss: 0.0276 - auc: 0.9997
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0285 - auc: 0.9997
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0278 - auc: 0.9997
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0293 - auc: 0.9997
## 28/49 [================>.............] - ETA: 1s - loss: 0.0284 - auc: 0.9997
## 29/49 [================>.............] - ETA: 1s - loss: 0.0275 - auc: 0.9997
## 30/49 [=================>............] - ETA: 1s - loss: 0.0269 - auc: 0.9997
## 31/49 [=================>............] - ETA: 0s - loss: 0.0266 - auc: 0.9998
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0258 - auc: 0.9998
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0257 - auc: 0.9998
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0250 - auc: 0.9998
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0245 - auc: 0.9998
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0259 - auc: 0.9998
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0254 - auc: 0.9998
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0250 - auc: 0.9998
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0302 - auc: 0.9997
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0322 - auc: 0.9996
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0317 - auc: 0.9996
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0311 - auc: 0.9997
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0311 - auc: 0.9996
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0315 - auc: 0.9996
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0311 - auc: 0.9997
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0310 - auc: 0.9997
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0312 - auc: 0.9996
```

```
## 48/49 [============================>.] - ETA: 0s - loss: 0.0310 - auc: 0.9997
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0306 - auc: 0.9997 - val_l
oss: 0.4383 - val_auc: 0.9904
## Epoch 23/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0121 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0132 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0170 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0197 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0211 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0179 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0179 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0183 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0167 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0152 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0141 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0134 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0131 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0131 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0173 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0164 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0157 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0154 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0149 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0162 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0156 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0156 - auc: 1.0000
## 23/49 [=============>................] - ETA: 1s - loss: 0.0150 - auc: 1.0000
## 24/49 [=============>................] - ETA: 1s - loss: 0.0148 - auc: 1.0000
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0146 - auc: 1.0000
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0143 - auc: 1.0000
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0148 - auc: 1.0000
## 28/49 [================>.............] - ETA: 1s - loss: 0.0147 - auc: 1.0000
## 29/49 [================>.............] - ETA: 1s - loss: 0.0147 - auc: 1.0000
## 30/49 [=================>............] - ETA: 1s - loss: 0.0144 - auc: 1.0000
## 31/49 [=================>............] - ETA: 0s - loss: 0.0171 - auc: 1.0000
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0169 - auc: 1.0000
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0165 - auc: 1.0000
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0160 - auc: 1.0000
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0175 - auc: 1.0000
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0176 - auc: 1.0000
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0173 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0170 - auc: 1.0000
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0167 - auc: 1.0000
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0163 - auc: 1.0000
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0160 - auc: 1.0000
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0157 - auc: 1.0000
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0159 - auc: 1.0000
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0160 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0158 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0158 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0156 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0154 - auc: 1.0000
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0154 - auc: 1.0000 - val_l
oss: 0.4077 - val_auc: 0.9912
## Epoch 24/25
```

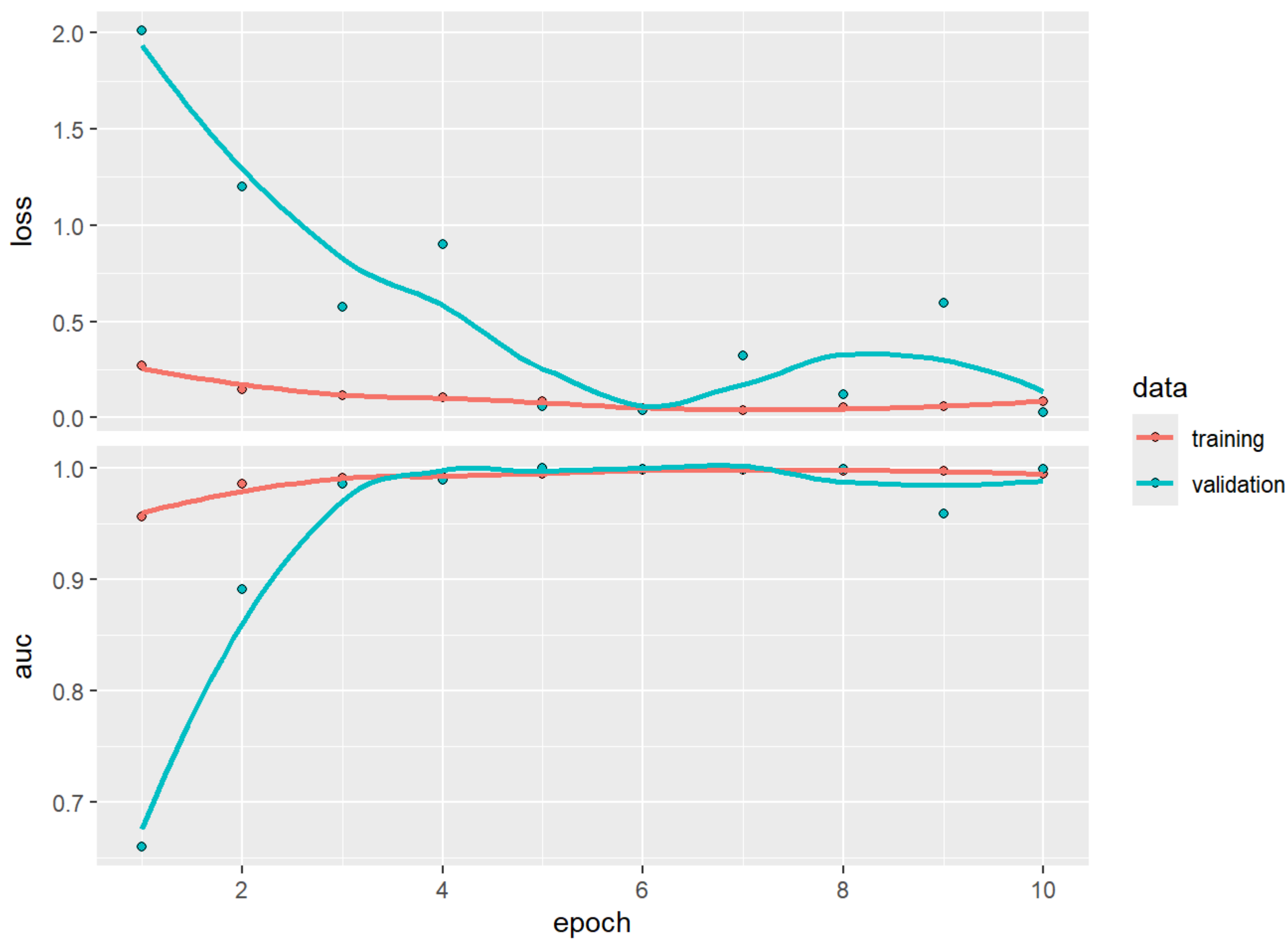
```
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0037 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0064 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0093 - auc: 1.0000
##  4/49 [=>............................] - ETA: 2s - loss: 0.0089 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0080 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0077 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0088 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0080 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0236 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0215 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0200 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0205 - auc: 0.9999
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0192 - auc: 0.9999
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0180 - auc: 0.9999
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0171 - auc: 0.9999
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0163 - auc: 0.9999
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0155 - auc: 0.9999
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0155 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0154 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0149 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0152 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0193 - auc: 0.9999
## 23/49 [=============>................] - ETA: 1s - loss: 0.0197 - auc: 0.9999
## 24/49 [=============>................] - ETA: 1s - loss: 0.0194 - auc: 0.9999
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0193 - auc: 0.9999
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0188 - auc: 0.9999
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0183 - auc: 0.9999
## 28/49 [================>.............] - ETA: 1s - loss: 0.0179 - auc: 0.9999
## 29/49 [================>.............] - ETA: 1s - loss: 0.0176 - auc: 0.9999
## 30/49 [=================>............] - ETA: 1s - loss: 0.0172 - auc: 0.9999
## 31/49 [=================>............] - ETA: 0s - loss: 0.0168 - auc: 0.9999
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0166 - auc: 0.9999
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0162 - auc: 0.9999
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0160 - auc: 0.9999
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0157 - auc: 0.9999
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0154 - auc: 1.0000
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0150 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0148 - auc: 1.0000
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0152 - auc: 1.0000
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0157 - auc: 1.0000
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0160 - auc: 0.9999
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0158 - auc: 0.9999
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0156 - auc: 0.9999
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0154 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0151 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0149 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0147 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0147 - auc: 1.0000
## 49/49 [==============================] - 3s 57ms/step - loss: 0.0148 - auc: 1.0000 - val_l
oss: 0.2065 - val_auc: 1.0000
## Epoch 25/25
##
##  1/49 [..............................] - ETA: 2s - loss: 0.0037 - auc: 1.0000
##  2/49 [>.............................] - ETA: 2s - loss: 0.0065 - auc: 1.0000
##  3/49 [>.............................] - ETA: 2s - loss: 0.0052 - auc: 1.0000
```

```
##  4/49 [=>............................] - ETA: 2s - loss: 0.0052 - auc: 1.0000
##  5/49 [==>...........................] - ETA: 2s - loss: 0.0050 - auc: 1.0000
##  6/49 [==>...........................] - ETA: 2s - loss: 0.0056 - auc: 1.0000
##  7/49 [===>..........................] - ETA: 2s - loss: 0.0050 - auc: 1.0000
##  8/49 [===>..........................] - ETA: 2s - loss: 0.0059 - auc: 1.0000
##  9/49 [====>.........................] - ETA: 2s - loss: 0.0075 - auc: 1.0000
## 10/49 [=====>........................] - ETA: 2s - loss: 0.0076 - auc: 1.0000
## 11/49 [=====>........................] - ETA: 2s - loss: 0.0078 - auc: 1.0000
## 12/49 [======>.......................] - ETA: 2s - loss: 0.0084 - auc: 1.0000
## 13/49 [======>.......................] - ETA: 1s - loss: 0.0097 - auc: 1.0000
## 14/49 [=======>......................] - ETA: 1s - loss: 0.0107 - auc: 1.0000
## 15/49 [========>.....................] - ETA: 1s - loss: 0.0106 - auc: 1.0000
## 16/49 [========>.....................] - ETA: 1s - loss: 0.0102 - auc: 1.0000
## 17/49 [=========>....................] - ETA: 1s - loss: 0.0105 - auc: 1.0000
## 18/49 [==========>...................] - ETA: 1s - loss: 0.0102 - auc: 1.0000
## 19/49 [==========>...................] - ETA: 1s - loss: 0.0104 - auc: 1.0000
## 20/49 [===========>..................] - ETA: 1s - loss: 0.0104 - auc: 1.0000
## 21/49 [===========>..................] - ETA: 1s - loss: 0.0102 - auc: 1.0000
## 22/49 [============>.................] - ETA: 1s - loss: 0.0103 - auc: 1.0000
## 23/49 [=============>................] - ETA: 1s - loss: 0.0101 - auc: 1.0000
## 24/49 [=============>................] - ETA: 1s - loss: 0.0098 - auc: 1.0000
## 25/49 [==============>...............] - ETA: 1s - loss: 0.0099 - auc: 1.0000
## 26/49 [==============>...............] - ETA: 1s - loss: 0.0096 - auc: 1.0000
## 27/49 [===============>..............] - ETA: 1s - loss: 0.0143 - auc: 1.0000
## 28/49 [================>.............] - ETA: 1s - loss: 0.0140 - auc: 1.0000
## 29/49 [================>.............] - ETA: 1s - loss: 0.0137 - auc: 1.0000
## 30/49 [=================>............] - ETA: 1s - loss: 0.0135 - auc: 1.0000
## 31/49 [=================>............] - ETA: 0s - loss: 0.0133 - auc: 1.0000
## 32/49 [==================>...........] - ETA: 0s - loss: 0.0130 - auc: 1.0000
## 33/49 [===================>..........] - ETA: 0s - loss: 0.0145 - auc: 1.0000
## 34/49 [===================>..........] - ETA: 0s - loss: 0.0144 - auc: 1.0000
## 35/49 [====================>.........] - ETA: 0s - loss: 0.0145 - auc: 1.0000
## 36/49 [=====================>........] - ETA: 0s - loss: 0.0144 - auc: 1.0000
## 37/49 [=====================>........] - ETA: 0s - loss: 0.0156 - auc: 1.0000
## 38/49 [======================>.......] - ETA: 0s - loss: 0.0154 - auc: 1.0000
## 39/49 [======================>.......] - ETA: 0s - loss: 0.0160 - auc: 1.0000
## 40/49 [=======================>......] - ETA: 0s - loss: 0.0159 - auc: 1.0000
## 41/49 [========================>.....] - ETA: 0s - loss: 0.0158 - auc: 1.0000
## 42/49 [========================>.....] - ETA: 0s - loss: 0.0157 - auc: 1.0000
## 43/49 [=========================>....] - ETA: 0s - loss: 0.0154 - auc: 1.0000
## 44/49 [=========================>....] - ETA: 0s - loss: 0.0152 - auc: 1.0000
## 45/49 [==========================>...] - ETA: 0s - loss: 0.0150 - auc: 1.0000
## 46/49 [===========================>..] - ETA: 0s - loss: 0.0150 - auc: 1.0000
## 47/49 [===========================>..] - ETA: 0s - loss: 0.0148 - auc: 1.0000
## 48/49 [============================>.] - ETA: 0s - loss: 0.0146 - auc: 1.0000
## 49/49 [==============================] - 3s 58ms/step - loss: 0.0144 - auc: 1.0000 - val_l
oss: 0.1394 - val_auc: 1.0000
```

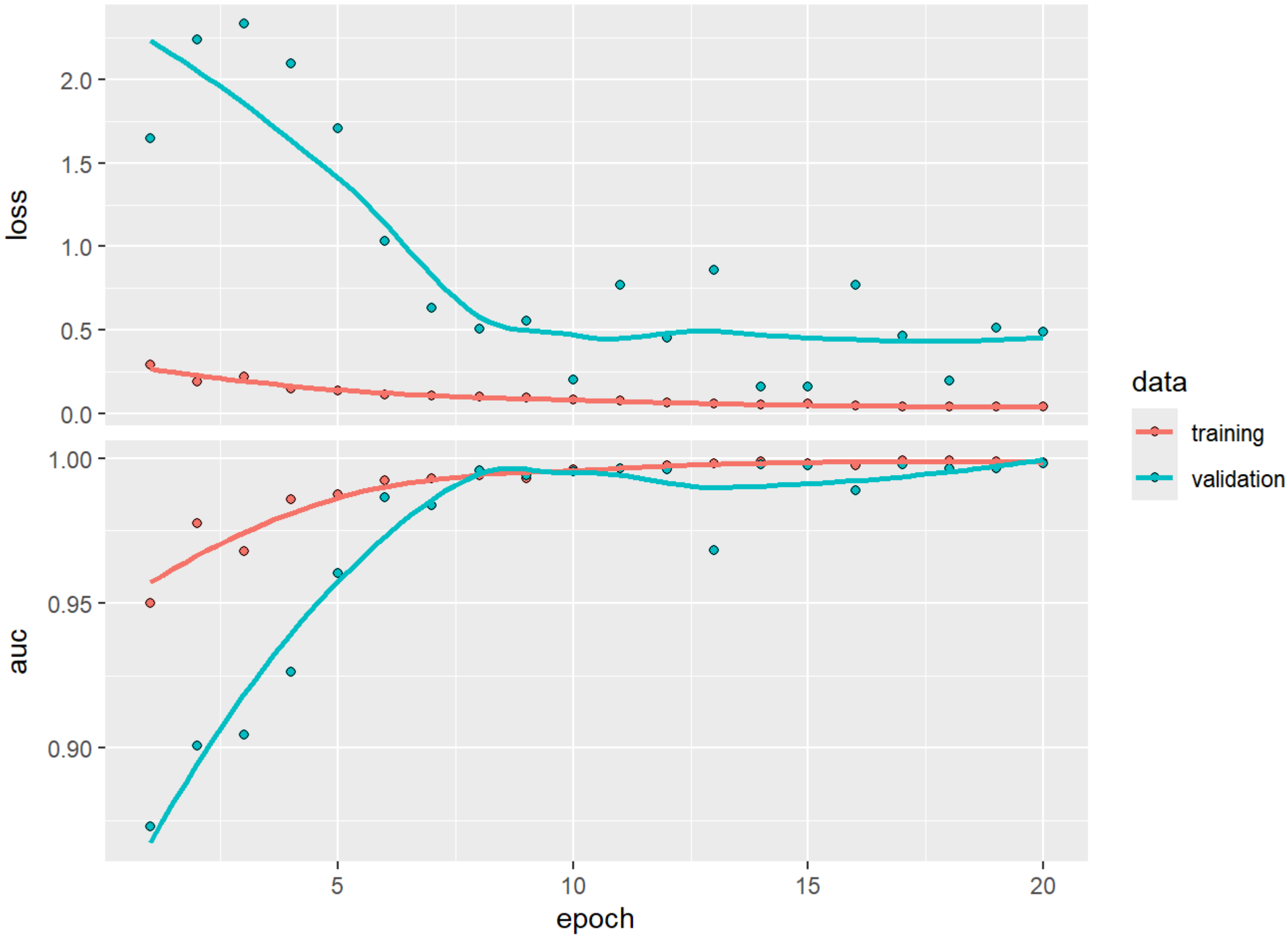
12. Plot training and validation AUC:

```
############### Turn to comments from the second run onwards ###############
# Plot training and validation loss and AUC:
for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    msg <- paste0("Adjusted OvR model ", label_map[class[i]], " vs ", neg_labels, ": \n")
    cat(crayon::bold(msg))
    }
  else {
    # Other models
    msg <- paste0("Adjusted OvR model ", label_map[class[i]], " vs ", "(", neg_labels, "):
\n")
    cat(crayon::bold(msg))
    }

  print(plot(history_list[[i]]))
  cat("\n")
}
```

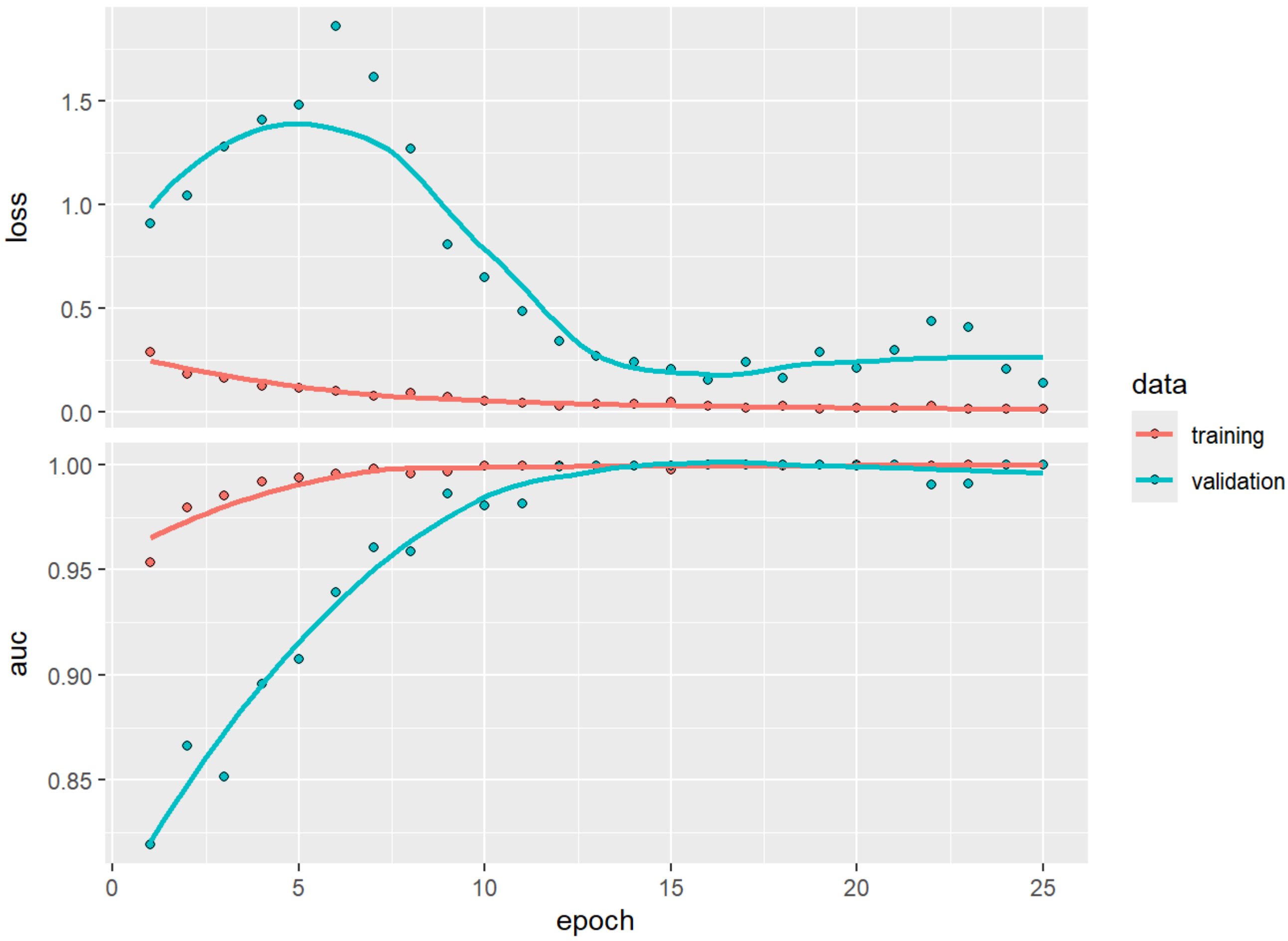
```
## Adjusted OvR model NoT vs (Gl, Mn, Pt):
```




```
##
## Adjusted OvR model Gl vs (Mn, Pt):
```




```
##
## Adjusted OvR model Mn vs Pt:
```



13. Using the saved best model weights, evaluate the model's AUC on training and test set:

```
output_train_ovr <- data.frame()
output_test_ovr <- data.frame()
labels_test_pred_ovr <- c()

for(i in 1:(length(class) - 1)){
  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")
  neg_labels_ <- paste0(label_map[remaining_classes[-1]], collapse = "-")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    mod <- paste0(label_map[class[i]], " vs ", neg_labels)
    }
  else {
    # Other models
    mod <- paste0(label_map[class[i]], " vs ", "(", neg_labels, ")")
  }

  # Load the saved best model weights to the model
  weights_filepath <- paste0("braintumor_best_model_weights_", label_map[class[i]], "_vs_", n
eg_labels_, ".h5")
  my_models_ovr[[i]] %>% load_model_weights_tf(weights_filepath)

  ### Evaluate the model on training set:
  eval_train <- evaluate(my_models_ovr[[i]], X_train_ovr[[i]], labels_train_ovr[[i]], verbose
= 0)     # Compute training loss and AUC

  ### Evaluate the model on test set:
  labels_test_pred <- predict(my_models_ovr[[i]], X_test_ovr[[i]], verbose = 0)
  labels_test_pred_ovr <- append(labels_test_pred_ovr, list(labels_test_pred))

  eval_test <- evaluate(my_models_ovr[[i]], X_test_ovr[[i]], labels_test_ovr[[i]], verbose =
0)      # Compute test loss and AUC

  output_train_ovr <- rbind(output_train_ovr, data.frame(Model = mod, Loss = eval_train[1], A
UC = eval_train[2]))
  output_test_ovr <- rbind(output_test_ovr, data.frame(Model = mod, Loss = eval_test[1], AUC
= eval_test[2]))
}

labels_test_pred_ovr <- lapply(labels_test_pred_ovr, as.vector)

rownames(output_train_ovr) <- NULL
rownames(output_test_ovr) <- NULL

cat("Loss and AUC on the training set for each Adjusted OvR model: \n")
```

```
## Loss and AUC on the training set for each Adjusted OvR model:
```

```
print(output_train_ovr)
```

```
##                  Model       Loss       AUC
## 1 NoT vs (Gl, Mn, Pt) 0.01086769 0.9999817
## 2      Gl vs (Mn, Pt) 0.06404608 0.9999385
## 3            Mn vs Pt 0.01743406 1.0000000
```

```
cat("\n")
```

```
cat("Loss and AUC on the test set for each Adjusted OvR model: \n")
```

```
## Loss and AUC on the test set for each Adjusted OvR model:
```

```
print(output_test_ovr)
```

```
##                  Model       Loss       AUC
## 1 NoT vs (Gl, Mn, Pt) 0.02913978 0.9980397
## 2      Gl vs (Mn, Pt) 0.15163811 0.9954776
## 3            Mn vs Pt 0.28011245 0.9732502
```

B/ Choosing decision threshold by ROC curve and decision curve:

1. **Criterias to determine the decision threshold for the CNN model:**

```
mod_names <- c()

for (i in 1:(length(class) - 1)) {
  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    mod_name <- paste0(label_map[class[i]], " vs ", neg_labels)
  }
  else {
    # Other models
    mod_name <- paste0(label_map[class[i]], " vs ", "(", neg_labels, ")")
  }

  mod_names <- c(mod_names, mod_name)
}
```

- **Youden index criterion:**

```
# Optimal thresholds obtained by maximizing youden index
j_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  j <- tpr - fpr
  j <- sort(j, decreasing = TRUE, index.return = TRUE)
  j_ordered <- data.frame(j = j[1], thresholds = thresholds[rank(j$ix)])
  for (k in 1:nrow(j_ordered)){
    if (is.finite(j_ordered$thresholds[k])) {
      return(j_ordered$thresholds[k])
      break
      }
  }
}

cj_ovr <- c()

for (i in 1:(length(class) - 1)){
  cj <- j_threshold(labels_test_ovr[[i]], labels_test_pred_ovr[[i]])
  cj_ovr <- append(cj_ovr, cj)
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
names(cj_ovr) <- mod_names

cj_ovr
```

```
## NoT vs (Gl, Mn, Pt)      Gl vs (Mn, Pt)           Mn vs Pt
##          0.75128877          0.82267696         0.09279775
```

- **Closest to (0,1) criterion:**

```
# Optimal threshold obtained by maximizing sensitivity and specificity to the extent that bot
h are closest to 100%
d_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  d <- sapply(1:length(fpr), function(i) sqrt((1 - tpr[i])^2 + (fpr[i])^2))
  d <- sort(d, decreasing = FALSE, index.return = TRUE)
  d_ordered <- data.frame(d = d[1], thresholds = thresholds[rank(d$ix)])
  for (k in 1:nrow(d_ordered)){
    if (is.finite(d_ordered$thresholds[k])) {
      return(d_ordered$thresholds[k])
      break
      }
  }
}

cd_ovr <- c()

for (i in 1:(length(class) - 1)){
  cd <- d_threshold(labels_test_ovr[[i]], labels_test_pred_ovr[[i]])
  cd_ovr <- append(cd_ovr, cd)
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
names(cd_ovr) <- mod_names

cd_ovr
```

```
## NoT vs (Gl, Mn, Pt)      Gl vs (Mn, Pt)           Mn vs Pt
##          0.75128877          0.82267696         0.04816283
```

- **Symetric point criterion:**

```
# Optimal threshold obtained by finding the point where sensitivity and specificity are equiv
alent
s_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  s <- abs(fpr + tpr - 1)
  s <- sort(s, decreasing = FALSE, index.return = TRUE)
  s_ordered <- data.frame(s = s[1], thresholds = thresholds[rank(s$ix)])
  for (k in 1:nrow(s_ordered)){
    if (is.finite(s_ordered$thresholds[k])) {
      return(s_ordered$thresholds[k])
      break
      }
  }
}

cs_ovr <- c()

for (i in 1:(length(class) - 1)){
  cs <- s_threshold(labels_test_ovr[[i]], labels_test_pred_ovr[[i]])
  cs_ovr <- append(cs_ovr, cs)
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
names(cs_ovr) <- mod_names

cs_ovr
```

```
## NoT vs (Gl, Mn, Pt)      Gl vs (Mn, Pt)           Mn vs Pt
##          0.75128877          0.80522388         0.02507351
```

2. Plot ROC curve, decision curve and optimal thresholds for the CNN model:

```
# Plot ROC curve
thresholds_cnn_ovr <- c()
plot_list <- c()

for (i in 1:(length(class) - 1)){
  roc_obj <- pROC::roc(labels_test_ovr[[i]], labels_test_pred_ovr[[i]])
  thresholds_cnn <- roc_obj$thresholds
  thresholds_cnn_ovr <- append(thresholds_cnn_ovr, list(thresholds_cnn))
  roc_data <- data.frame(fpr = rev(1 - roc_obj$specificities), tpr = rev(roc_obj$sensitivitie
s))

  for (k in 1:length(roc_obj$thresholds)){
    if (roc_obj$thresholds[k] == cj_ovr[i]){
      specificity_j <- roc_obj$specificities[k]
      sensitivity_j <- roc_obj$sensitivities[k]
    }
    if (roc_obj$thresholds[k] == cd_ovr[i]){
      specificity_d <- roc_obj$specificities[k]
      sensitivity_d <- roc_obj$sensitivities[k]
    }
    if (roc_obj$thresholds[k] == cs_ovr[i]){
      specificity_s <- roc_obj$specificities[k]
      sensitivity_s <- roc_obj$sensitivities[k]
    }
  }

  diagonal_data <- data.frame(x = seq(0, 1, length.out = 100), y = 1 - seq(0, 1, length.out =
100))
  thresholds_data <- data.frame(x1 = 0, x2 = 1 - specificity_d, y1 = 1, y2 = sensitivity_d)
  sp_se_j <- data.frame(sp = specificity_j, se = sensitivity_j)
  sp_se_d <- data.frame(sp = specificity_d, se = sensitivity_d)
  sp_se_s <- data.frame(sp = specificity_s, se = sensitivity_s)

  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    plot_title <- paste0(label_map[class[i]], " vs ", neg_labels)
    }
  else {
    # Other models
    plot_title <- paste0(label_map[class[i]], " vs ", "(", neg_labels, ")")
    }

  rc <- ggplot(data = roc_data, aes(x = fpr, y = tpr)) +
  geom_line(linetype = "solid", linewidth = 1, col = "blue") +
  geom_line(data = diagonal_data, aes(x = x, y = y), linetype = "dashed", linewidth = 1, col
= "orange") +
  geom_segment(data = thresholds_data, aes(x = x1, y = y1, xend = x2, yend = y2), linetype =
"dotdash", linewidth = 1, col = "pink") +
  geom_point(data = sp_se_j, aes(x = 1 - sp, y = se, col = "c_j"), size = 4, show.legend = TR
UE) +
  geom_point(data = sp_se_d, aes(x = 1 - sp, y = se, col = "c_d"), size = 4, show.legend = TR
UE) +
```

```
  geom_point(data = sp_se_s, aes(x = 1 - sp, y = se, col = "c_s"), size = 4, show.legend = TR
UE) +
  labs(x = "False positive rate", y = "True positive rate") + ggtitle(plot_title) +
  scale_color_manual(name = "Threshold: ", labels = unname(TeX(c("At $\\hat{c}_{J}$", "At
$\\hat{c}_{D}$", "At $\\hat{c}_{S}$"))), values=c("red", "purple", "green")) +
  theme_bw() +
  theme(plot.title = element_text(face="bold", hjust = 0.5, size = 13))
  plot_list[[i]] <- rc
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

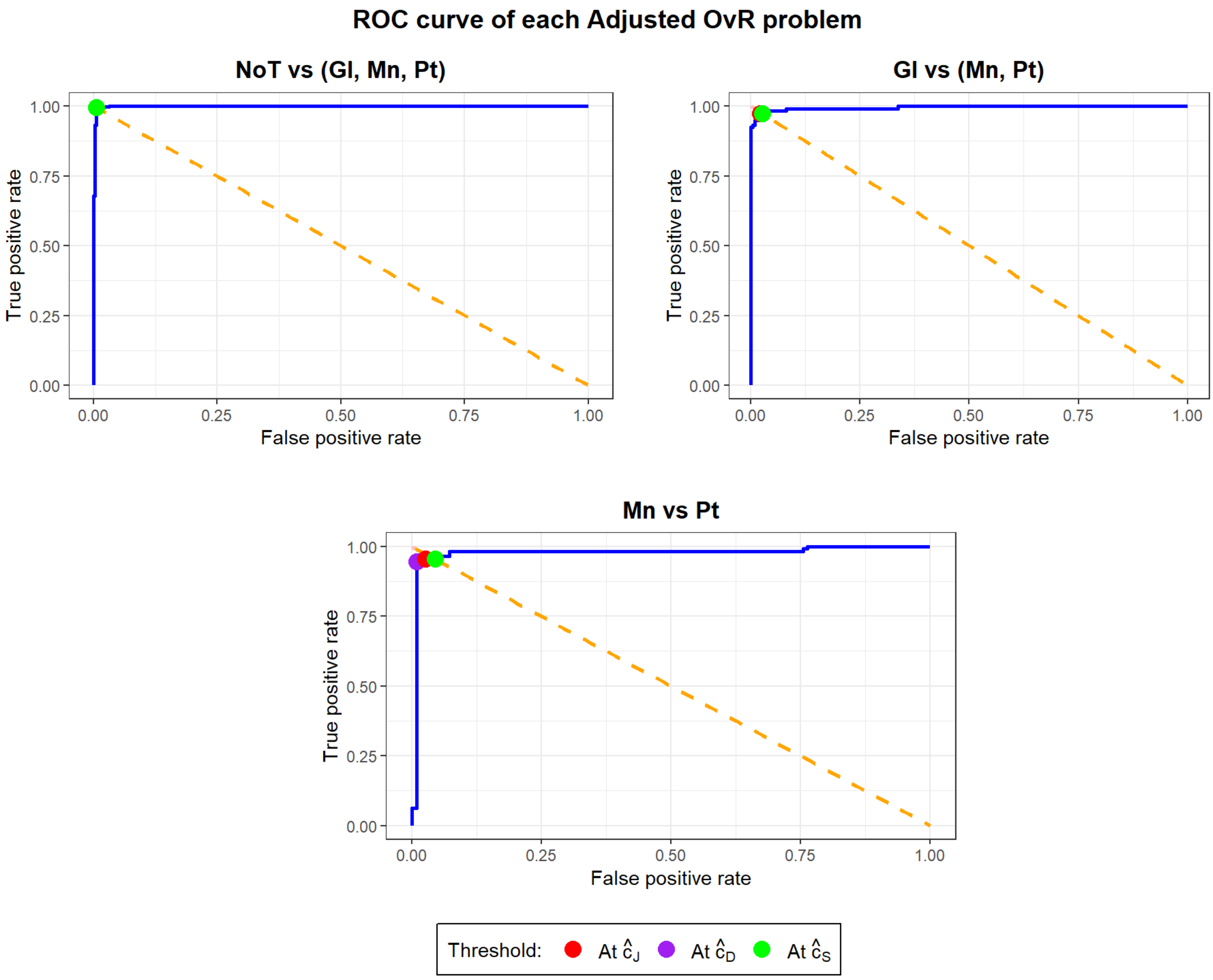
```
names(plot_list) <- c("A", "B", "C")

layout <- "AABB
           #CC#"

# Combine all plots in the list into a grid layout (2 columns)
combined_plot <- wrap_plots(plot_list, design = layout, ncol = 2)

legend <- get_legend(plot_list[[1]])
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```

```
# Arrange the combined plot with a shared legend and title
final_plot <- combined_plot + plot_layout(guides = "collect") &
  theme(plot.margin = unit(c(2, 1, 0, 1), "lines"), legend.position = "bottom", legend.backgr
ound = element_rect(color = "black", size = 0.5, linetype = "solid"), legend.text = element_t
ext(size = 11), legend.box.margin = unit(c(0.6, 1, 1, 1), "lines"))   # Add 2 lines at the to
p (plot.margin = unit(c(top,right,bottom,left), "lines")
```

```
## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(final_plot)
grid.text("ROC curve of each Adjusted OvR problem", x = 0.5, y = 0.94, gp = gpar(fontsize = 1
4, fontface = "bold"))
```



```
# Plot decision curve

# Net benefit function
nb <- function(y_true, y_pred, c, prev){
  roc_obj <- pROC::roc(y_true, y_pred)
  for (i in 1:length(roc_obj$thresholds)){
    if (roc_obj$thresholds[i] == c){
      se_c <- roc_obj$sensitivities[i]
      sp_c <- roc_obj$specificities[i]
      break
    }
  }
  phi_c <- se_c * prev - (1 - sp_c) * (1 - prev) * c/(1 - c)
  return(phi_c)
}

plot_list <- c()

for (i in 1:(length(class) - 1)){
  # Calculate disease prevalence in the population:
  # p = number of diseased / total population size (calculate from whole dataset (labels) or
just test set (labels_test)?)
  p <- sum(labels_test_ovr[[i]]) / length(labels_test_ovr[[i]])

  df_cnn <- data.frame(labels_test_pred_ = labels_test_pred_ovr[[i]], labels_test_ = labels_t
est_ovr[[i]])
  nb_j <- data.frame(cj = cj_ovr[i], nbj = nb(labels_test_ovr[[i]], labels_test_pred_ovr
[[i]], cj_ovr[i], p))
  nb_d <- data.frame(cd = cd_ovr[i], nbd = nb(labels_test_ovr[[i]], labels_test_pred_ovr
[[i]], cd_ovr[i], p))
  nb_s <- data.frame(cs = cs_ovr[i], nbs = nb(labels_test_ovr[[i]], labels_test_pred_ovr
[[i]], cs_ovr[i], p))

  remaining_classes <- class[i:length(class)]
  neg_labels <- paste0(label_map[remaining_classes[-1]], collapse = ", ")

  if (i == length(class) - 1) {
    # Last model - only two classes left
    plot_title <- paste0(label_map[class[i]], " vs ", neg_labels)
    }
  else {
    # Other models
    plot_title <- paste0(label_map[class[i]], " vs ", "(", neg_labels, ")")
    }

  # For the Decision curve plot, use the thresholds values from ROC curve plot to make sure t
hat the optimal thresholds show on the decision curve correctly
  dca_obj <- dcurves::dca(labels_test_ ~ labels_test_pred_, data = df_cnn, label = list(label
s_test_pred_ = "CNN model"), thresholds = thresholds_cnn_ovr[[i]], prevalence = p)
  dc <- plot(dca_obj, smooth = FALSE) + geom_line(linewidth = 1) +
    ggtitle(plot_title) +
    geom_point(data = nb_j, aes(x = cj, y = nbj, col = "c_j"), size = 4, show.legend = TRUE)
+
    geom_point(data = nb_d, aes(x = cd, y = nbd, col = "c_d"), size = 4, show.legend = TRUE)
+
```

```
    geom_point(data = nb_s, aes(x = cs, y = nbs, col = "c_s"), size = 4, show.legend = TRUE)
+
    scale_color_manual(labels = unname(TeX(c("Treat All", "Treat None", "CNN model", "At $\\h
at{c}_{J}$", "At $\\hat{c}_{D}$", "At $\\hat{c}_{S}$"))), values=c("orange", "pink", "blue",
"red", "purple", "green")) +
    guides(colour = guide_legend(byrow = TRUE, override.aes = list(linetype = c("solid", "sol
id", "solid", rep("blank", 3)), shape = c(NA, NA, NA, rep(16, 3))))) + theme(plot.title = ele
ment_text(face="bold", hjust = 0.5, size = 13))
  plot_list[[i]] <- dc
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Assuming '1' is [Event] and '0' is [non-Event]
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Assuming '1' is [Event] and '0' is [non-Event]
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

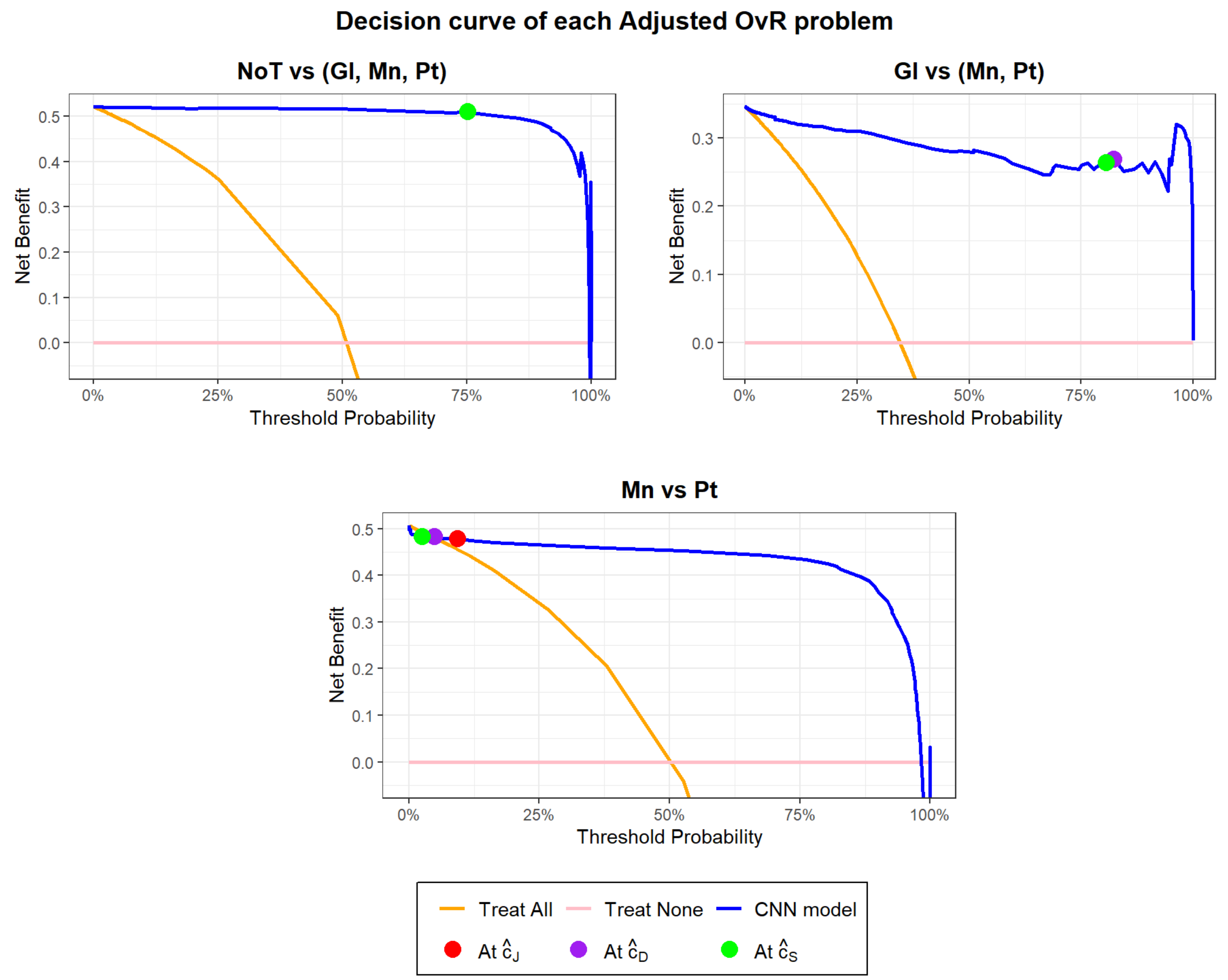
```
## Setting direction: controls < cases
```

```
## Assuming '1' is [Event] and '0' is [non-Event]
```

```
names(plot_list) <- c("A", "B", "C")

layout <- "AABB
           #CC#"

# Combine all plots in the list into a grid layout (2 columns)
combined_plot <- wrap_plots(plot_list, design = layout, ncol = 2)

legend <- get_legend(plot_list[[1]])
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```

```
# Arrange the combined plot with a shared legend and title
final_plot <- combined_plot + plot_layout(guides = "collect") &
  theme(plot.margin = unit(c(2, 1, 0, 1), "lines"), legend.position = "bottom", legend.backgr
ound = element_rect(color = "black", size = 0.5, linetype = "solid"), legend.text = element_t
ext(size = 11), legend.box.margin = unit(c(0.6, 1, 1, 1), "lines"))   # Add 2 lines at the to
p (plot.margin = unit(c(top,right,bottom,left), "lines")

print(final_plot)
grid.text("Decision curve of each Adjusted OvR problem", x = 0.5, y = 0.94, gp = gpar(fontsiz
e = 14, fontface = "bold"))
```




```
best_thresholds <- c()
c_res <- c()

for (i in 1:(length(class) - 1)) {
  p <- sum(labels_test_ovr[[i]]) / length(labels_test_ovr[[i]])

  nb_j <- nb(labels_test_ovr[[i]], labels_test_pred_ovr[[i]], unname(cj_ovr[i]), p)
  nb_d <- nb(labels_test_ovr[[i]], labels_test_pred_ovr[[i]], unname(cd_ovr[i]), p)
  nb_s <- nb(labels_test_ovr[[i]], labels_test_pred_ovr[[i]], unname(cs_ovr[i]), p)
  nb_values <- c(cj = nb_j, cd = nb_d, cs = nb_s)

  best_threshold <- names(which.max(nb_values))   # Name of the threshold with highest net be
nefit
  best_thresholds <- c(best_thresholds, best_threshold)
  threshold_map <- list(cj = cj_ovr[i], cd = cd_ovr[i], cs = cs_ovr[i])
  c_res <- c(c_res, threshold_map[[best_threshold]])   # Get the optimal threshold's value
}
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
cat(crayon::bold("Optimal threshold of each Adjusted OvR model: \n"))
```

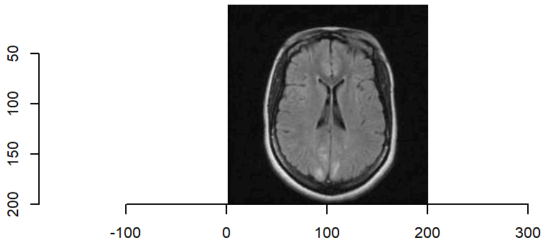
```
## Optimal threshold of each Adjusted OvR model:
```

```
for (i in 1:(length(class) - 1)) {
  cat(mod_names[i], ": ", best_thresholds[i], " = ", c_res[i], "\n", sep = "")
}
```

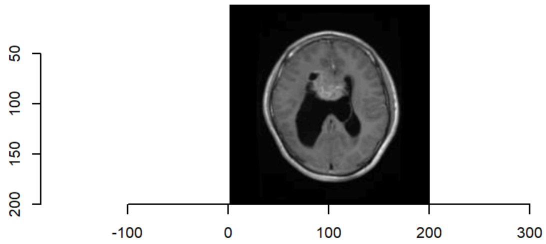
```
## NoT vs (Gl, Mn, Pt): cj = 0.7512888
## Gl vs (Mn, Pt): cj = 0.822677
## Mn vs Pt: cs = 0.02507351
```

```
## Example: Predicting the class of a new data point with CNN model
new_obs_ <- X_test[2:7]  # Some observations from the test set
new_obs <- do.call(abind, new_obs_)
new_obs <- aperm(new_obs, perm = c(4,1,2,3))
# In the procedure: You need to preprocess (Convert into the correct format, Resize, Normaliz
e, Reshape, ... in the same method and with the same values calculated from the preprocessing
step of the training set) the new data if needed
actual_classes <- labels_test[2:7]
class_preds <- c()
mods_data <- c()

for (i in 1:nrow(new_obs)) {
  pred_class <- NA  # Default if no model meets threshold
  labels_pred_ovr <- c()

  # Sequentially check each Adjusted OvR model
  for (j in 1:(length(class) - 1)) {
    # Predict probability
    labels_pred <- predict(my_models_ovr[[j]], new_obs[i,,, , drop = FALSE], verbose = 0)
    labels_pred <- round(labels_pred, digits = 7)
    labels_pred_ovr <- c(labels_pred_ovr, labels_pred)

    # Compare with threshold
    if (labels_pred >= c_res[j]) {  # thresholds[j] is the optimal threshold for the j-th Adj
usted OvR model
      pred_class <- class[j]  # Predict as the positive class of this model
      break  # Stop at first model that meets the threshold
    }
  }

  # If none met the threshold, assign to the last class
  if (is.na(pred_class)) {
    pred_class <- class[length(class)]
  }

  class_preds <- c(class_preds, pred_class)

  # Save all model's probabilities
  if (length(labels_pred_ovr) < (length(class) - 1)) {
    labels_pred_ovr <- c(labels_pred_ovr, rep(NA, (length(class) - 1) - length(labels_pred_ov
r)))   # Making predicted probabilities vector of all samples have the same length
  }

  mods_data[[i]] <- c(paste0("Observation ", i), labels_pred_ovr)
}

# Final result
par(mfrow = c(3, 2))  # Create subplots with 2 rows and 3 columns

for (i in 1:nrow(new_obs)){
  plot(as.cimg(new_obs[i,,,]))
  title(main =  paste0("Observation ", i, "\n Predicted class: ", class_preds[i], "; Actual c
lass: ", actual_classes[i]), cex.main = 1.4)
}
```

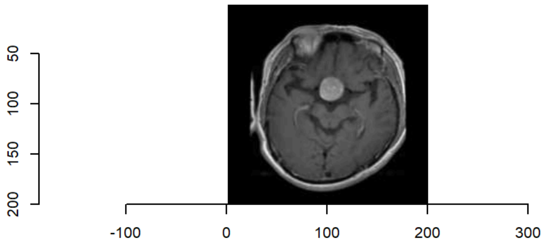
Observation 1
Predicted class: NoTumor; Actual class: NoTumor



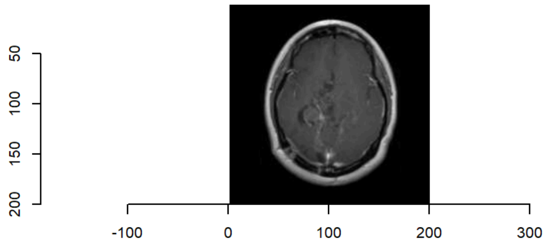
Observation 2
Predicted class: Glioma; Actual class: Glioma



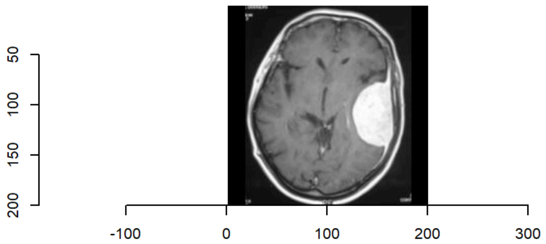
Observation 3
Predicted class: Pituitary; Actual class: Pituitary



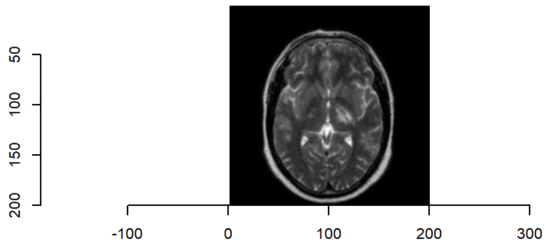
Observation 4
Predicted class: Glioma; Actual class: Glioma



Observation 5
Predicted class: Meningioma; Actual class: Meningioma



Observation 6
Predicted class: NoTumor; Actual class: NoTumor




```r
# Specifically, each model's predicted probabilities are as follows
mods_prob <- do.call(rbind, mods_data)
colnames(mods_prob) <- c("Observation", mod_names)
mods_prob <- as.data.frame(mods_prob)
print(mods_prob)
```

```
##     Observation NoT vs (Gl, Mn, Pt) Gl vs (Mn, Pt) Mn vs Pt
## 1 Observation 1           0.9989261           <NA>     <NA>
## 2 Observation 2           0.0020926      0.9980562     <NA>
## 3 Observation 3           0.0036063      0.0461544 1.85e-05
## 4 Observation 4           0.0007578      0.9992962     <NA>
## 5 Observation 5           0.0866392       0.016209  0.89604
## 6 Observation 6            0.997558           <NA>     <NA>
```